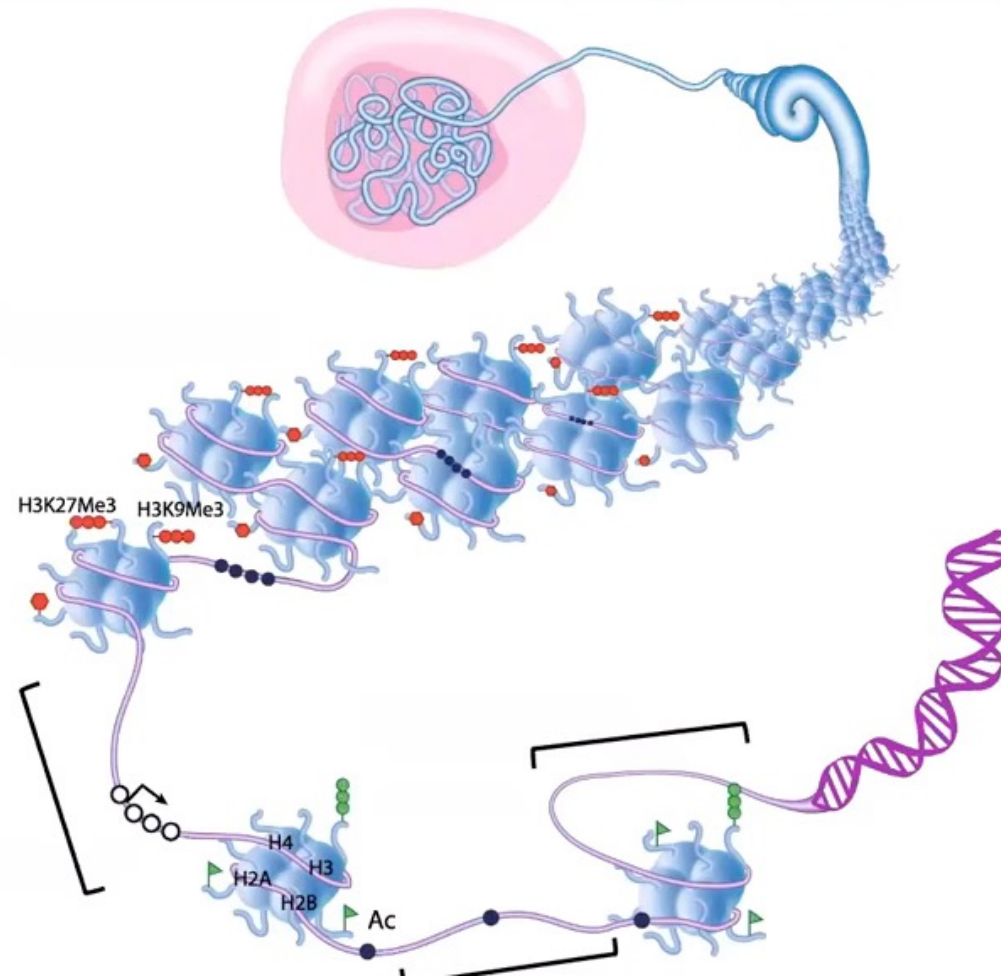


The detecting techniques for chromatin states

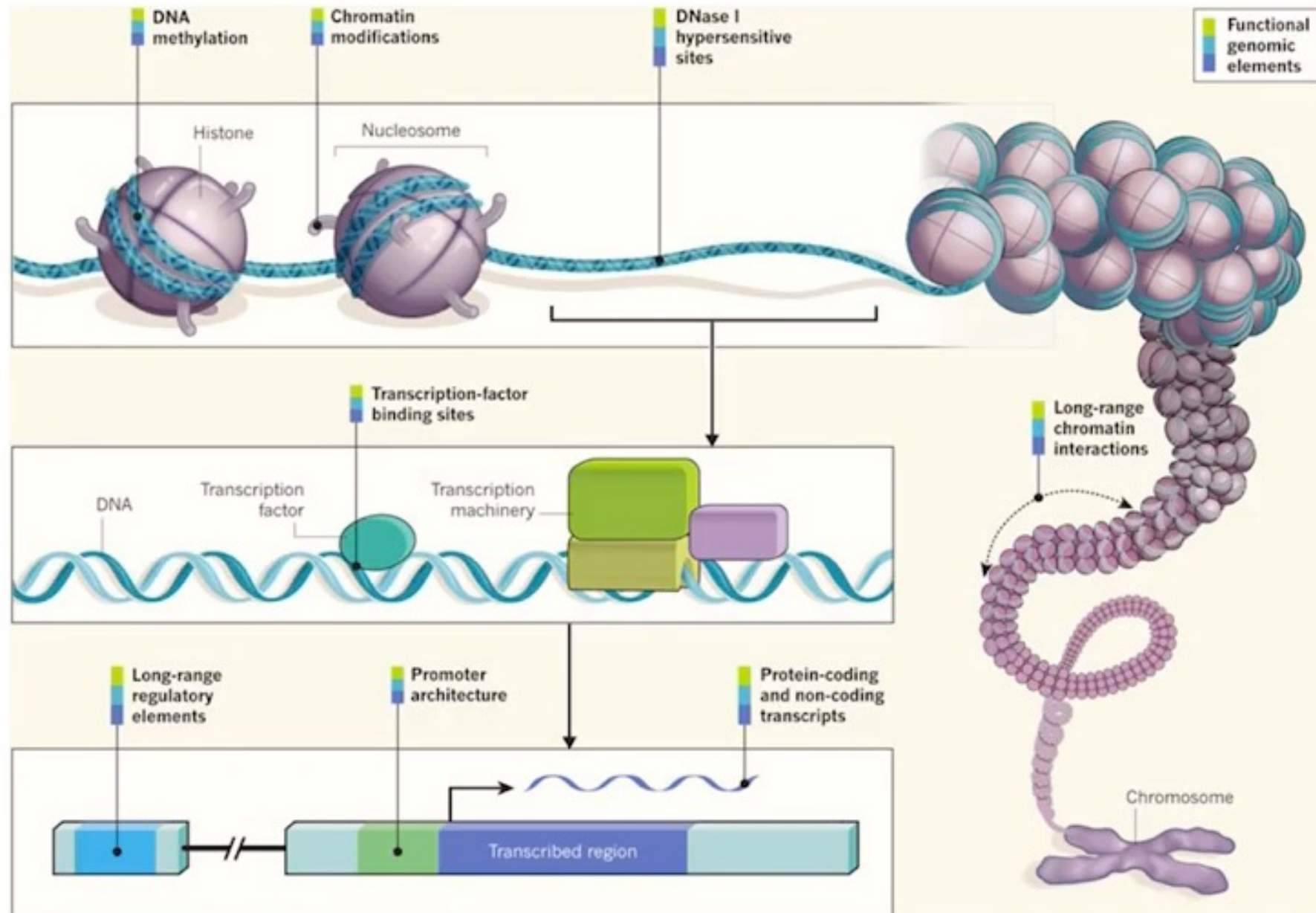
Mapping Open Chromatin

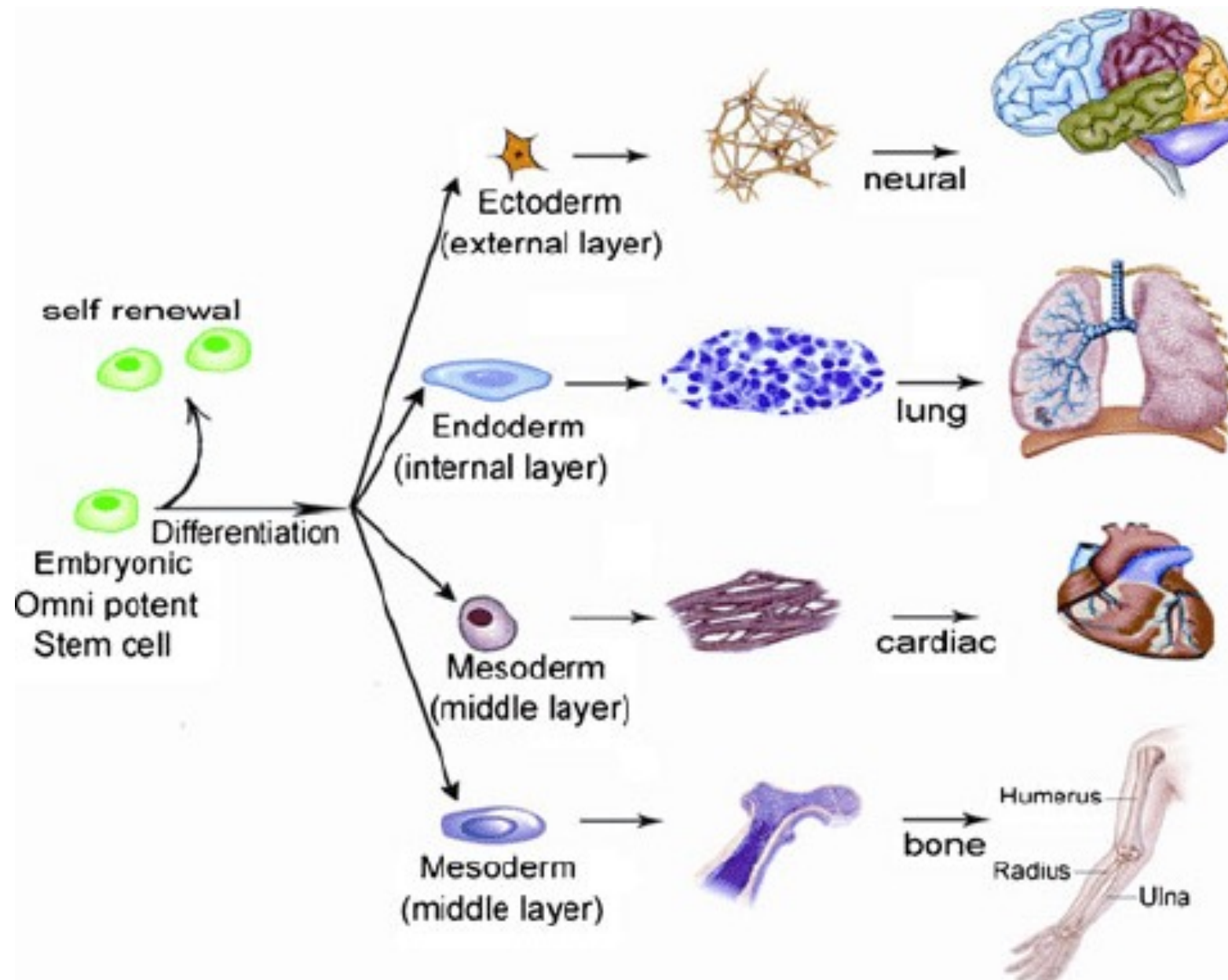


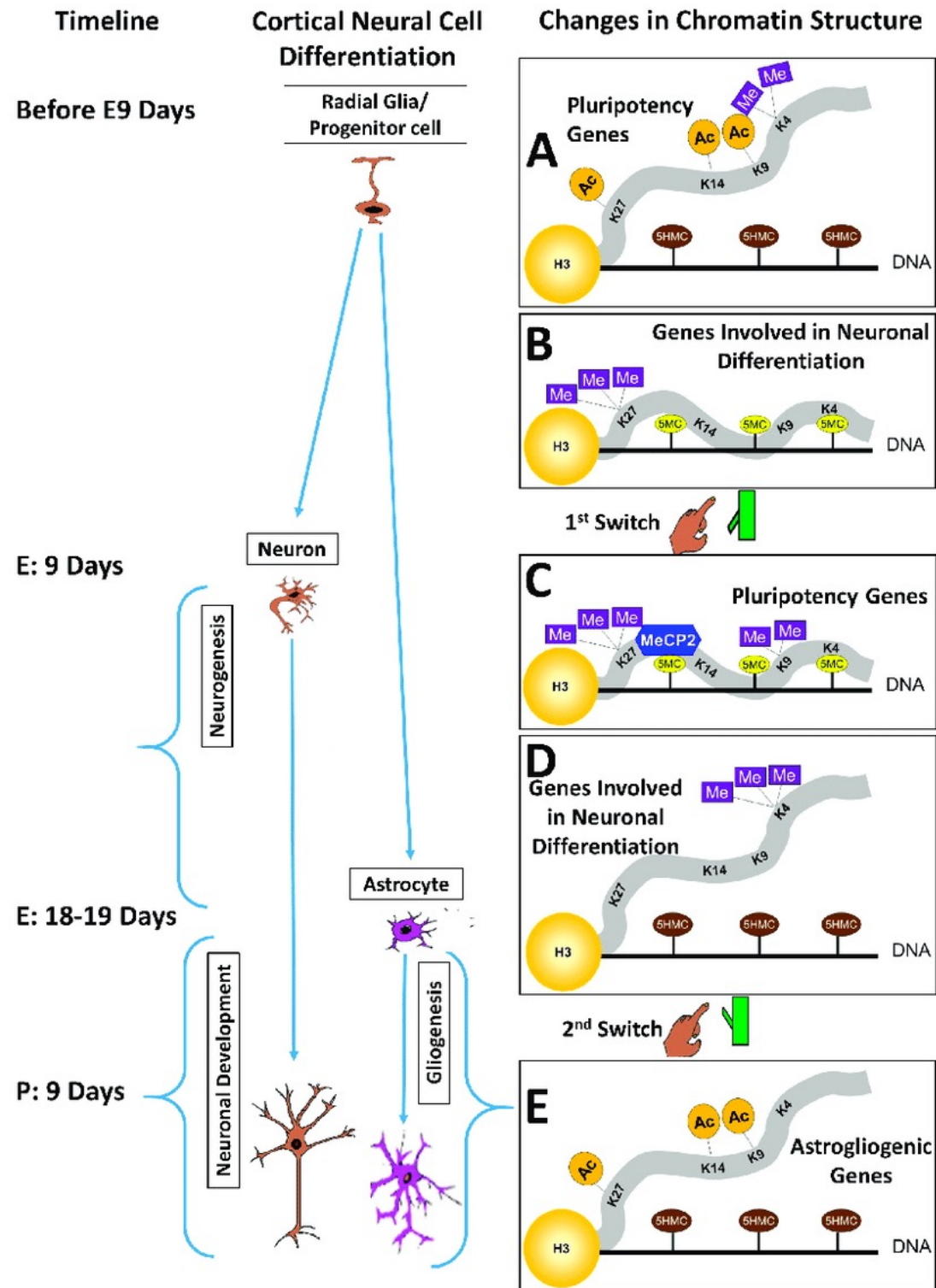
These open regions
can be sequenced
selectively

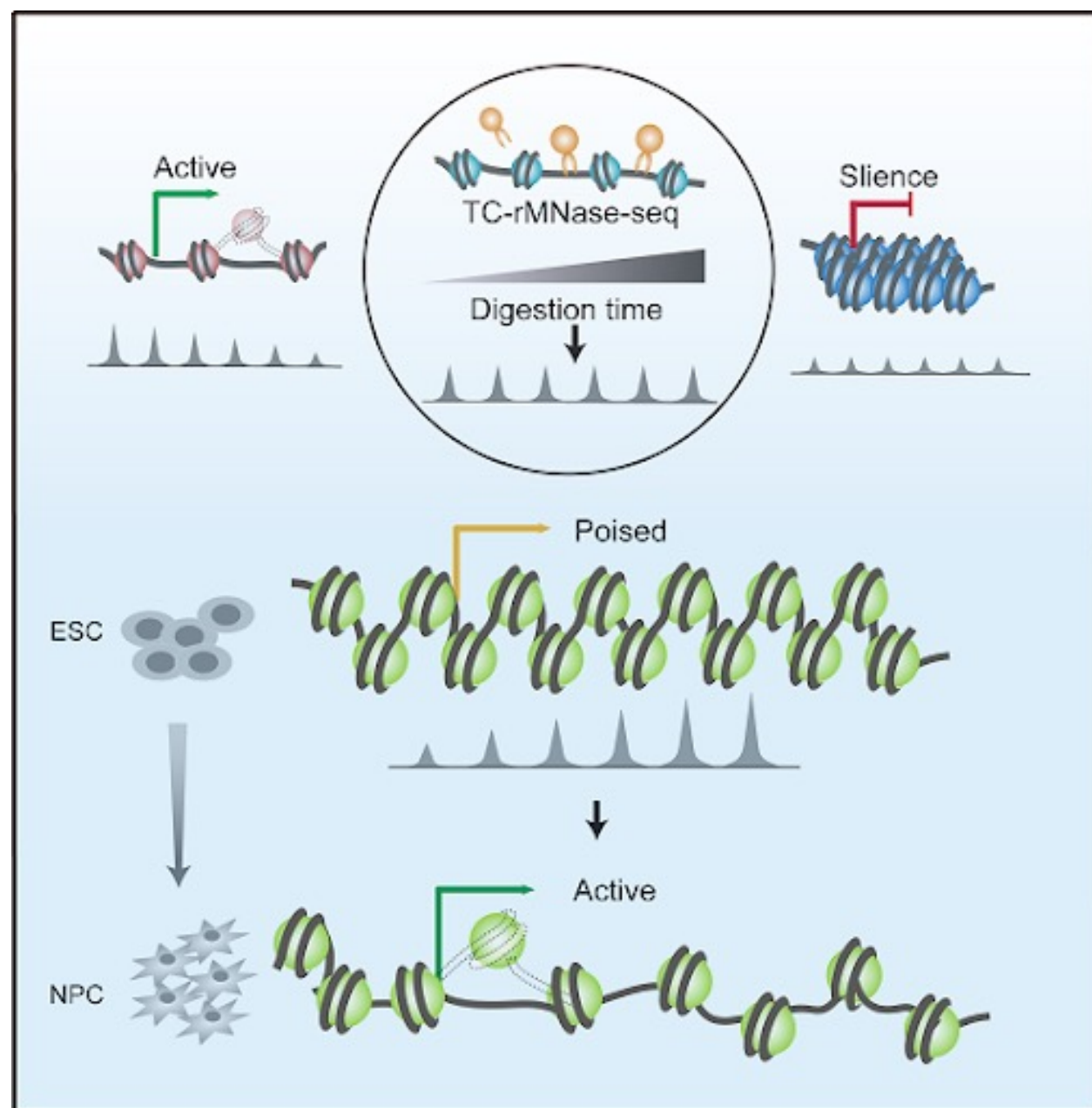
Holliday *et al.*, 2018,
Breast Cancer Research

Transcriptional regulation is encoded the genome



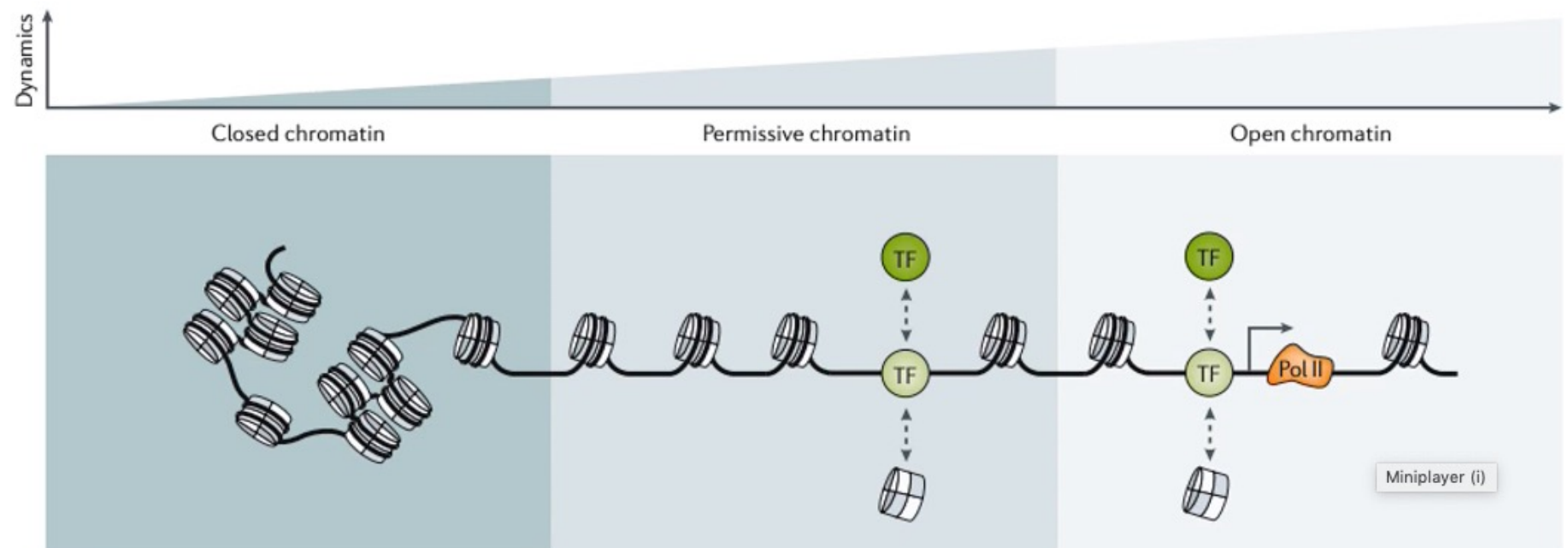




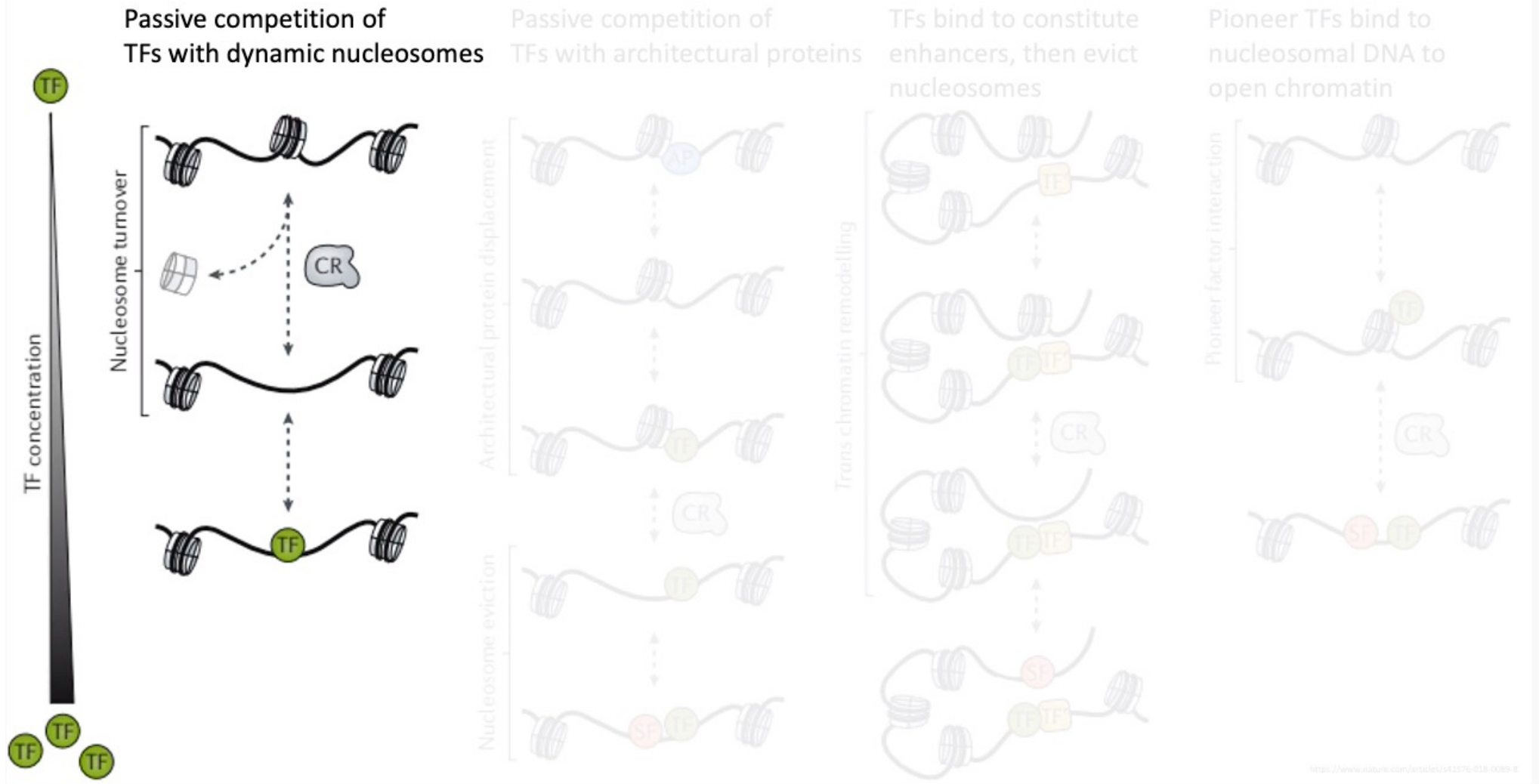


chromatin accessibility

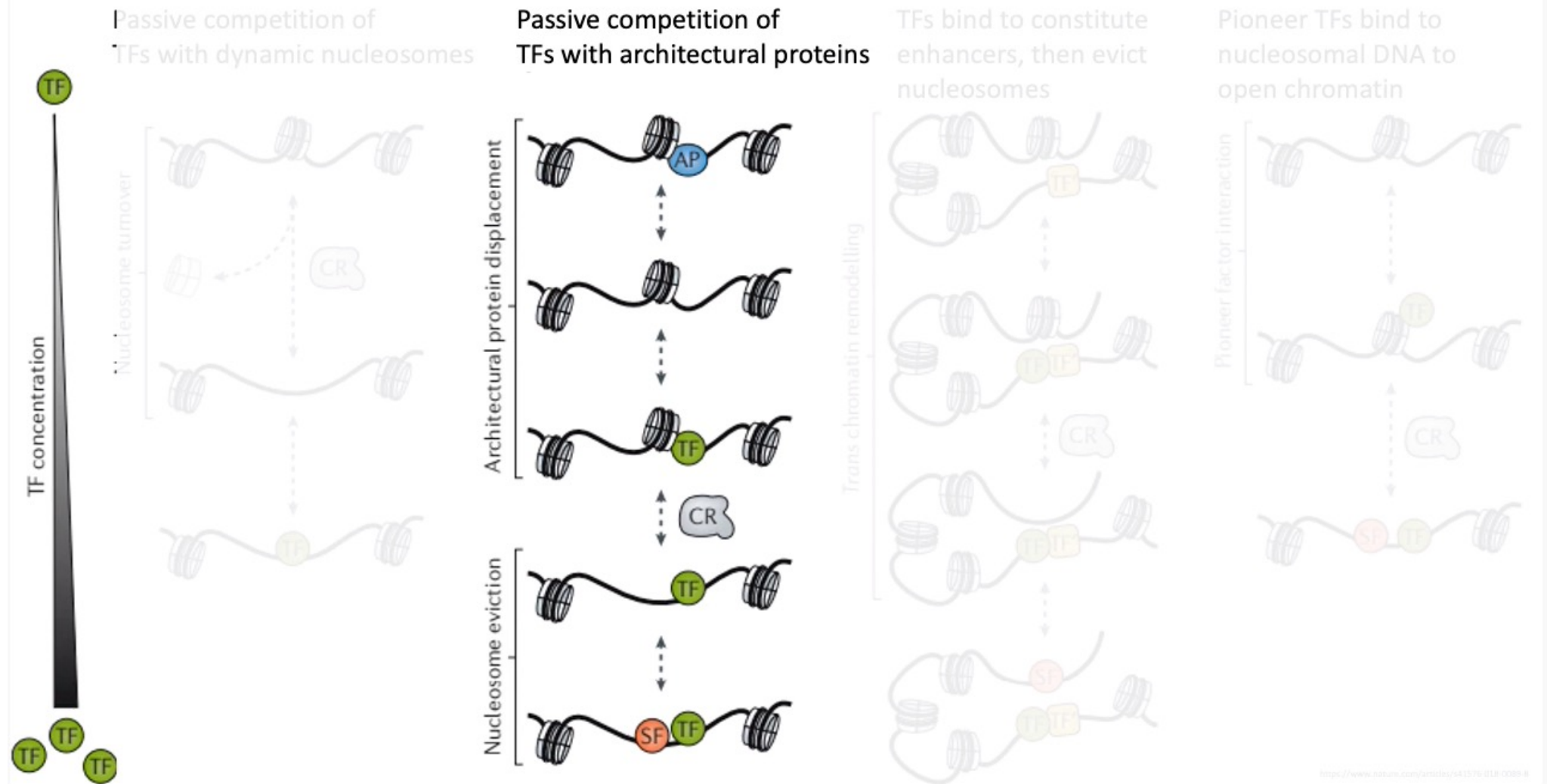
- Chromatin accessibility: % of time any given segment of the genome is occupied by a nucleosome
- Not necessarily heritable (like histone modifications)
 - But like histone modifications, they play a critical role in gene regulation
 - Accessibility can also change as a result of response to environment



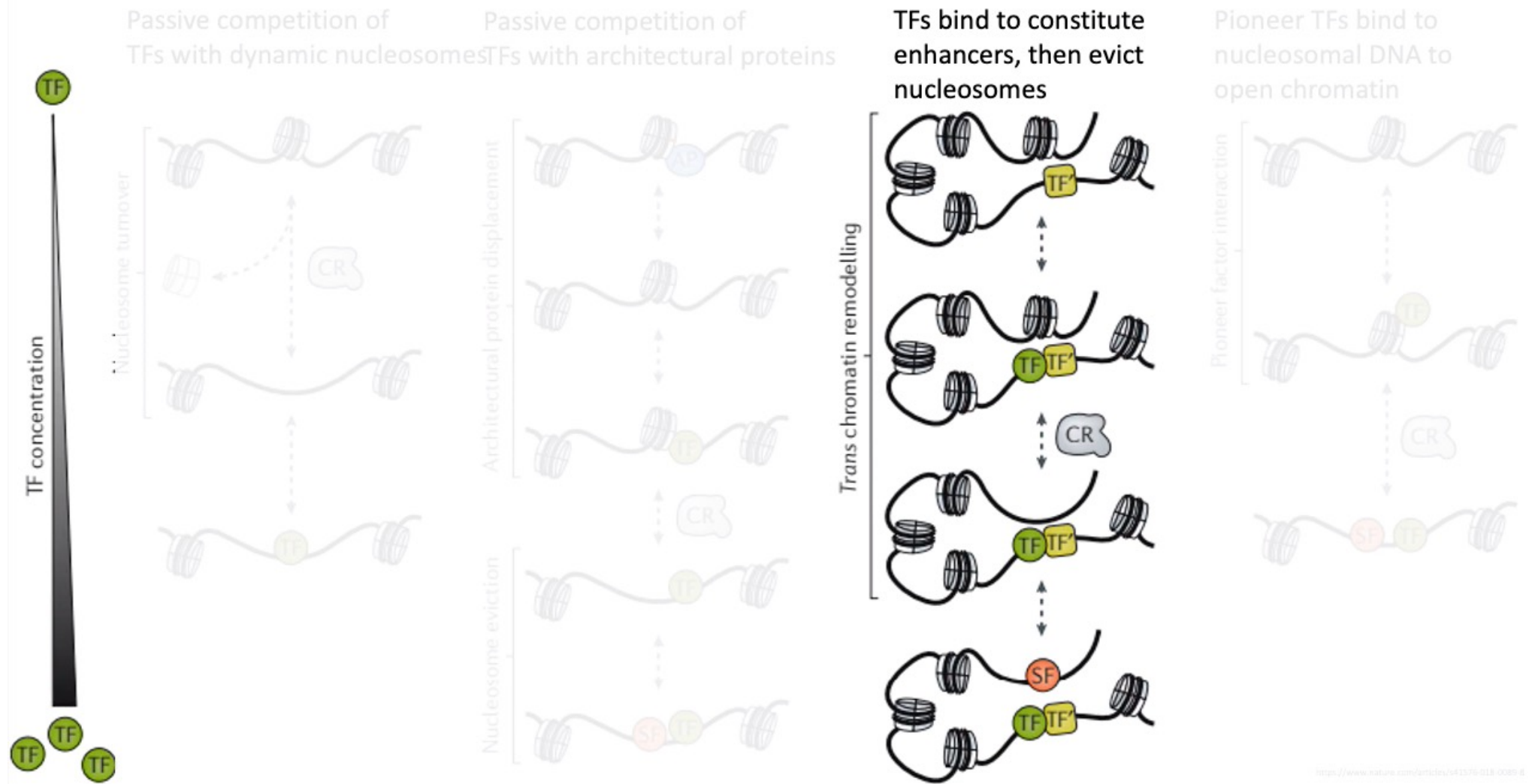
How is chromatin accessibility remodeled?



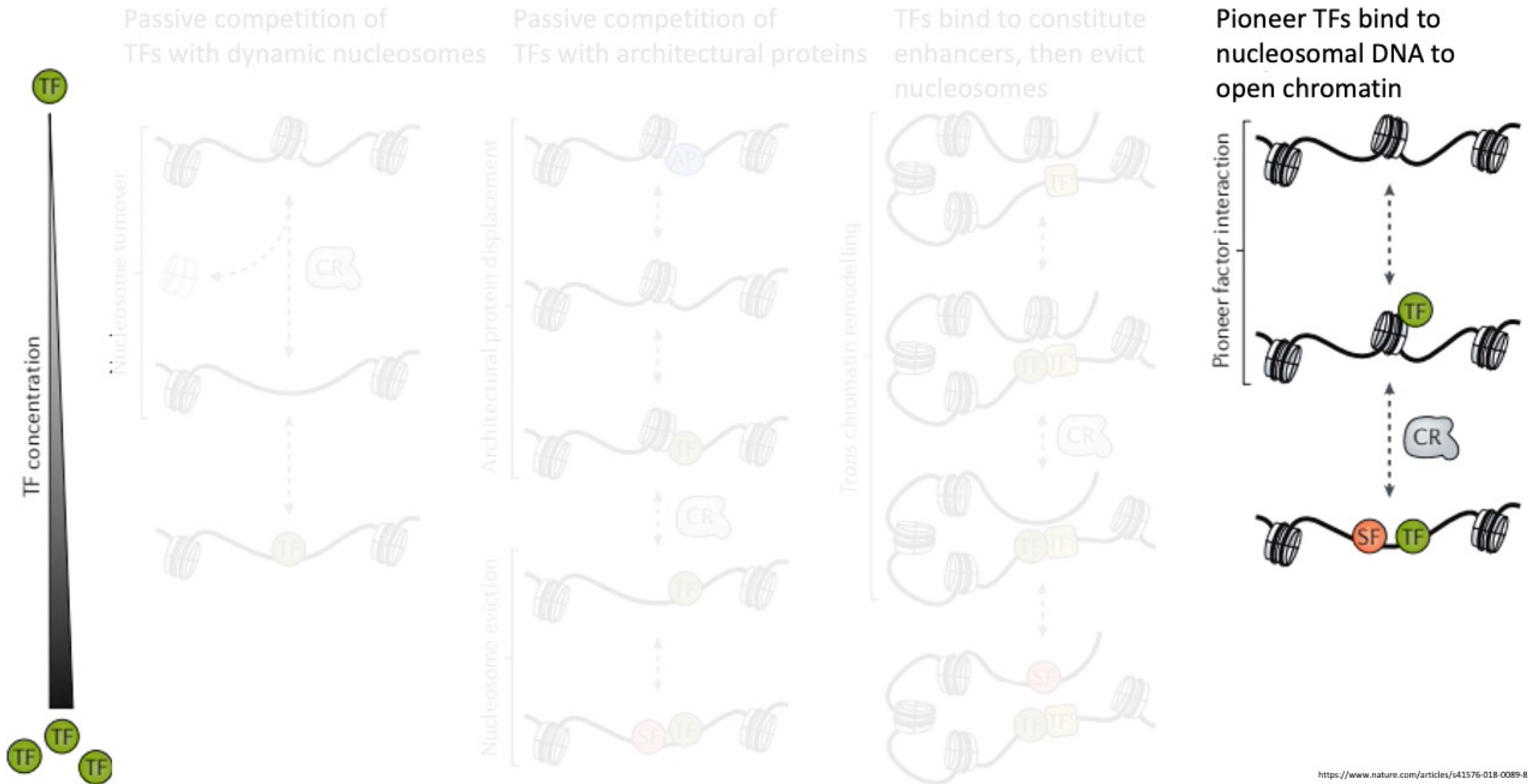
How is chromatin accessibility remodeled?



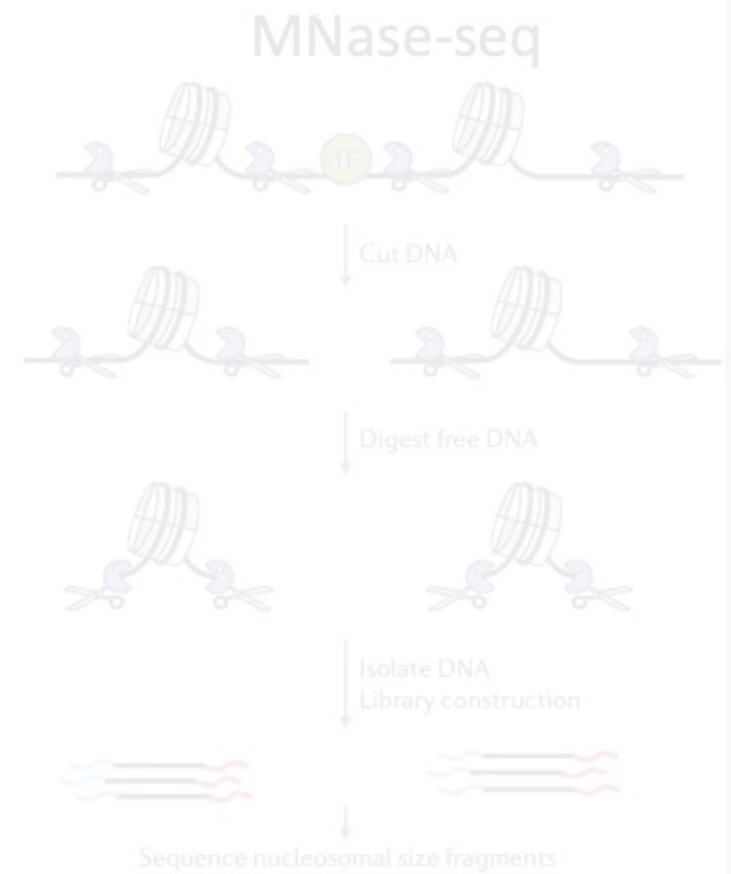
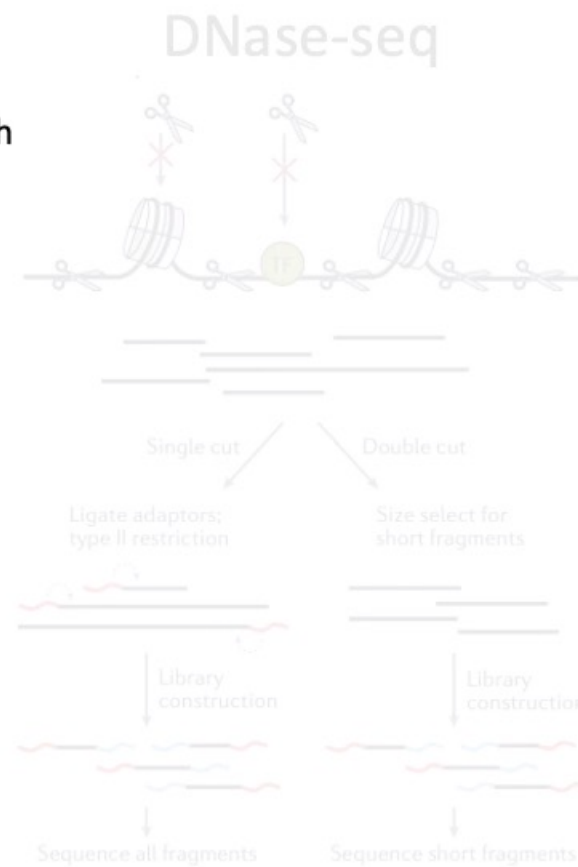
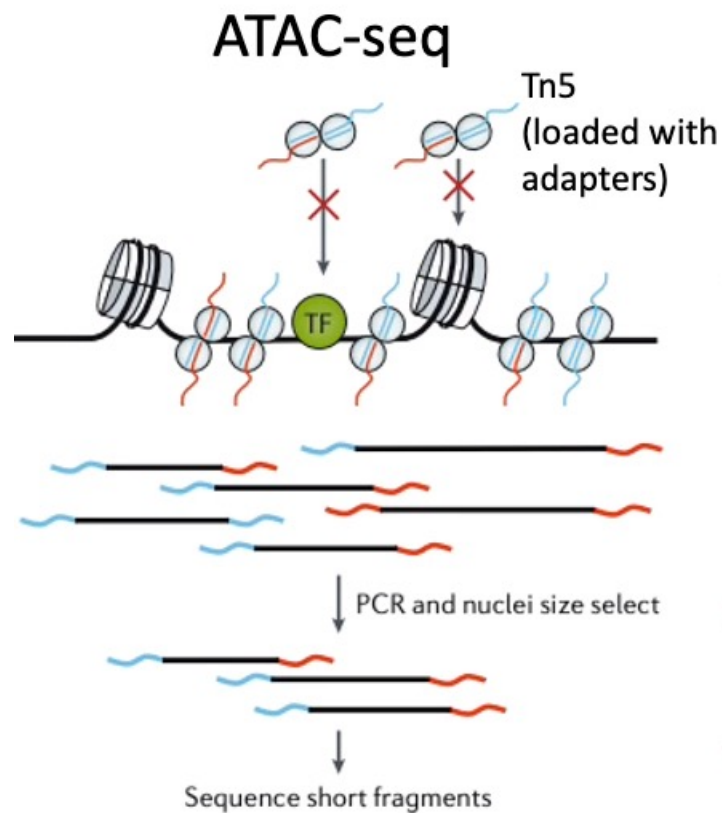
How is chromatin accessibility remodeled?



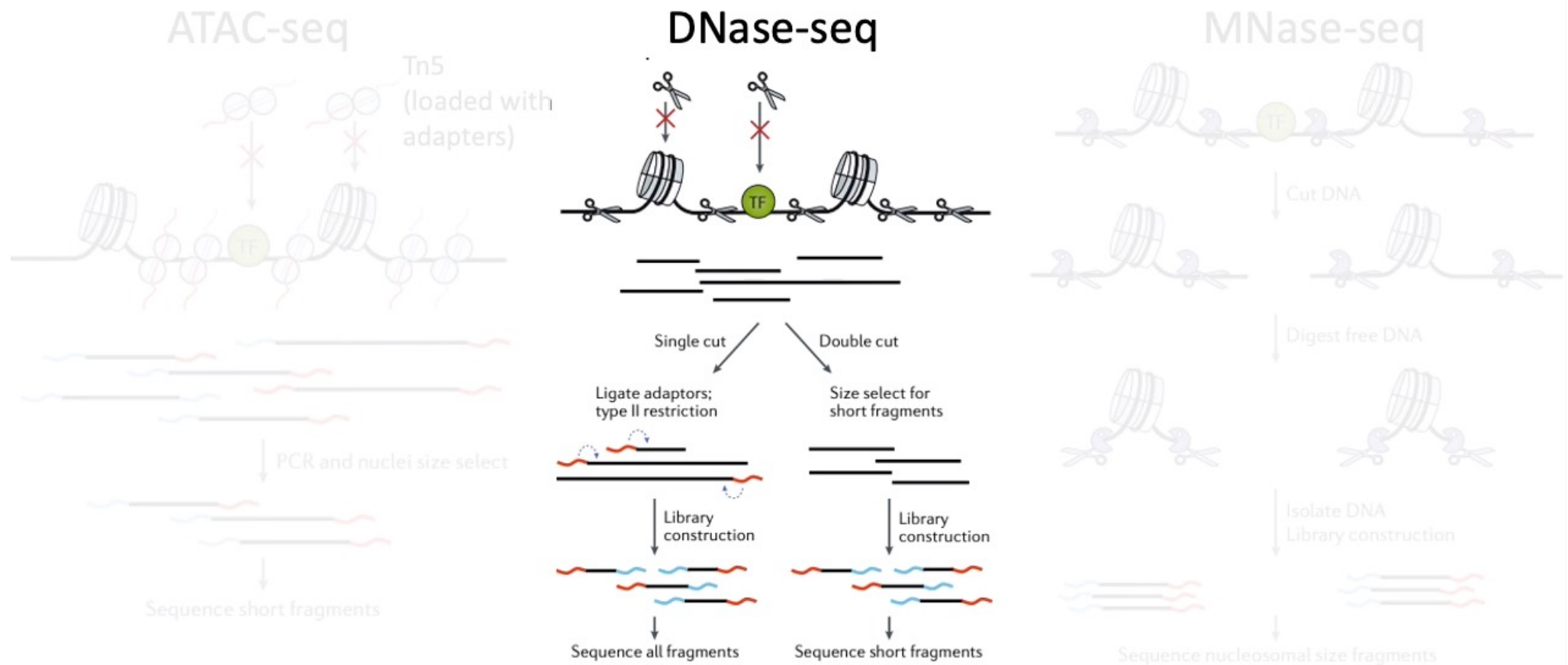
How is chromatin accessibility remodeled?



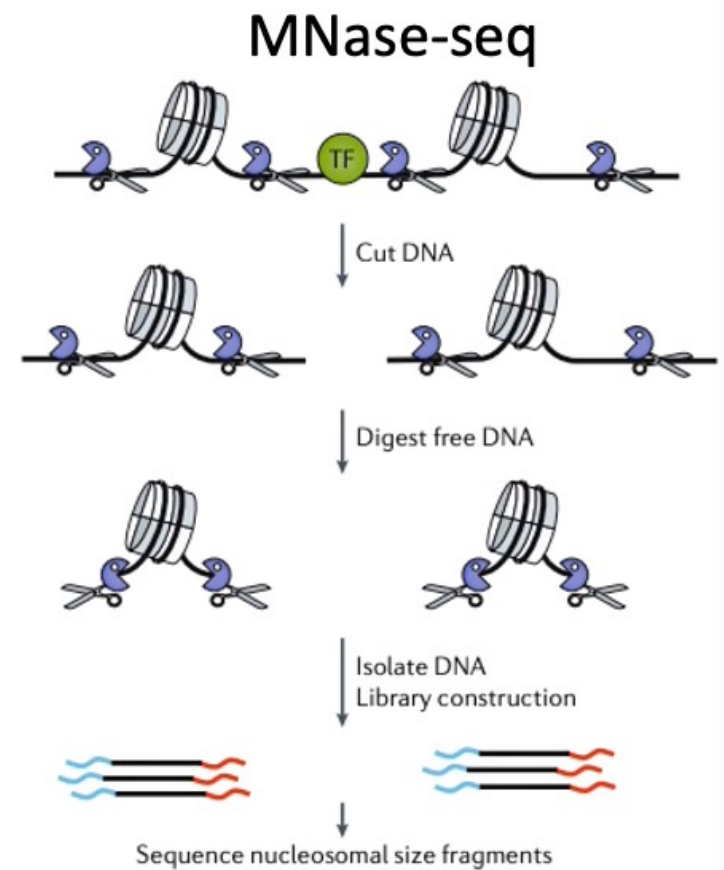
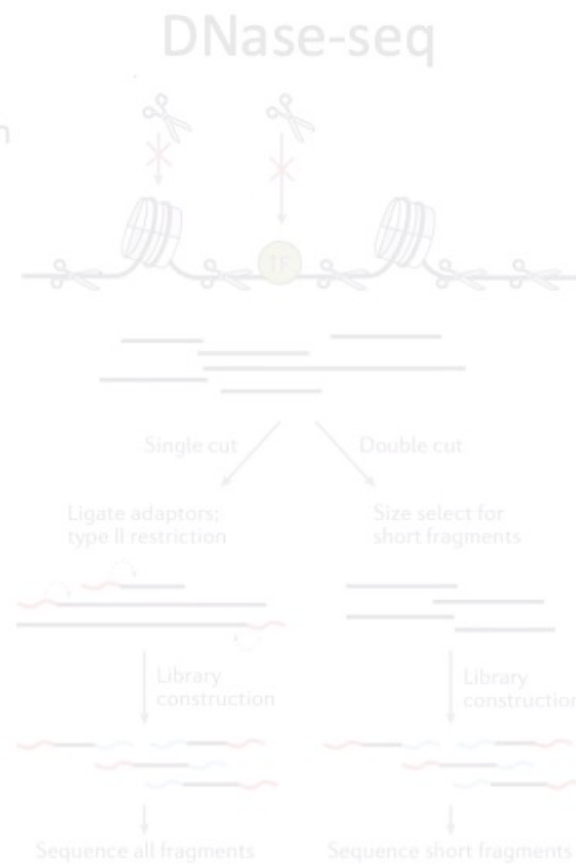
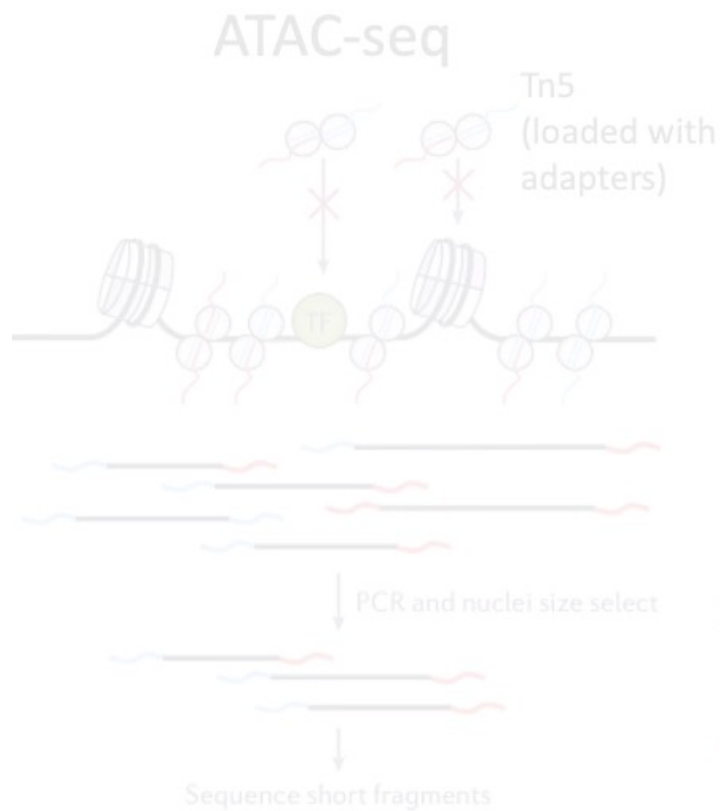
Chromatin accessibility assays

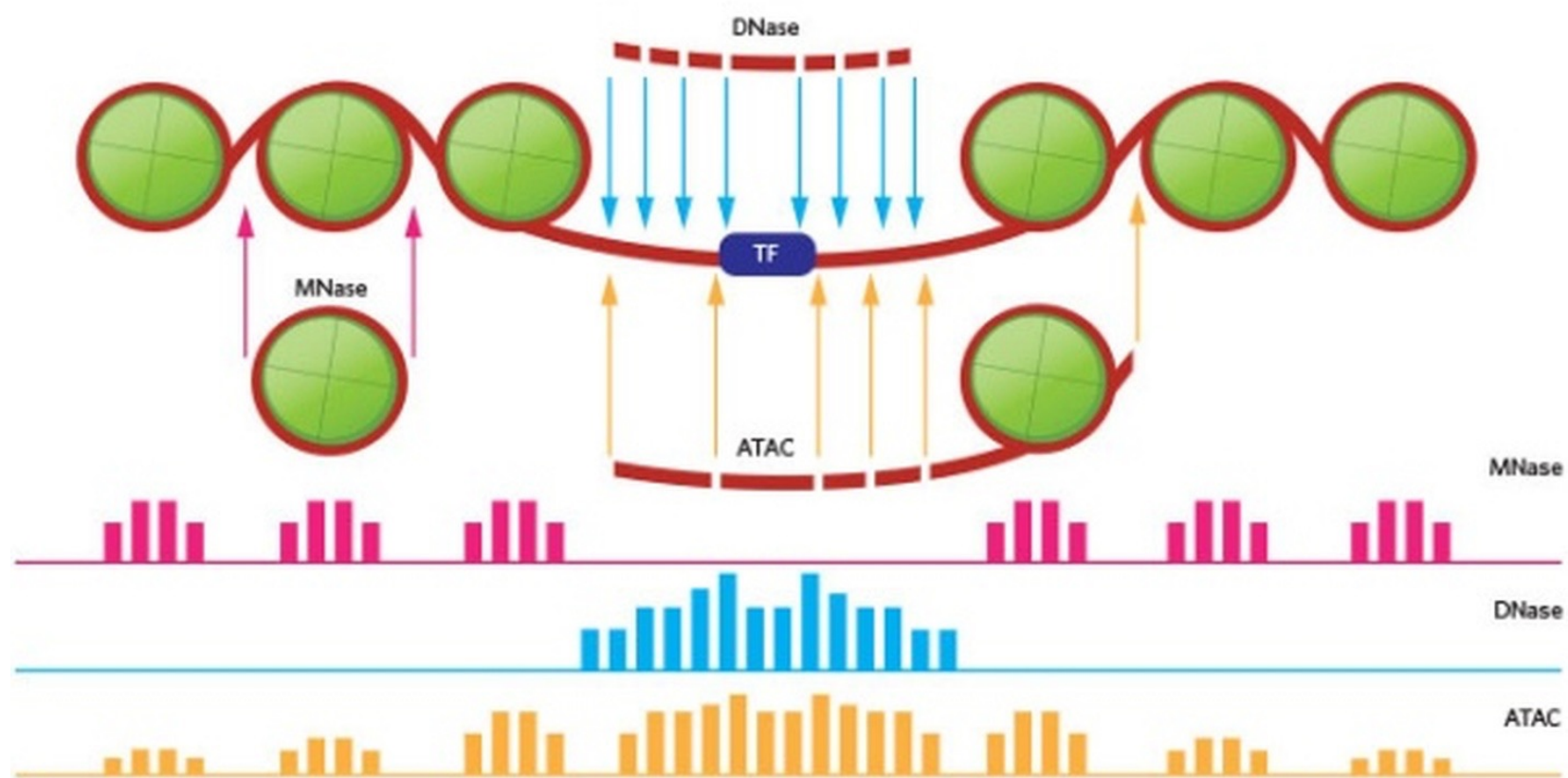


Chromatin accessibility assays

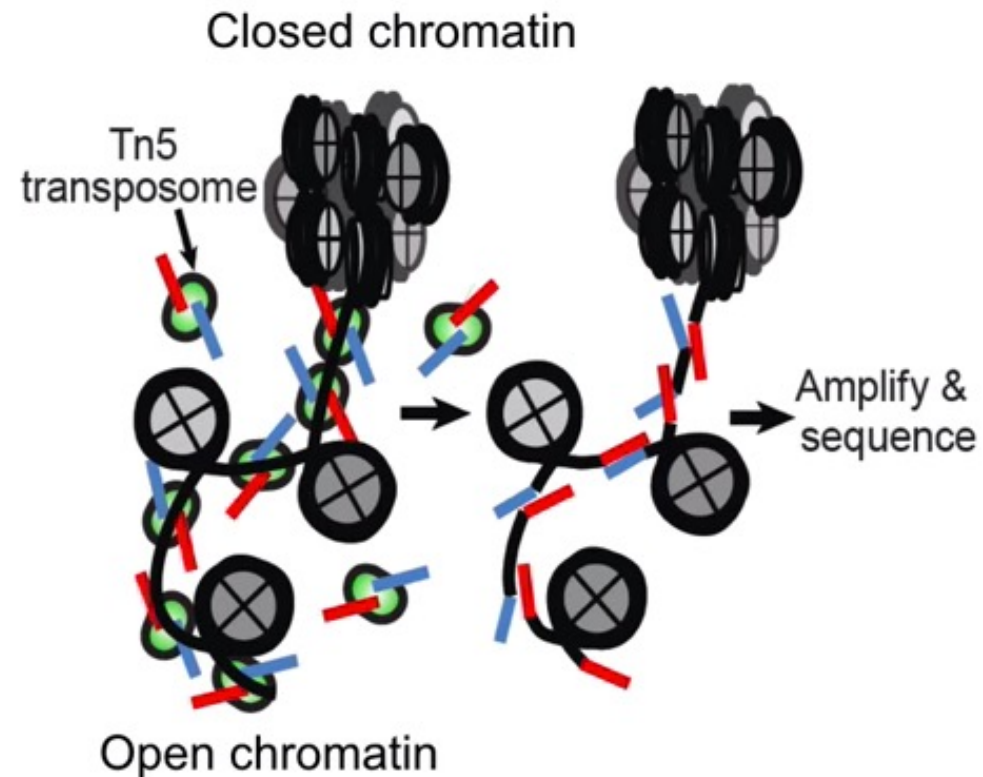
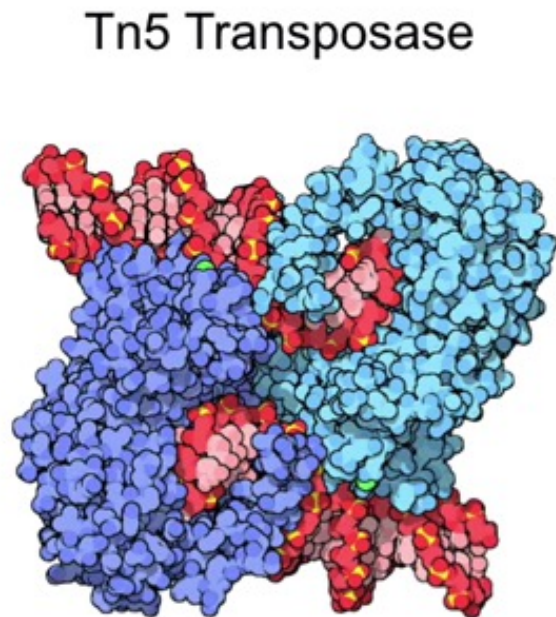


Chromatin accessibility assays



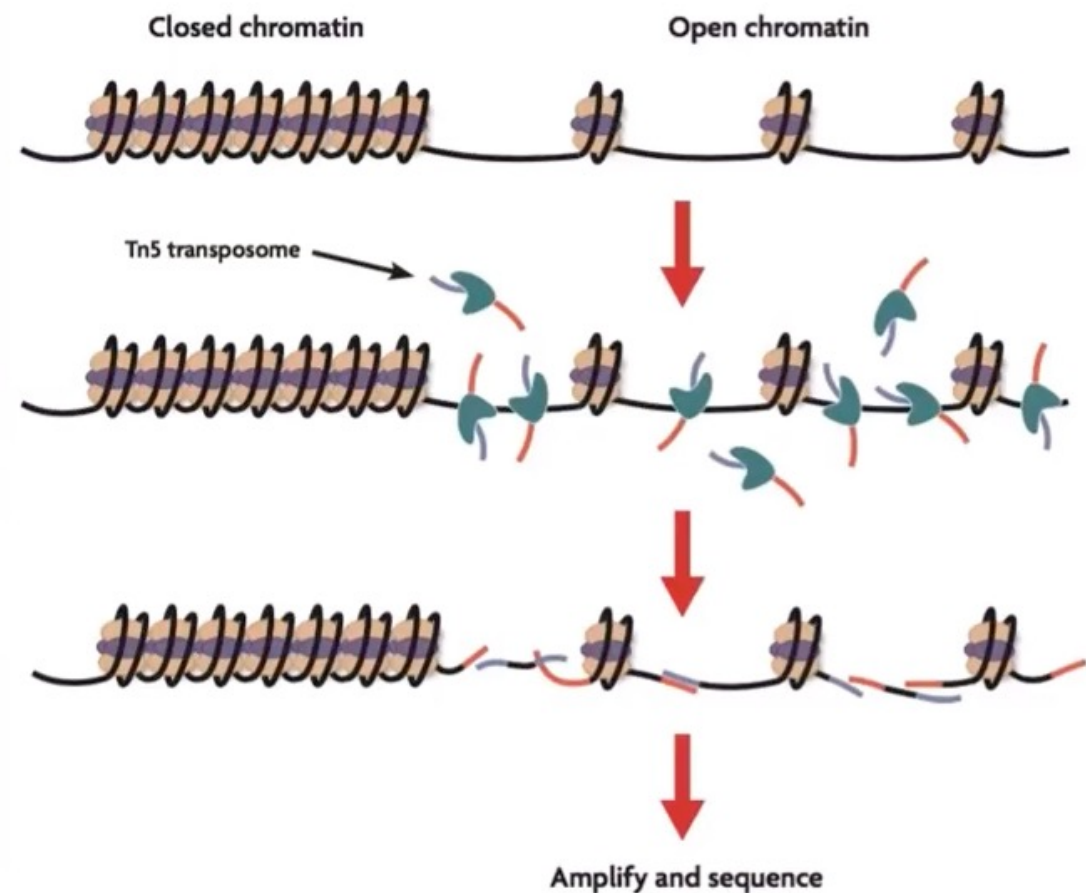


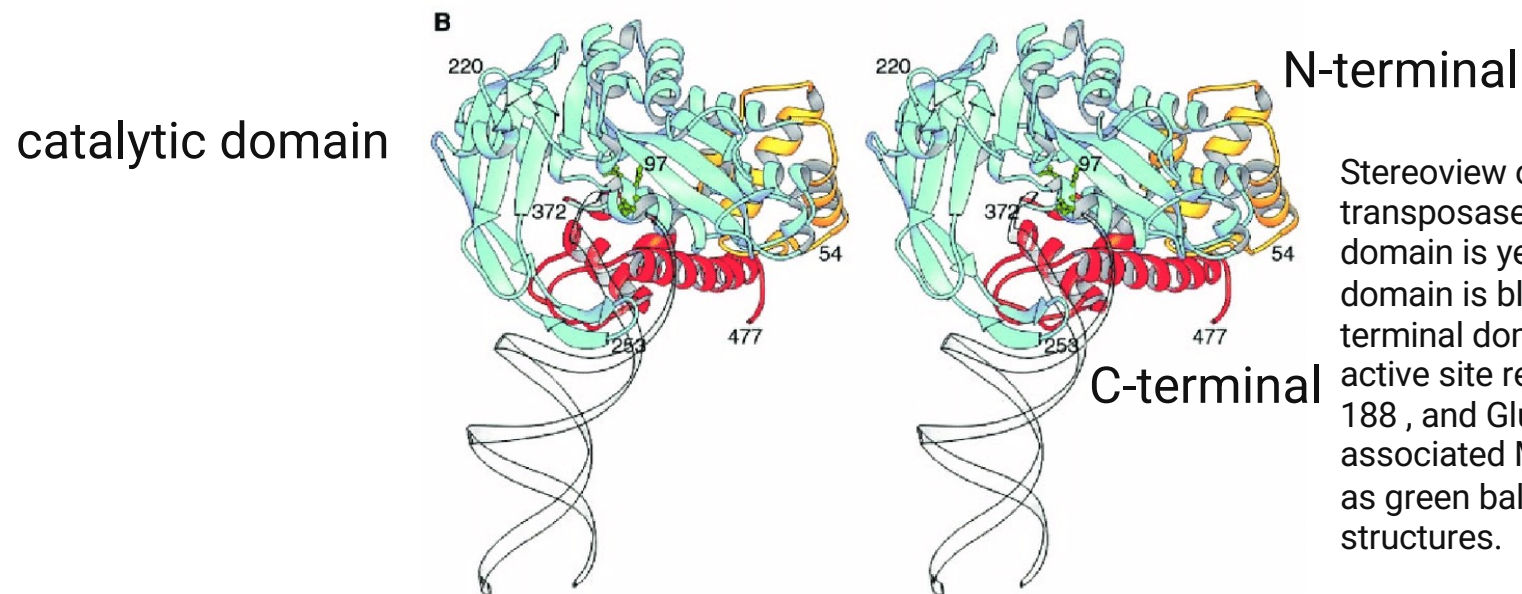
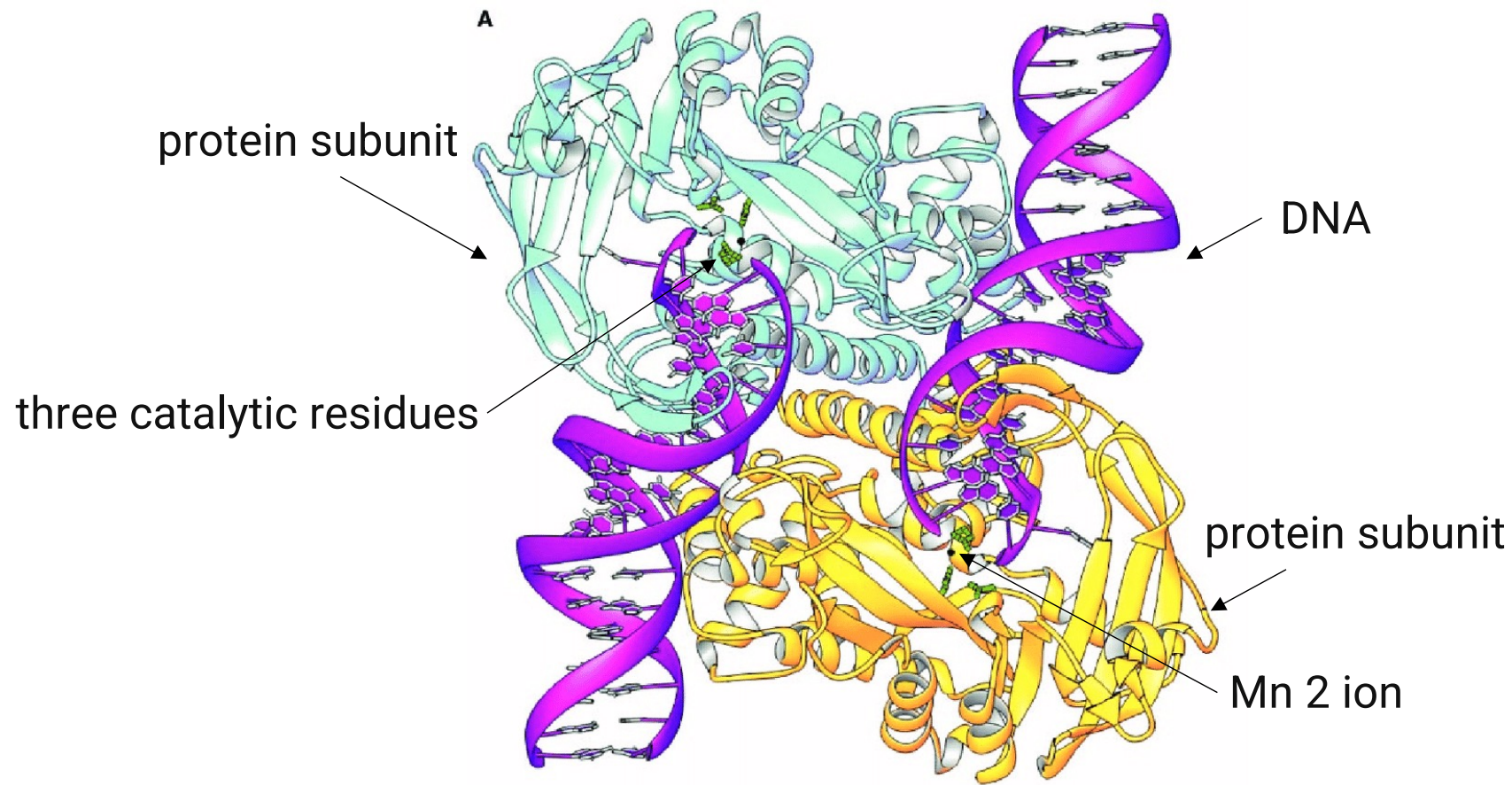
Assay of Transposase Accessible Chromatin



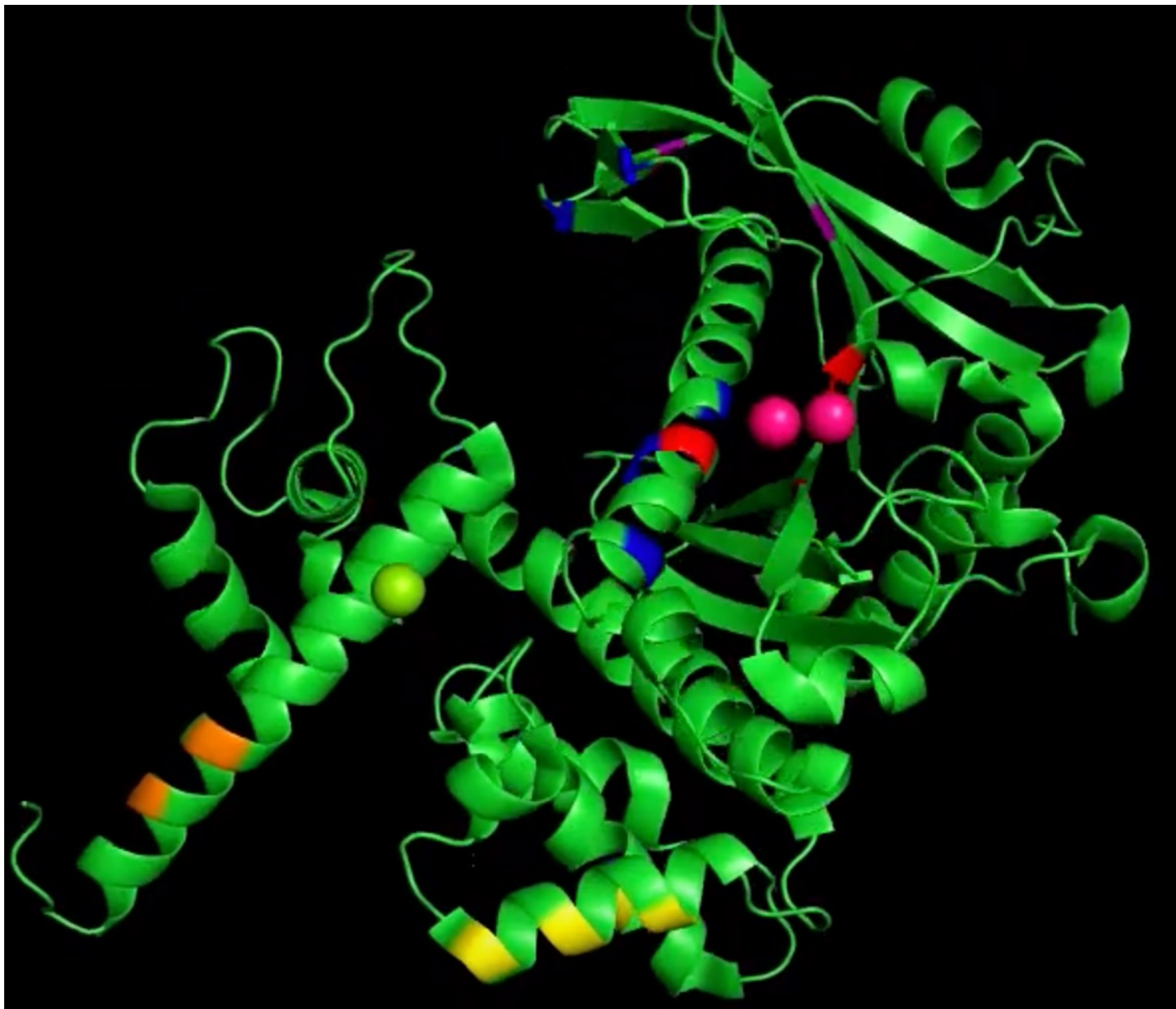
Assay for Transposase-Accessible Chromatin (ATAC)

1. Tn5 Transposase gains access to open/accessible regions of the genome
2. Transposase inserts known DNA sequences (adapters) into open regions
3. Libraries are amplified and sequenced

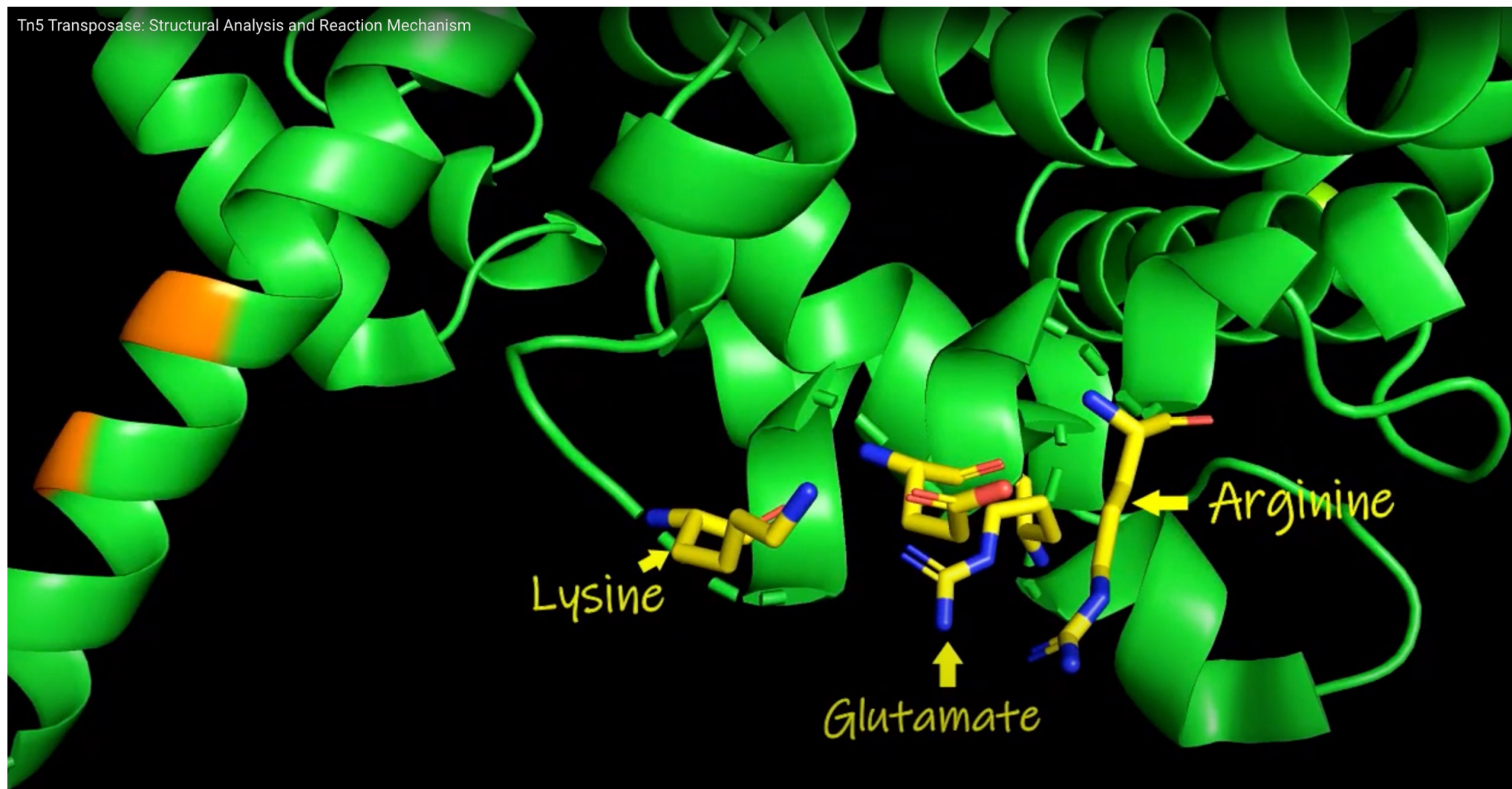




Stereoview of one monomer of transposase. The NH 2-terminal domain is yellow, the catalytic domain is blue, and the COOH-terminal domain is red. The active site residues Asp 97 , Asp 188 , and Glu 326 and the associated Mn 2 ion are shown as green ball-and-stick structures.

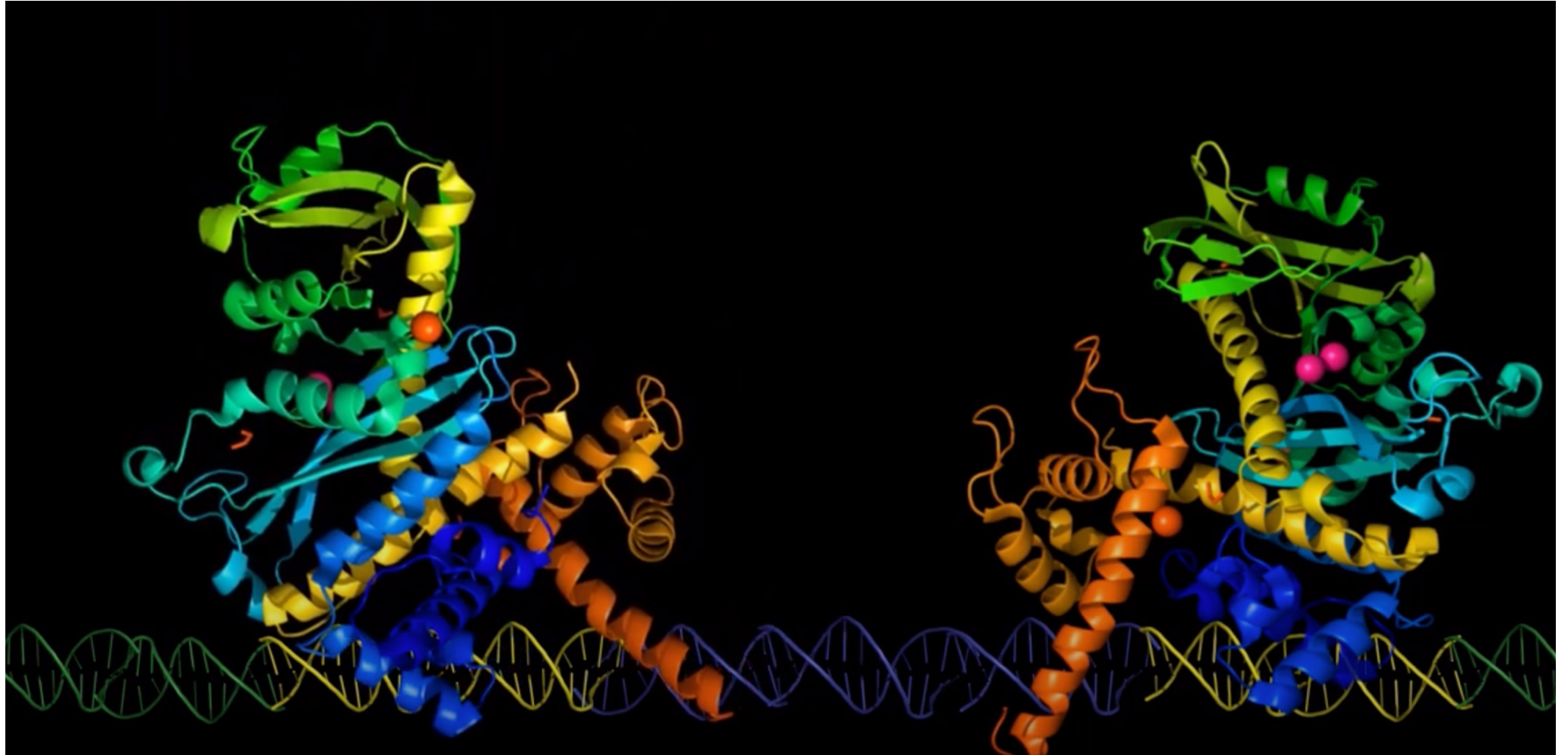


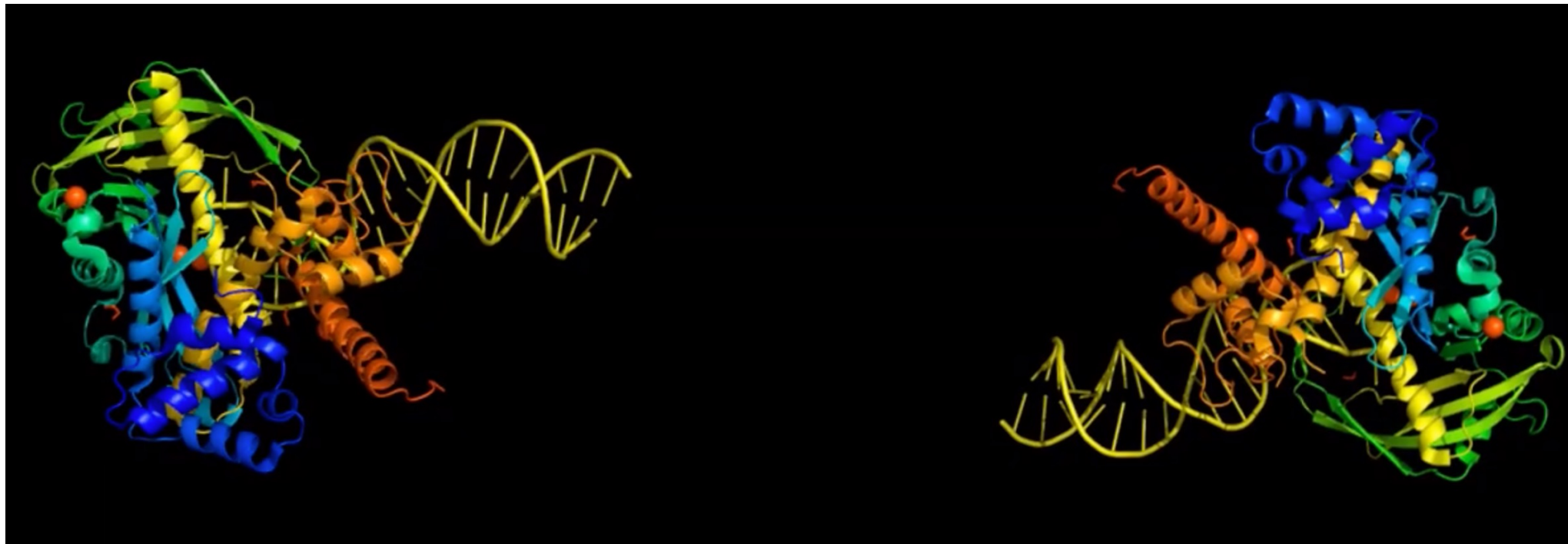
N-terminal

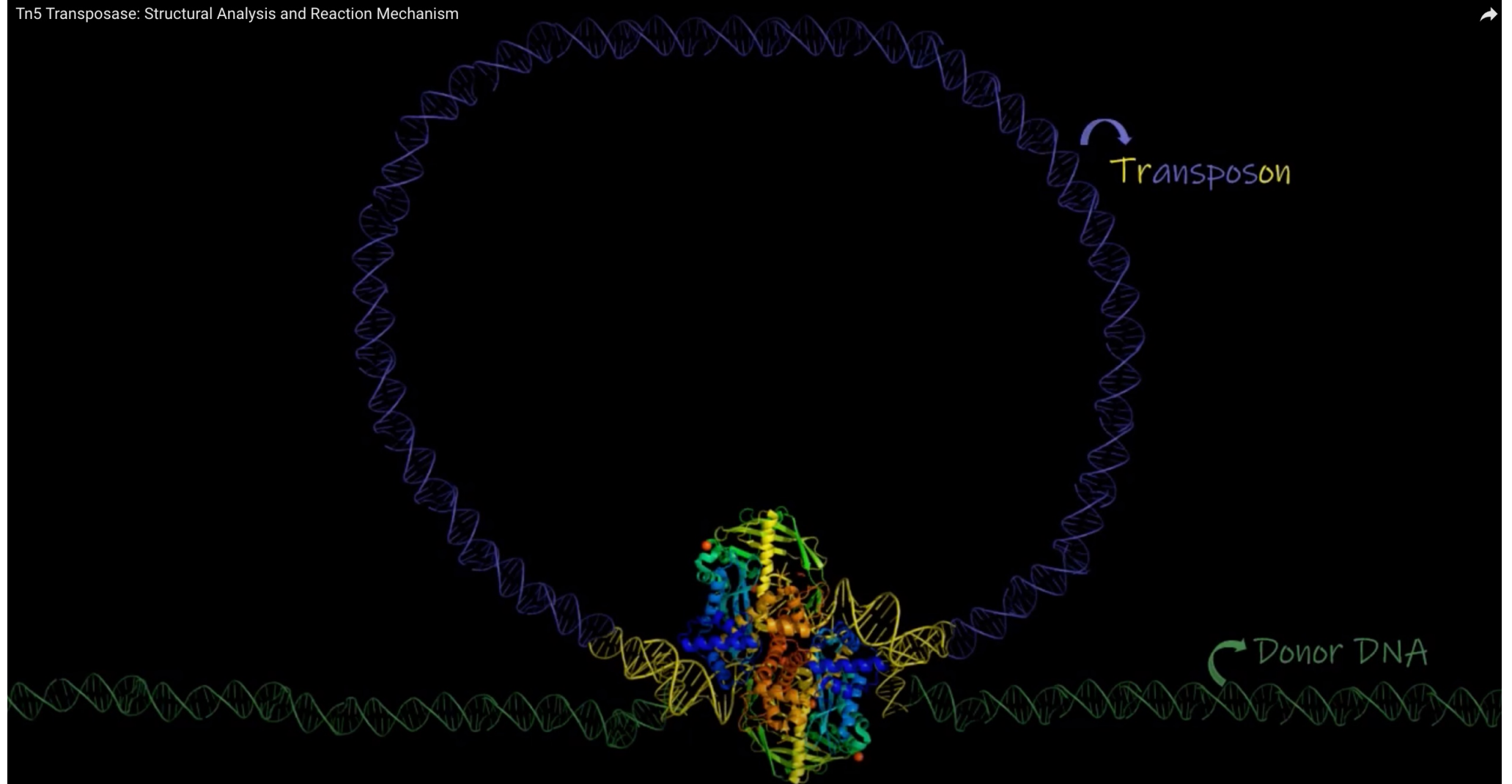


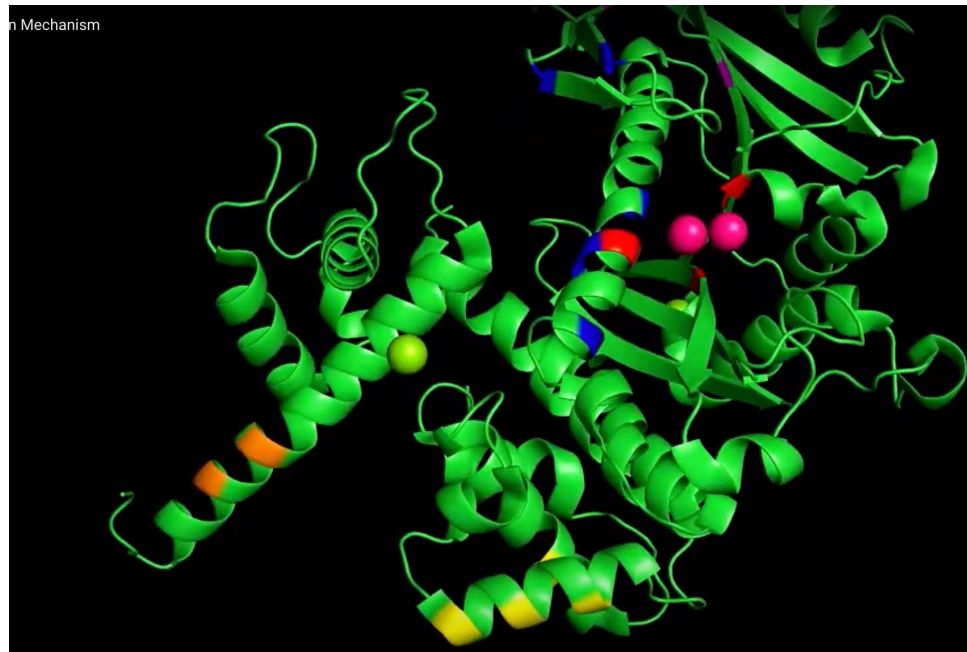
N-terminal



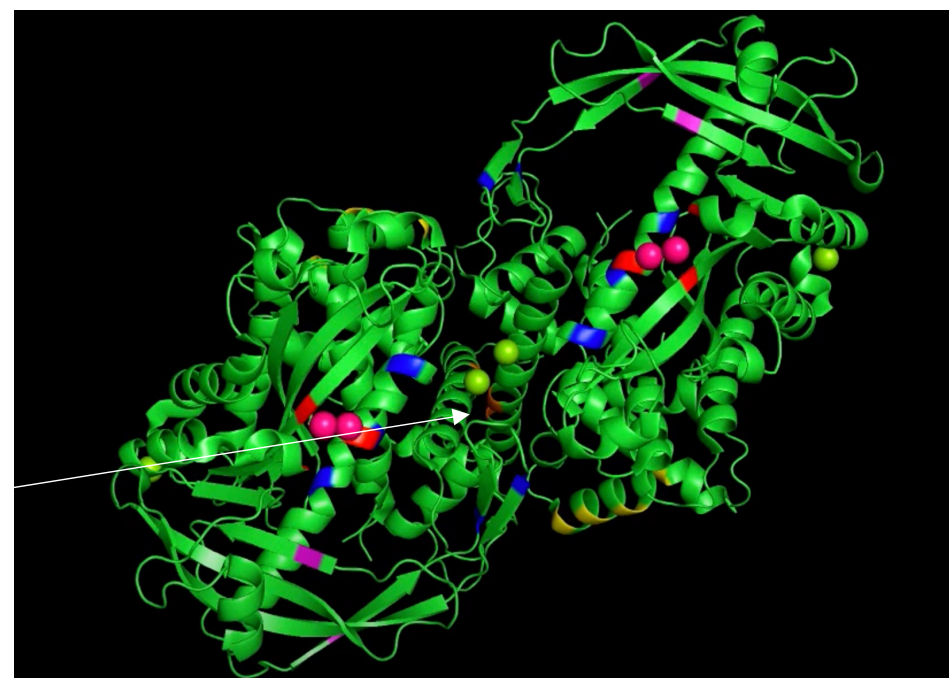
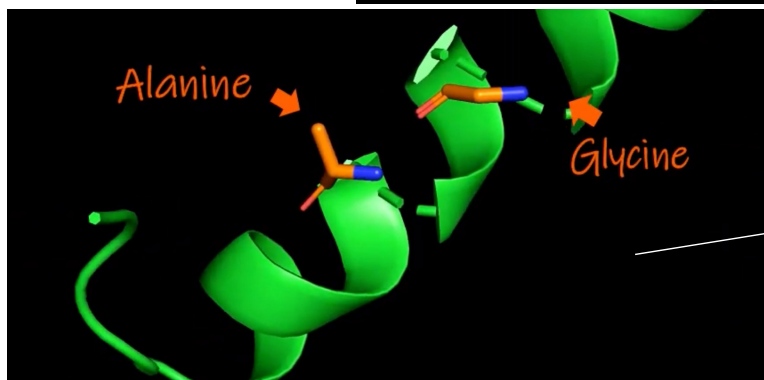
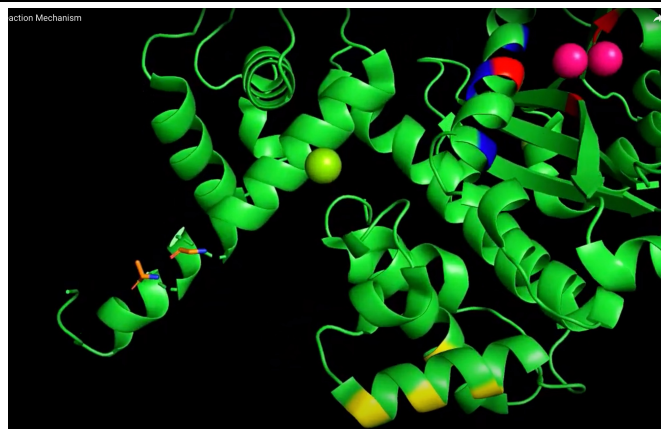


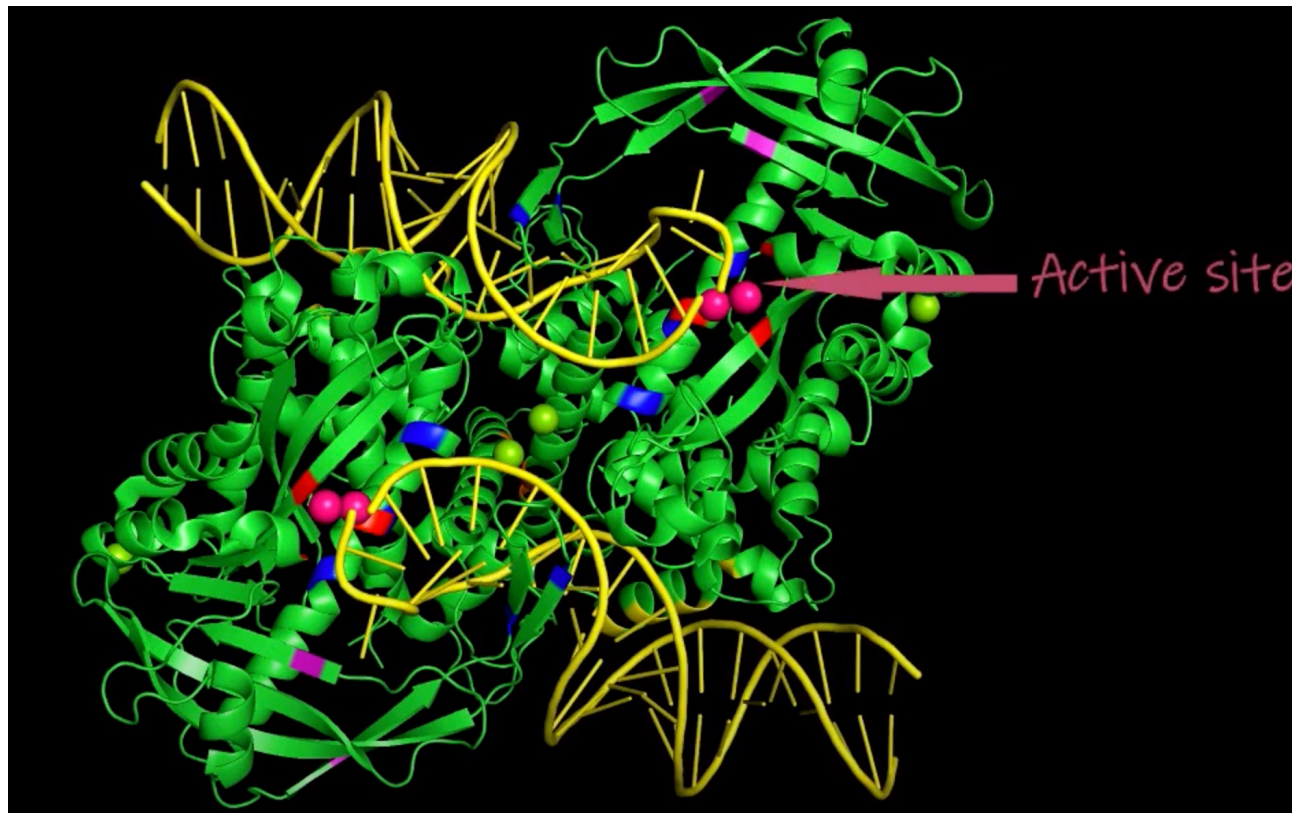
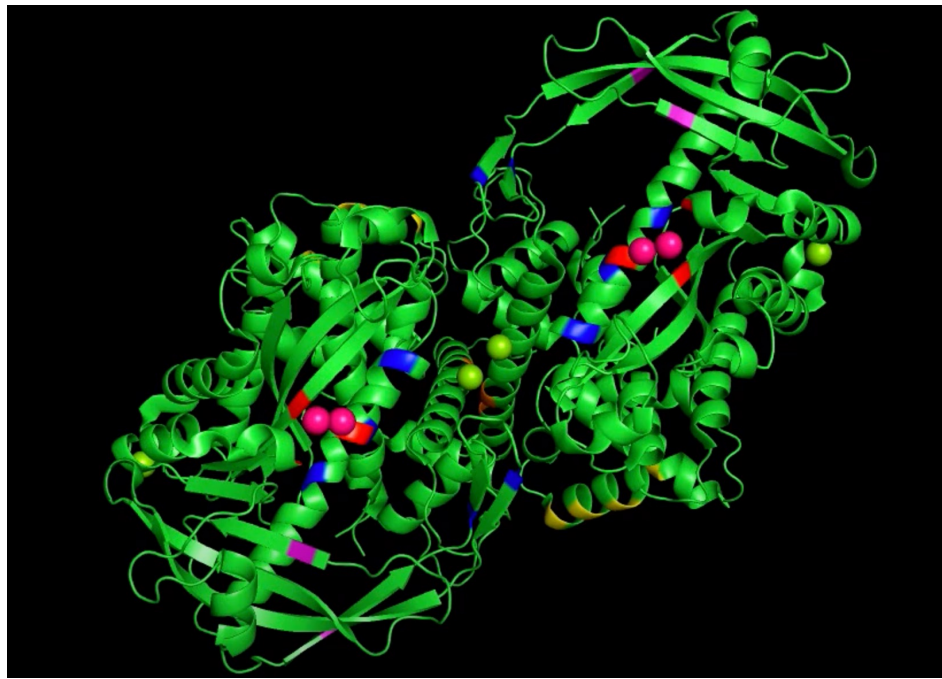


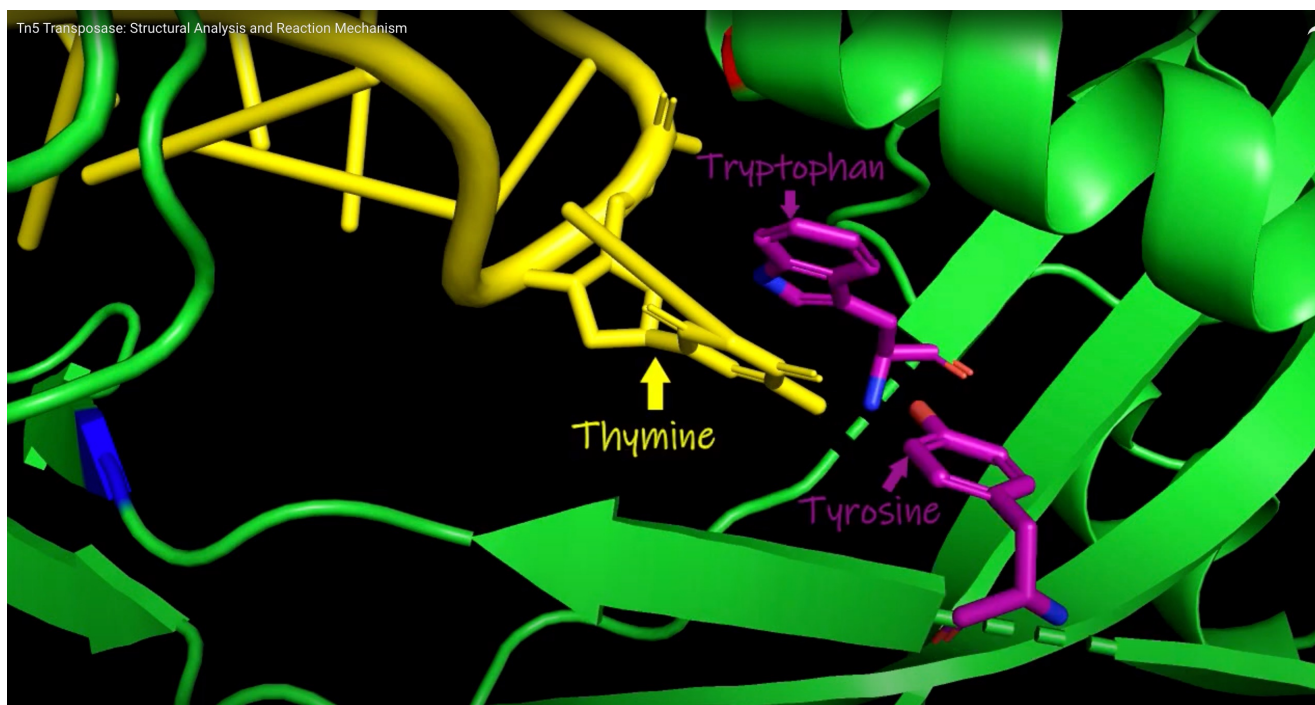


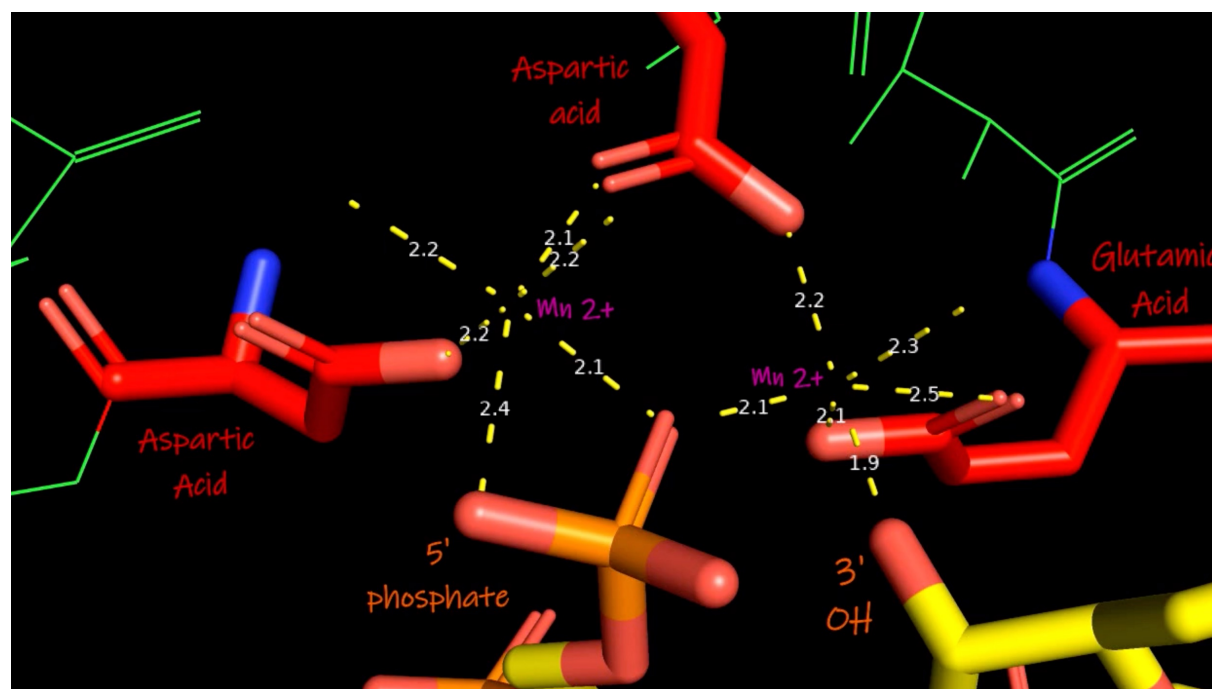
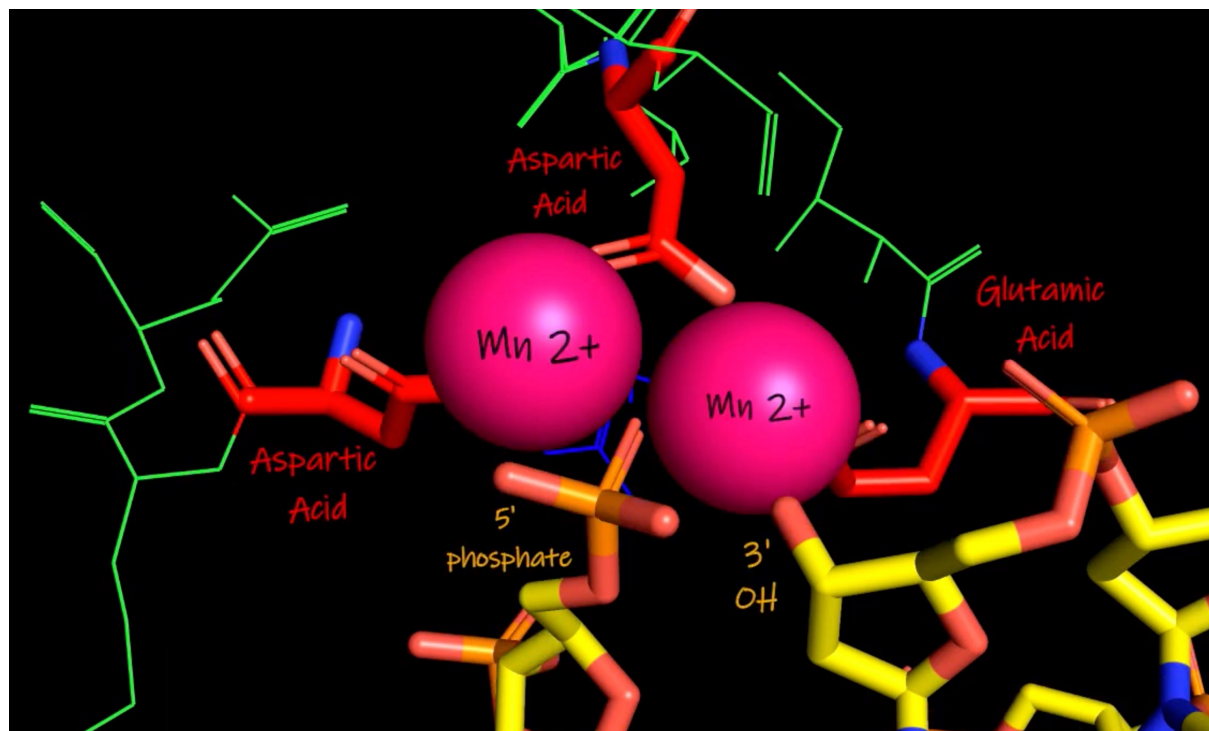


C-terminal

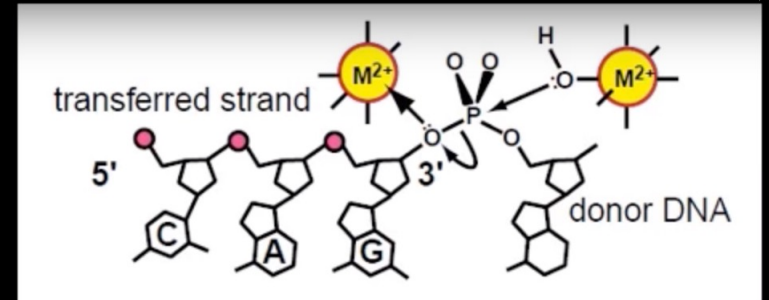
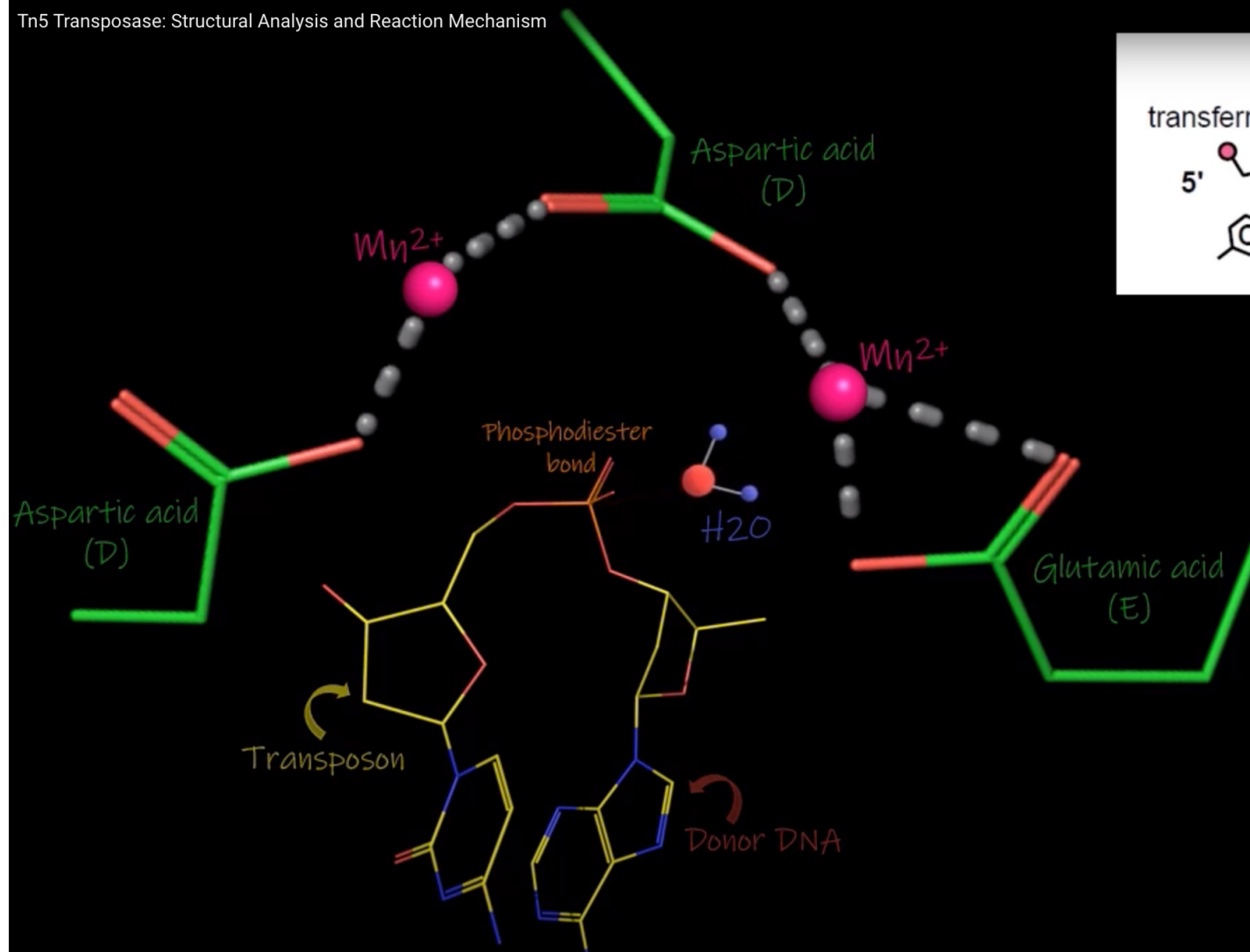




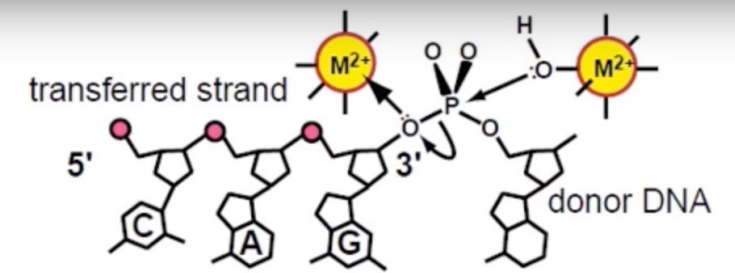
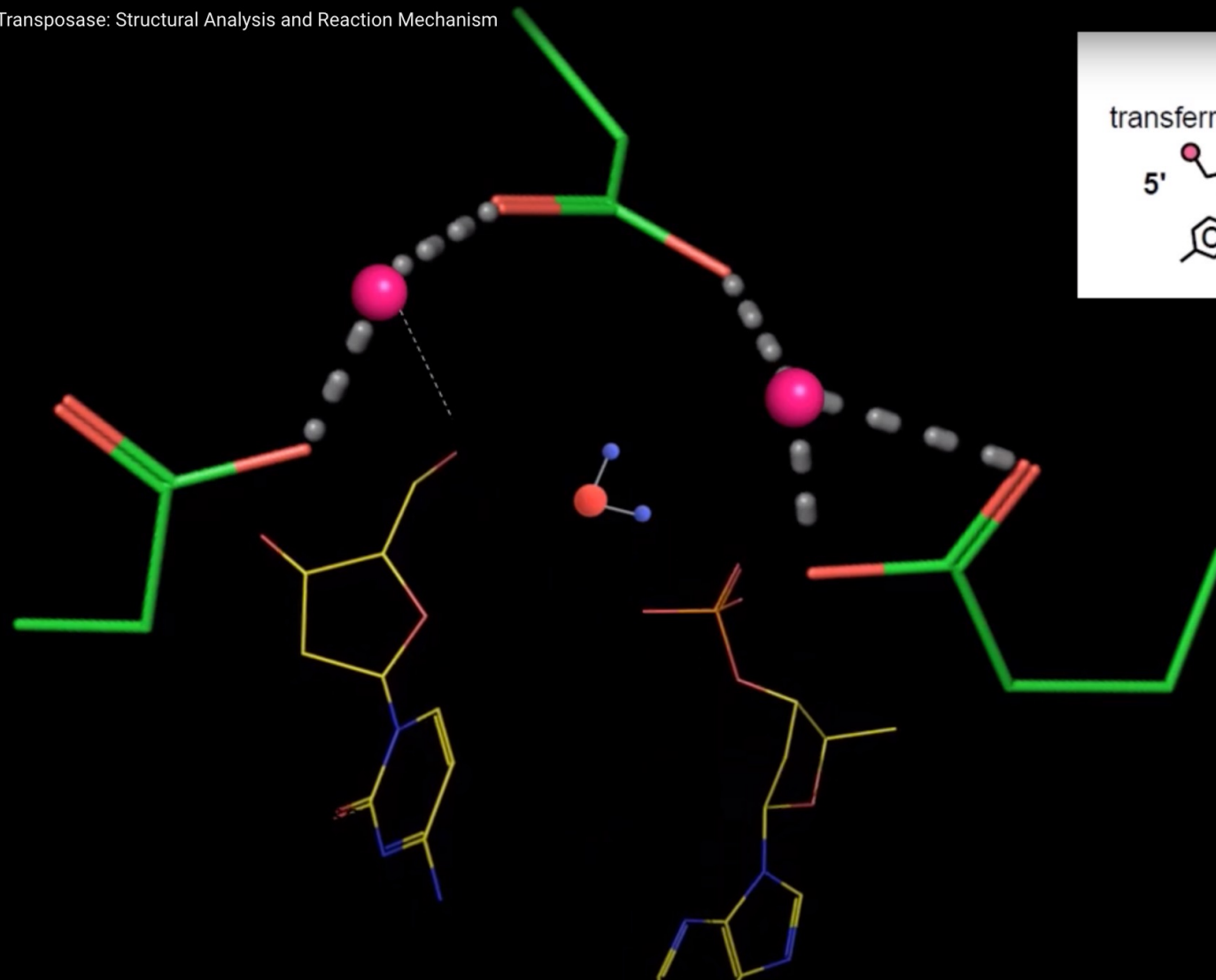


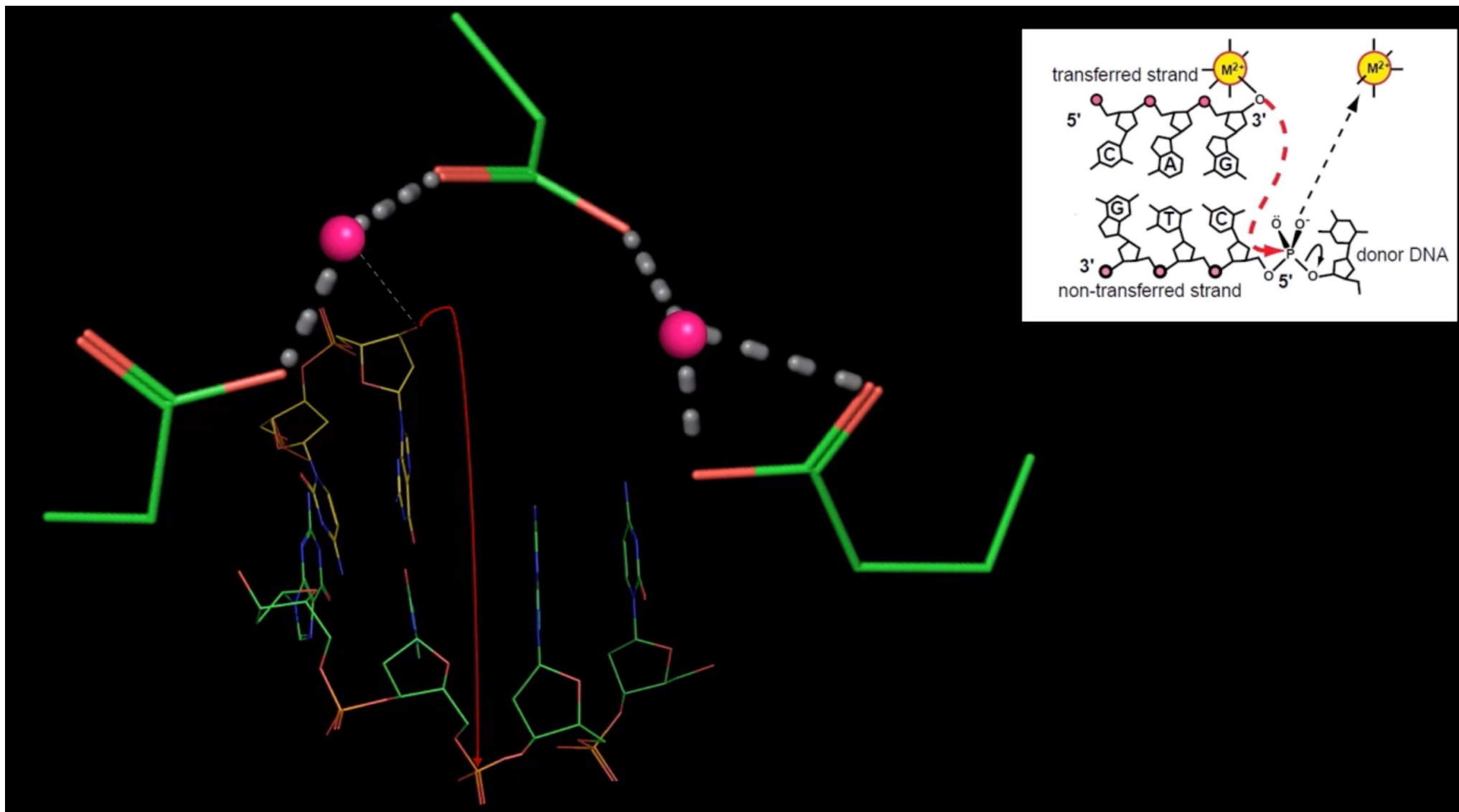


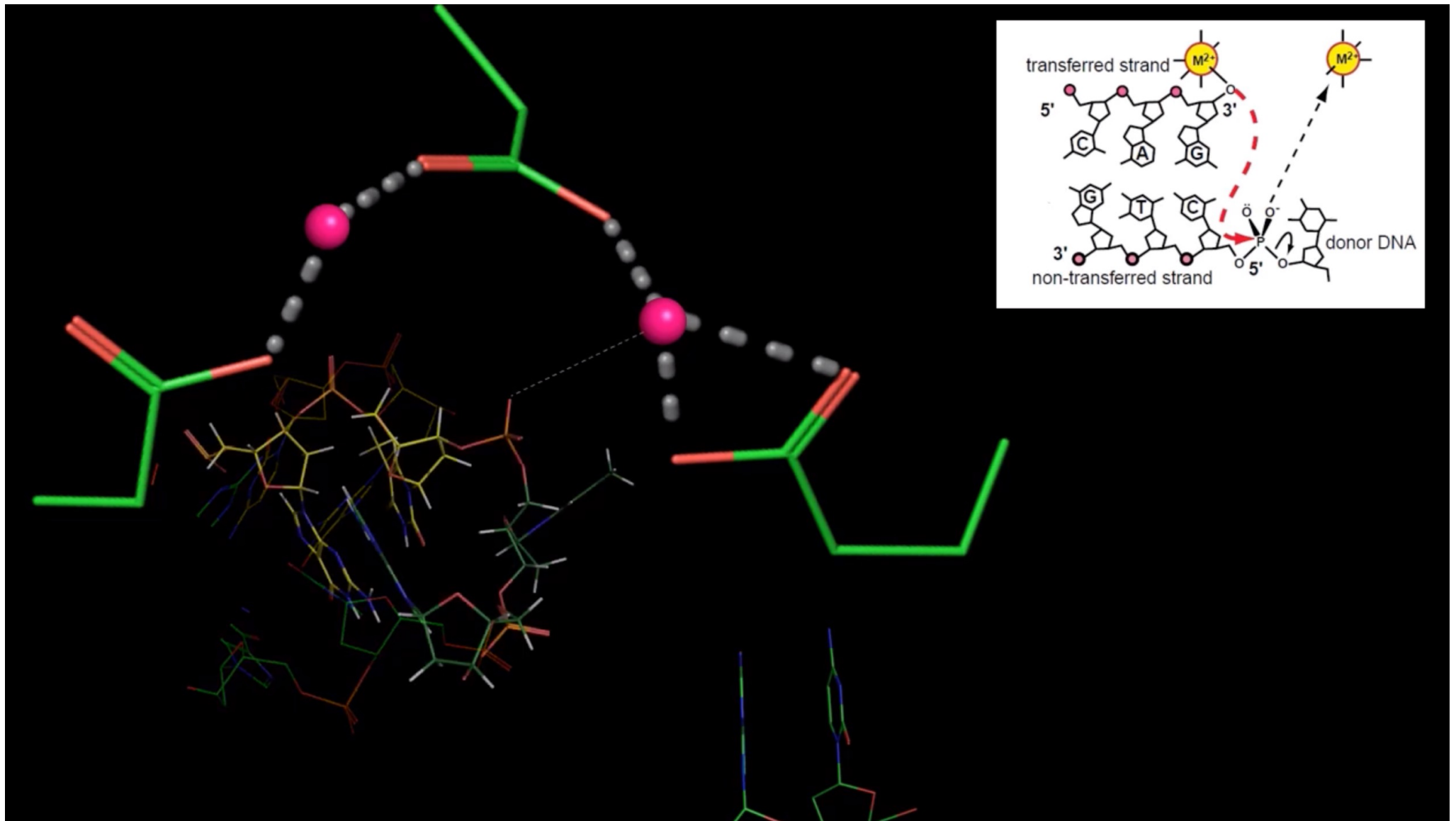
Tn5 Transposase: Structural Analysis and Reaction Mechanism



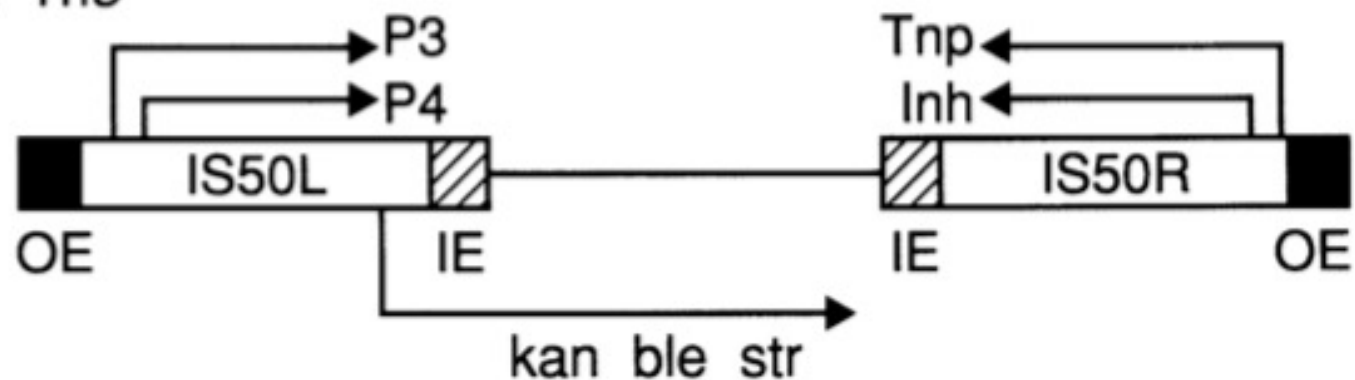
Tn5 Transposase: Structural Analysis and Reaction Mechanism



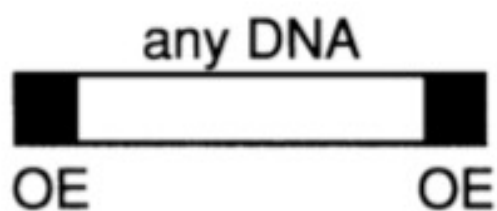




A Tn5



B Simplified transposon



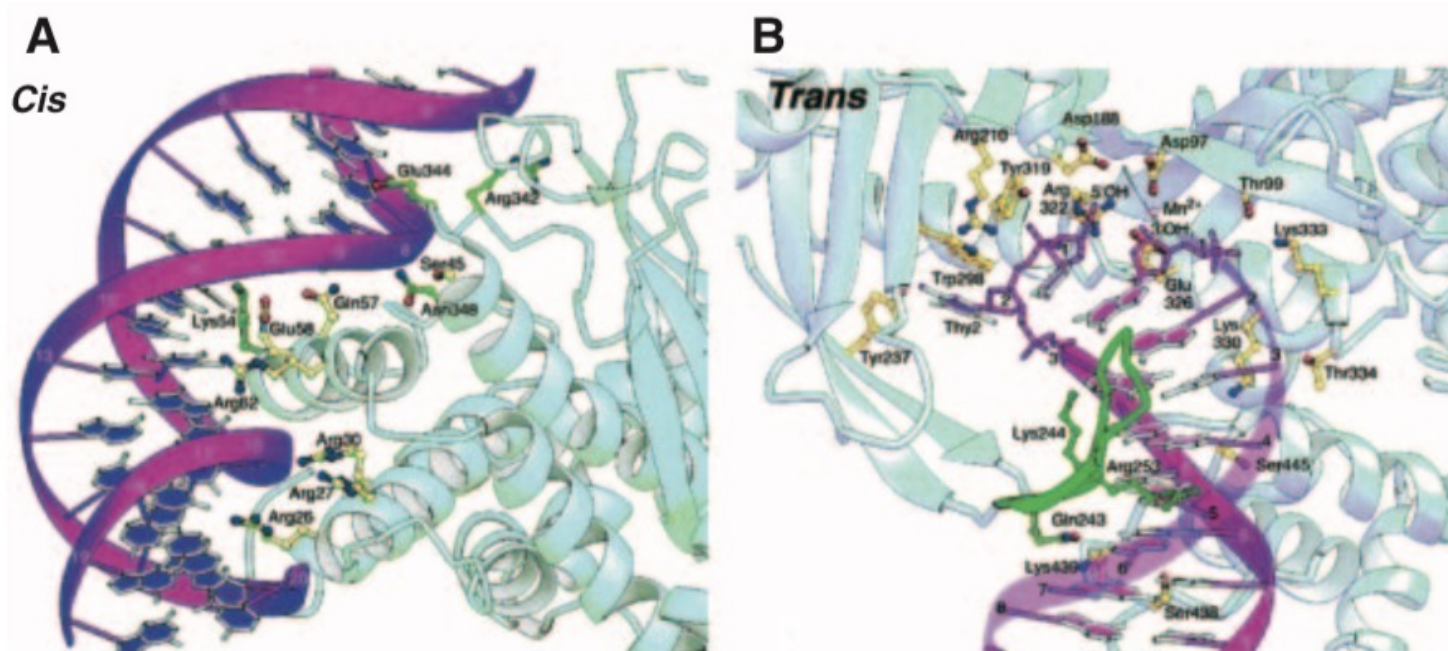
C End sequences

OE: CTG**A**CTCTT**A**TACACA**A**GT

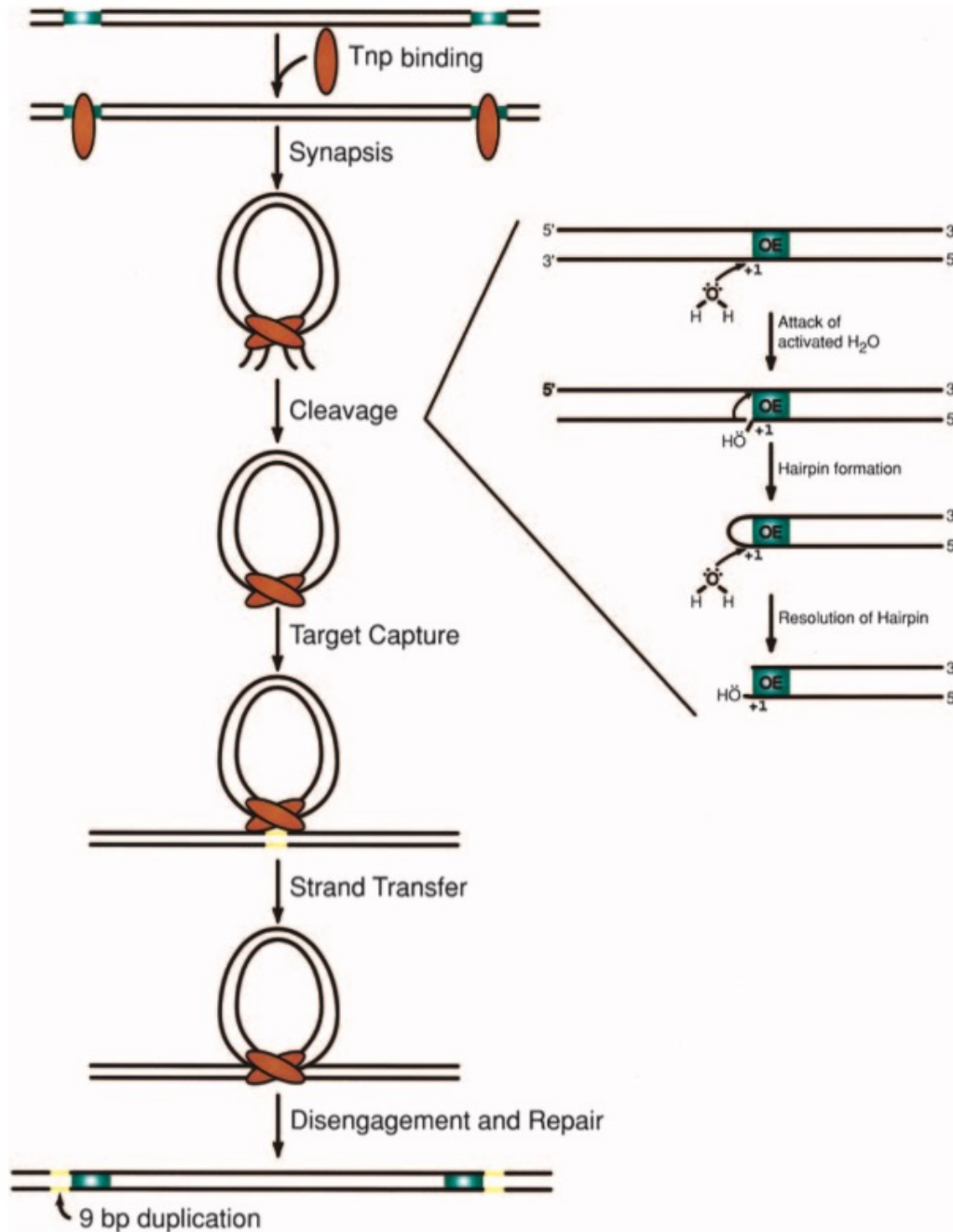
IE: CTG**T**CTCTT**G**ATCAG**A**TCT

ME: CTGTCTCTT**A**TACACA**A**TCT

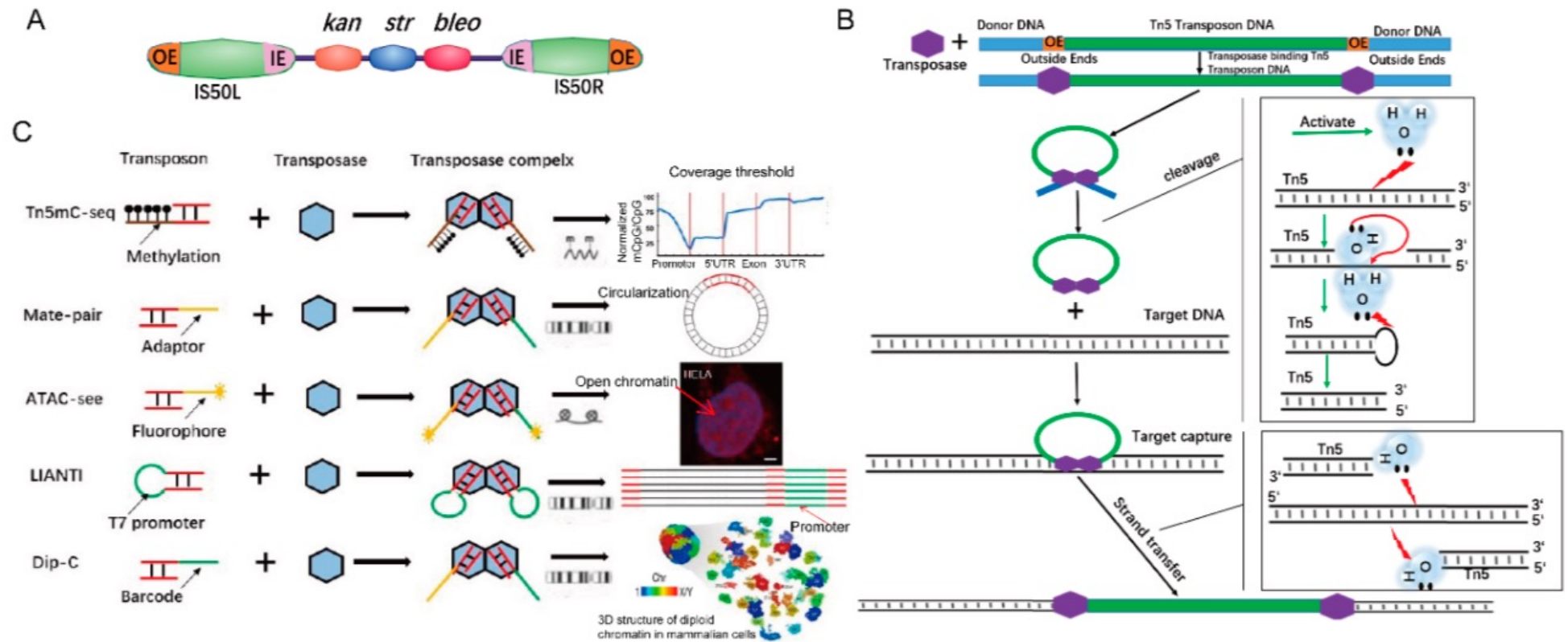




Details of the Tnp–DNA contacts in the synaptic complex. The *cis* (A) and the *trans* (B) DNA contacts presented previously in Davies *et al.* (2000) are shown. The site of E54K hyperactive mutation and residues 342, 344 and 348 are emphasized in green in the *cis* contacts and the 252–259 β -loop clamp (also in green) is emphasized in the *trans* contacts.

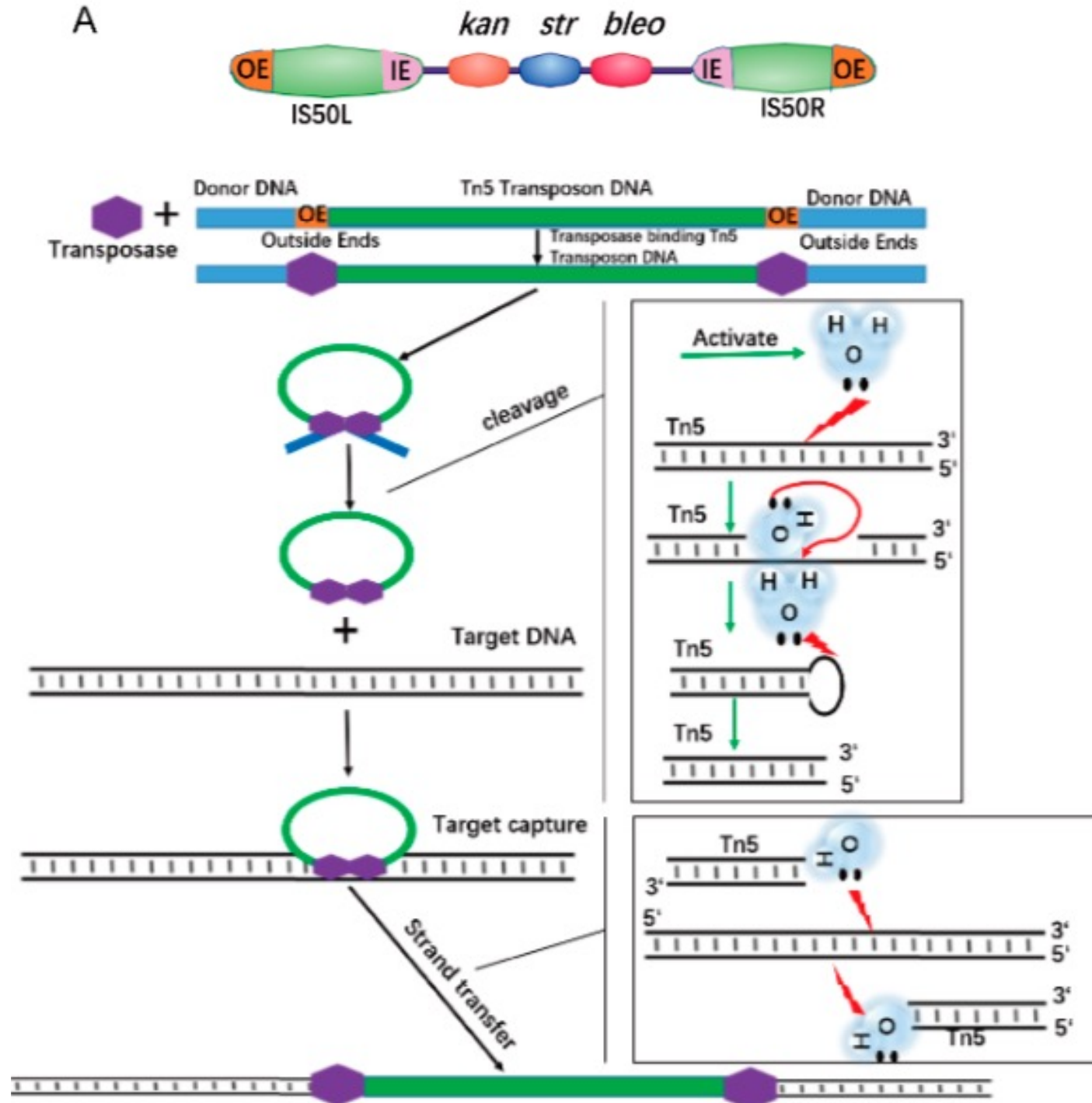


Tn5 transposition mechanism. Synaptic complexes are formed after Tnp binds to the ends and then dimerizes forming both Tnp-Tnp and *trans* Tnp-DNA contacts. In the presence of Mg^{++} or Mn^{++} , three catalytic steps occur resulting in cleavage of transposon-donor DNA bonds. The released synaptic complex then captures target DNA and strand transfer (that also requires the bound Mg^{++}) occurs. See Reznikoff (2002) for more details.

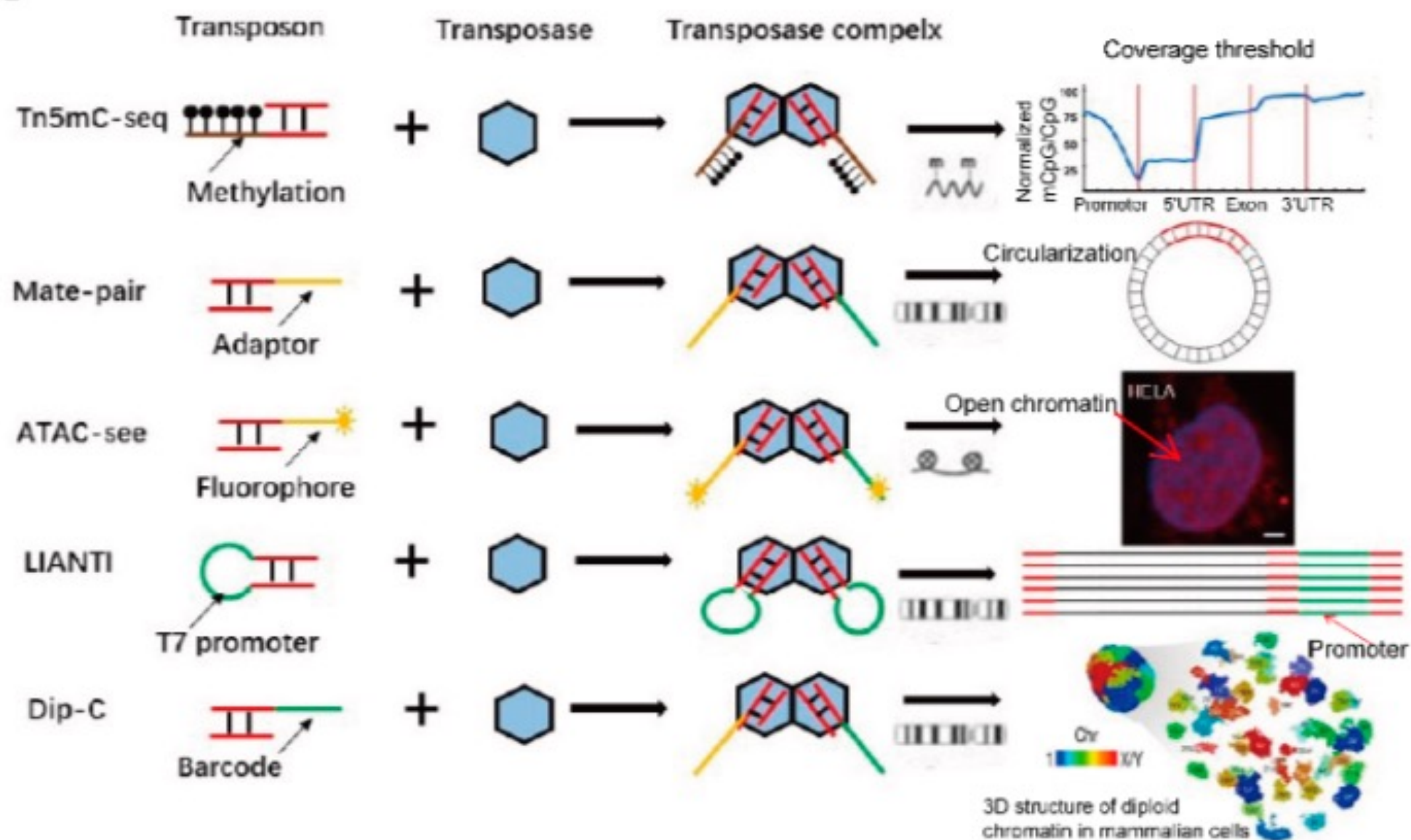


Tn5 transposon structure and transposition mechanism. (A). Tn5 transposon structure consisting of a core sequence that encodes three antibiotics and two inverted IS50 sequences. The outside ends (OEs) bind to Tn5 transposases. (B). Scheme of the Tn5 transposition mechanism [11]. (C). Tn5 adaptor modification is used in epigenetics, genomic structure, and chromatin visualization. Tn5mC-seq is used in research studies on DNA methylation. Mate-pair applied to the amplification of long DNA fragments. Assay for Transposase-Accessible Chromatin with high throughput sequencing (ATAC-Seq) is applied to open chromatin (scale bar 1:600). Linear Amplification via Transposon Insertion (LIANTI) is applied to genomic variation. Dip-C is used in reconstructing the 3D structure of the genome.

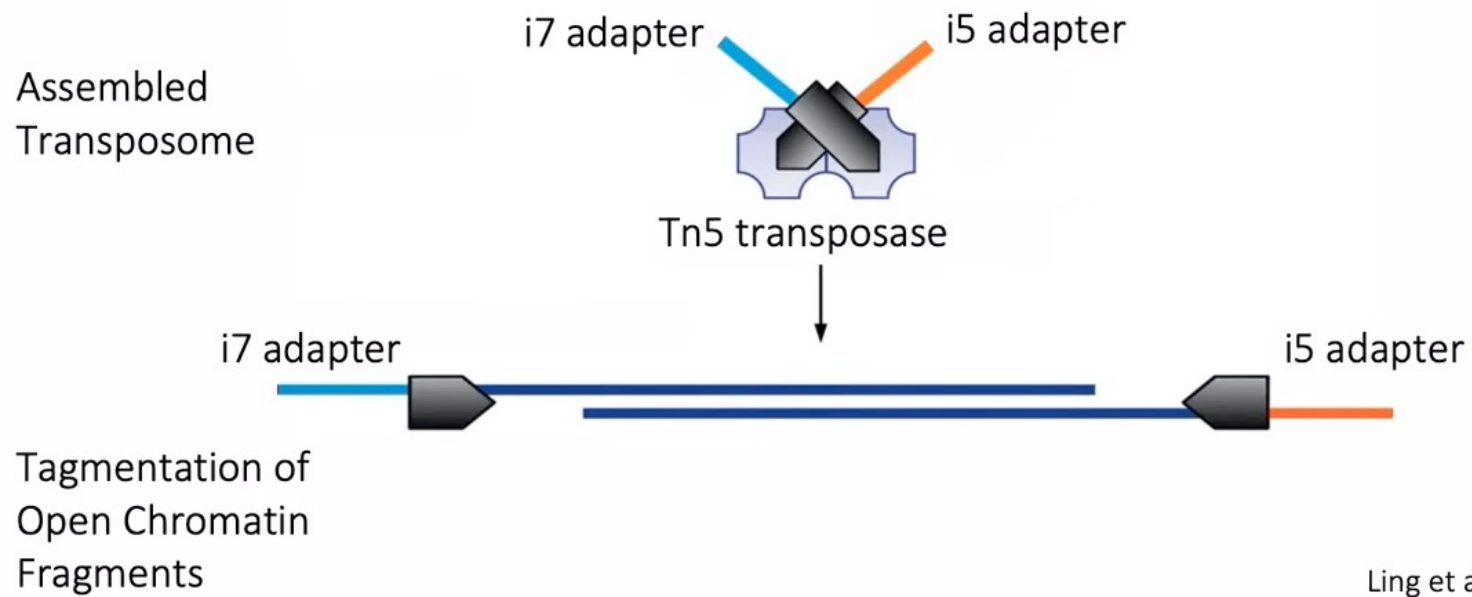
A



C

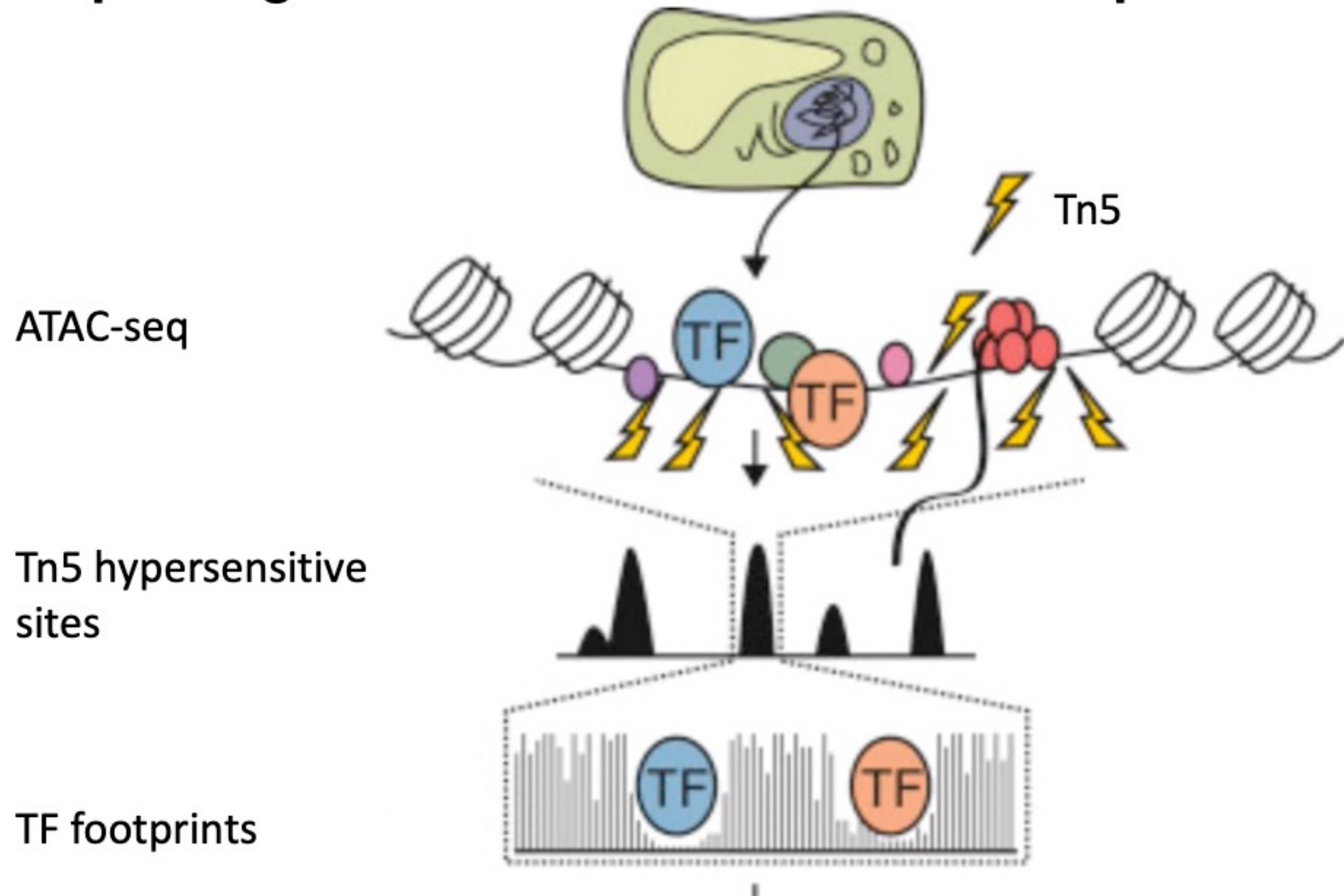


Tn5 Transposase



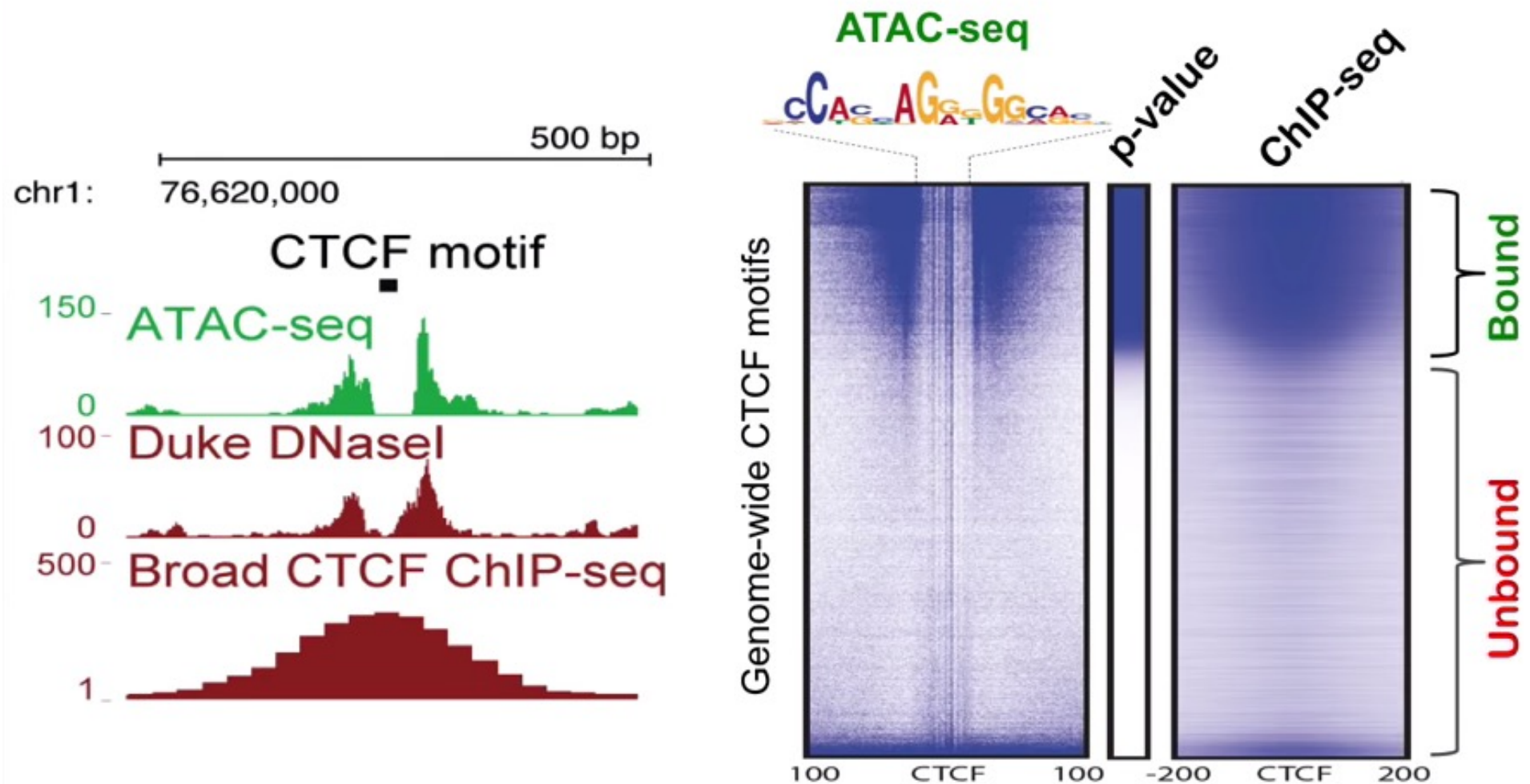
Ling et al., 2015, PNAS

Tn5 footprinting can reveal locations of transcription factors



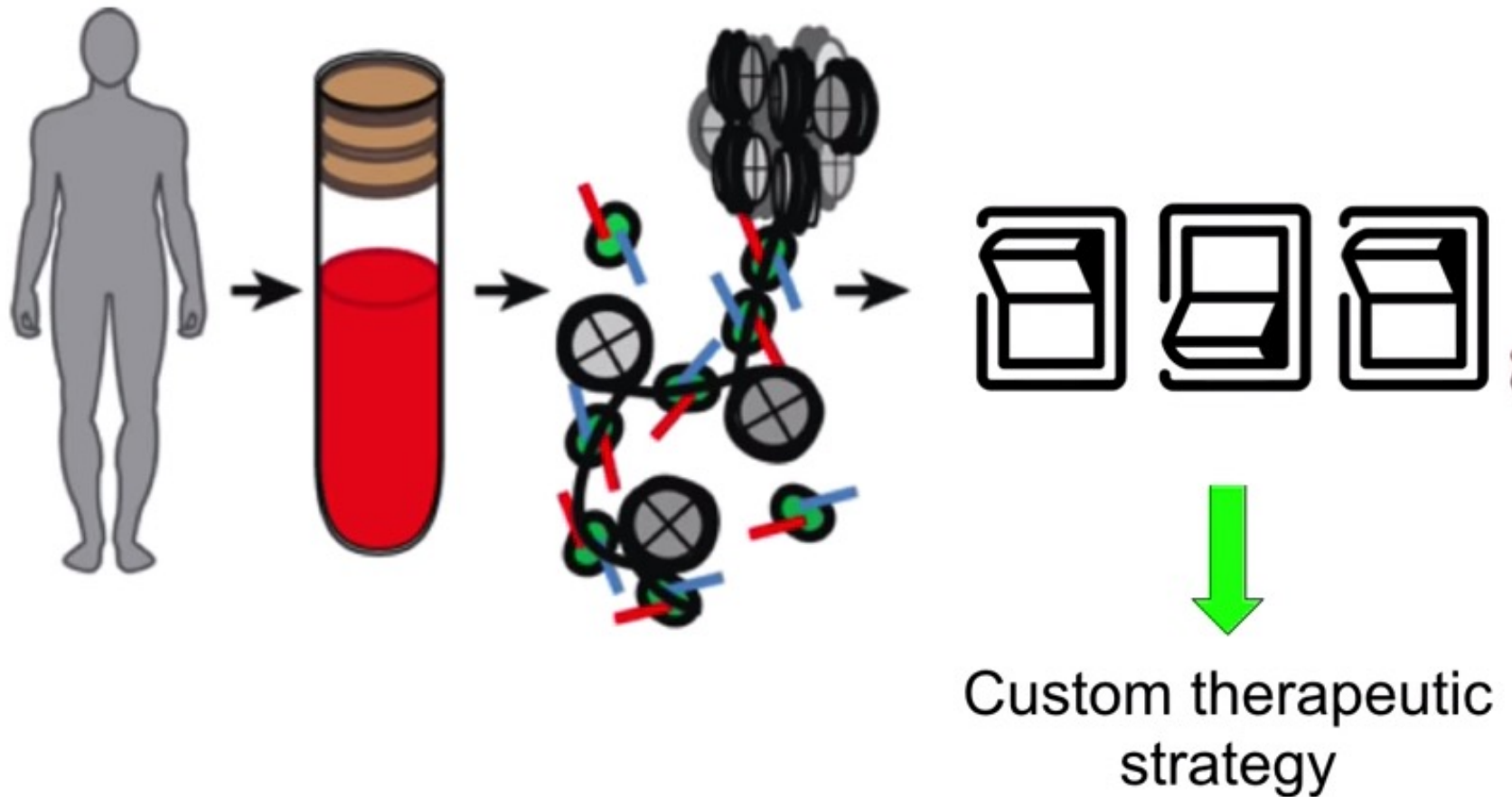
modified from <http://www.sciencedirect.com/science/article/pii/S2214662815000122>

ATAC-seq reveals DNA-protein interactions

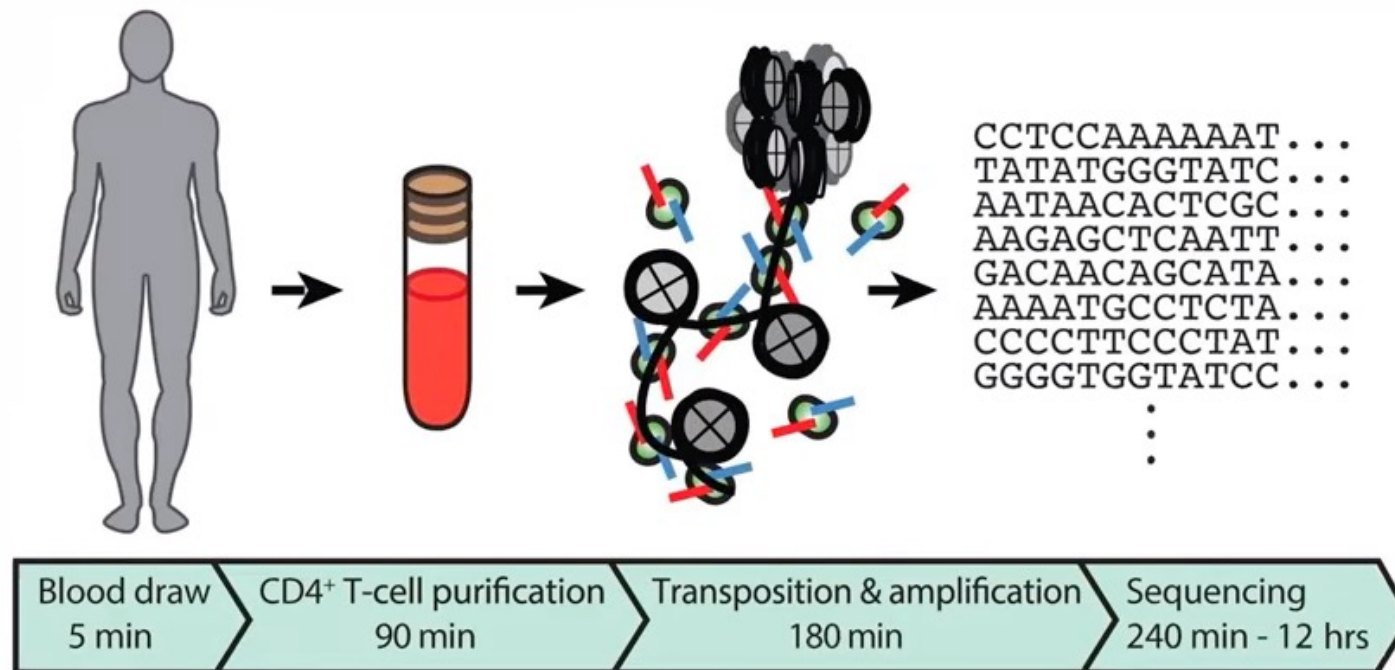


Binding locations of hundreds of factors read out at once,
in allele-specific fashion.

Personal GPS for navigating regulome

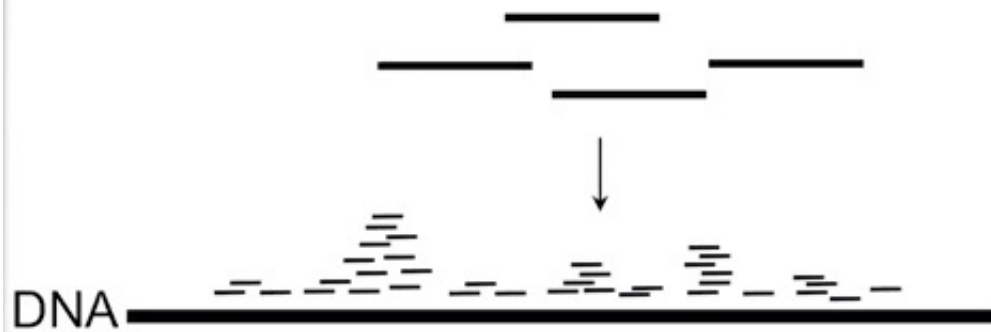


ATAC-seq Enables Real-Time Personal Epigenomics



Adapted from Buenrostro et al., 2013, Nat Methods

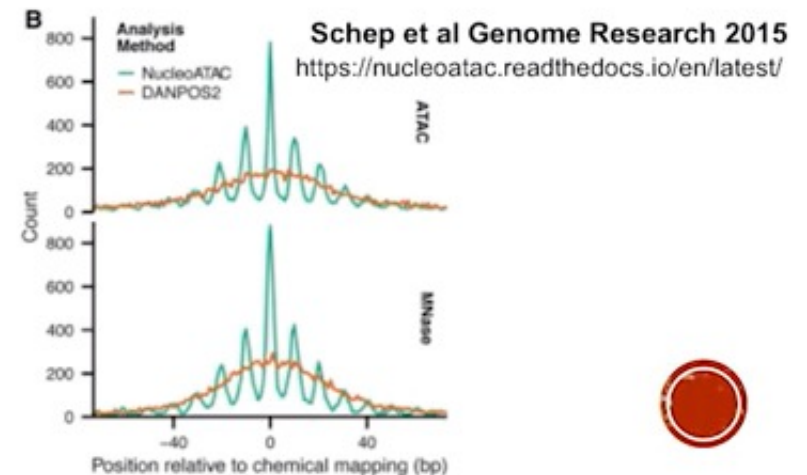
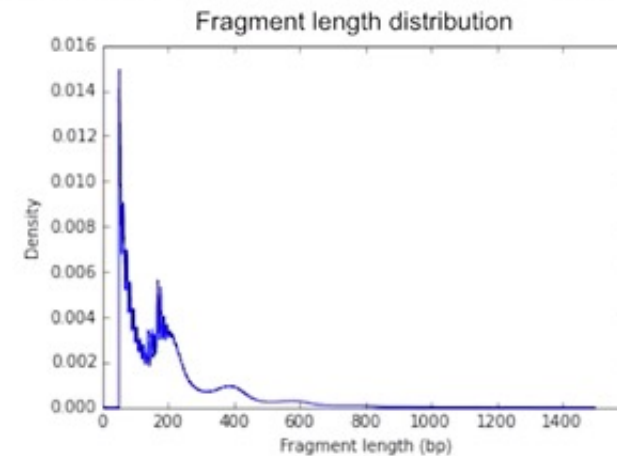
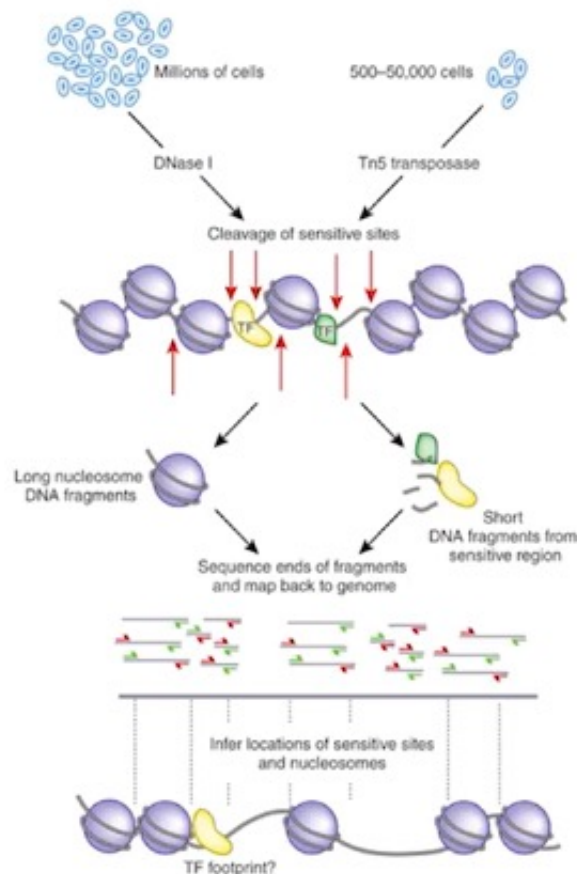
Introduction to an ATAC-seq basic analysis outline: Fragment distribution and nucleosome mapping



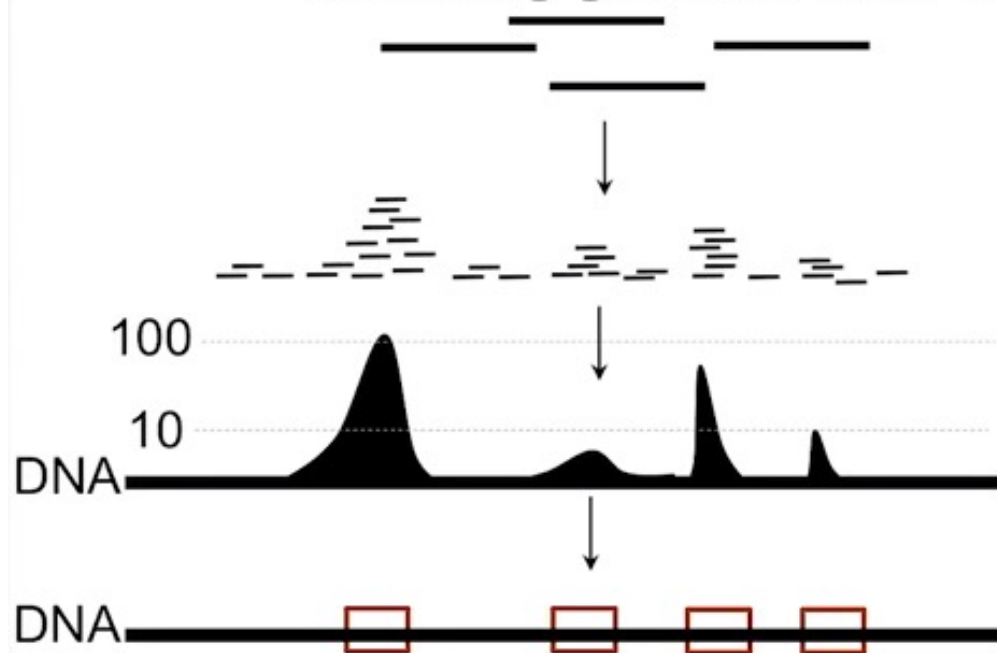
1

Sequencing reads/tags

Alignment to genome
(Bowtie, samtools, Picard, bwa etc)



Introduction to an ATAC-seq basic analysis outline: Detecting genome-wide chromatin accessibility dynamics



1

Sequencing reads/tags

2

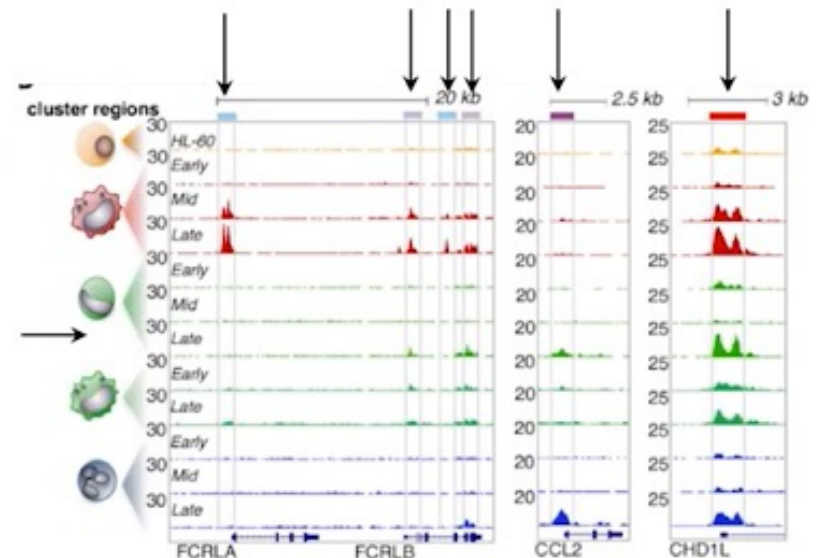
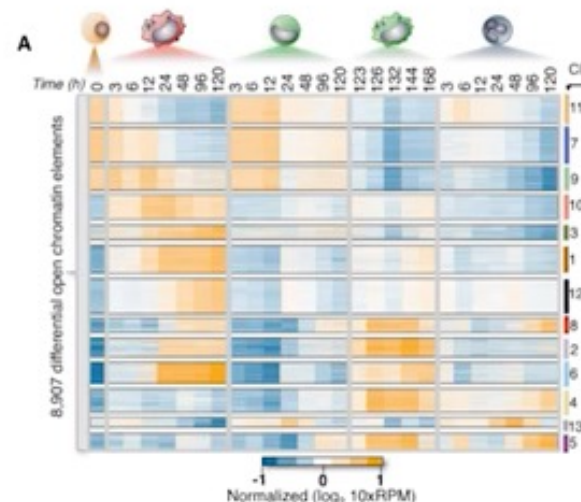
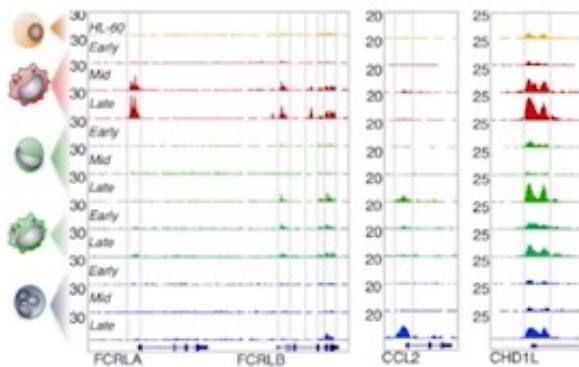
Alignment to genome
(Bowtie, samtools, Picard, bwa etc)

Signal construction
Visualization (IGV, UCSC etc)

ATAC-seq peak calling
(MACS2, HOMER, IDR etc)

ATAC-seq peaks

ATAC-seq peak clustering

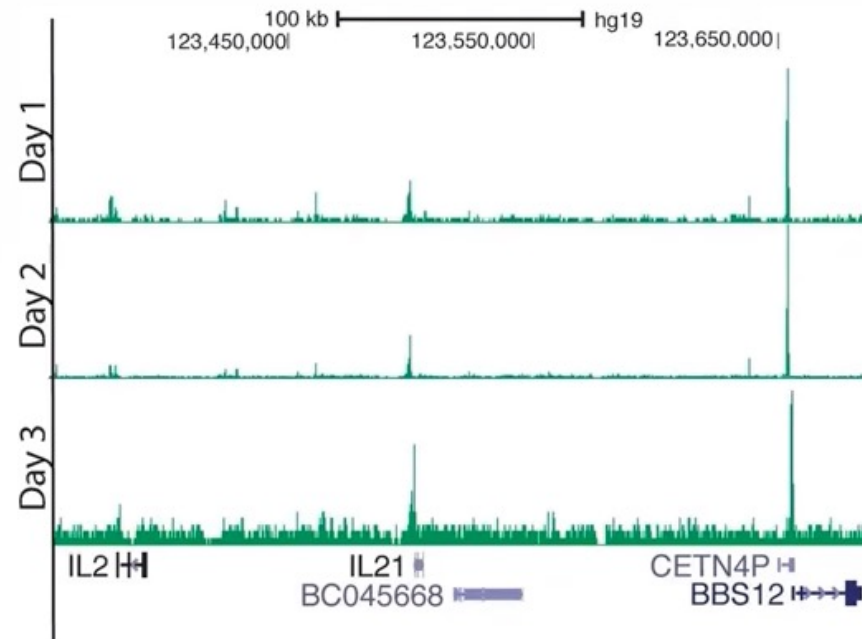


Resolving patterns of chromatin
accessibility with peak clusters



ATAC-seq Enables Real-Time Personal Epigenomics

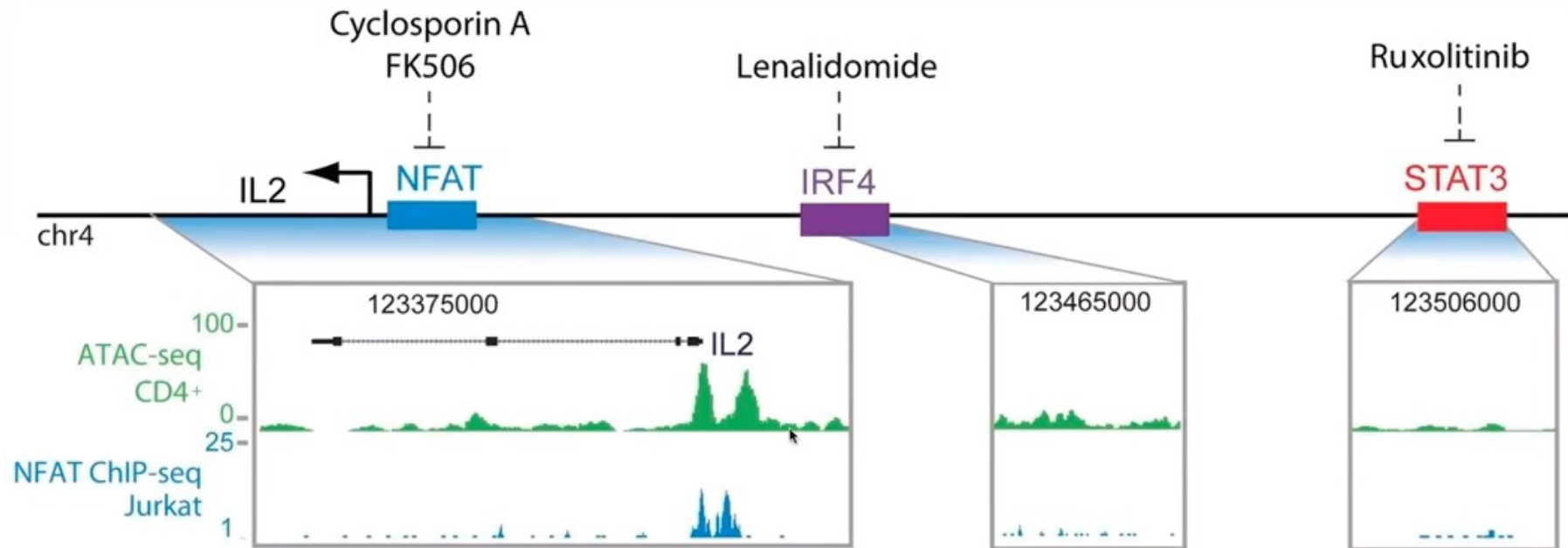
IL-2 is a key cytokine that drives T-cell growth and functions in inflammatory and autoimmune diseases



Buenrostro et al., 2013, Nat Methods

ATAC-seq Data Can Prioritize Candidate Drug Targets

Cyclosporin A is a good candidate for this patient, given that the NFAT binding site in the IL-2 promoter is aberrantly open



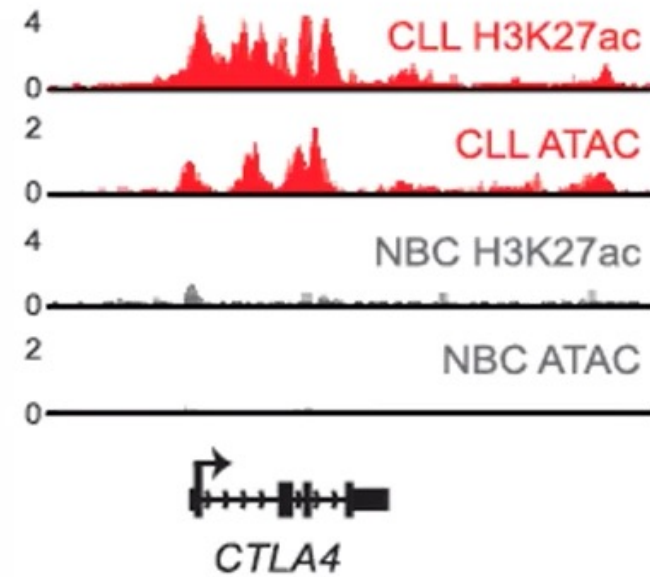
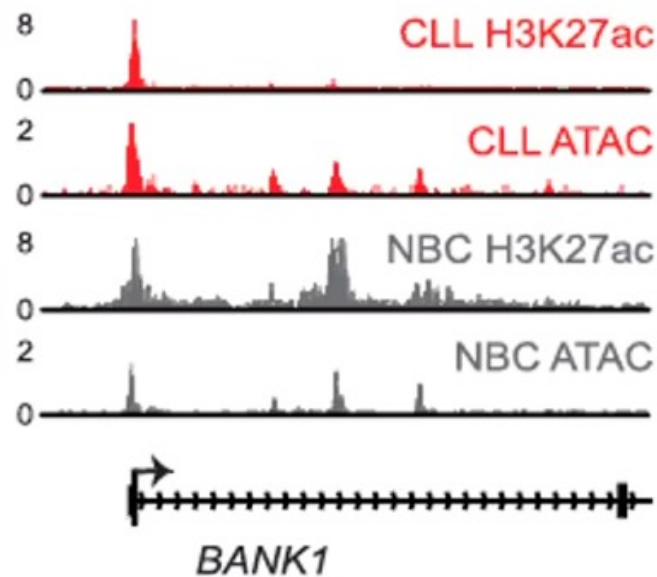
Buenrostro et al., 2013, Nat Methods

Data Integration: ATAC-seq Enhances Other NGS Assays

ATAC-seq correlates with ChIP-seq for H3K27ac

Chronic
Lymphocytic
Leukemia

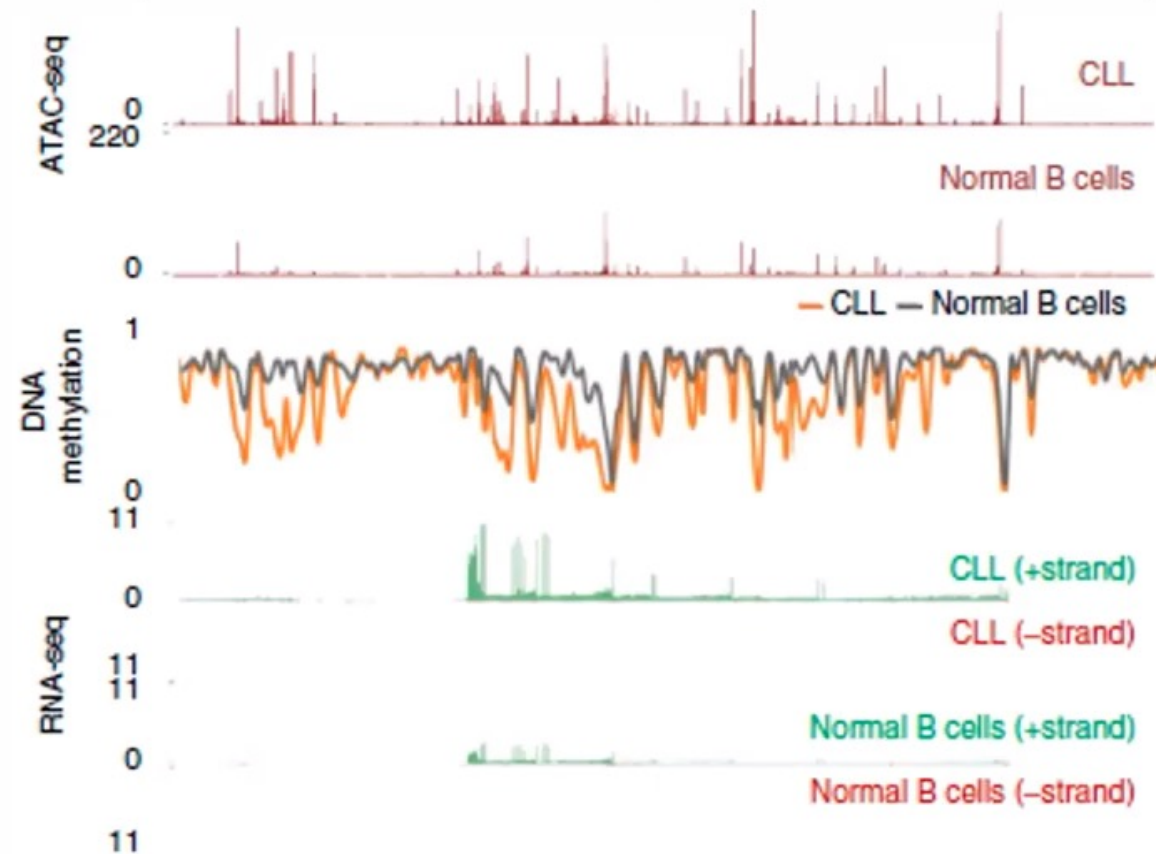
Normal
Blood
Cells



Ott et al., 2018, Cancer Cell

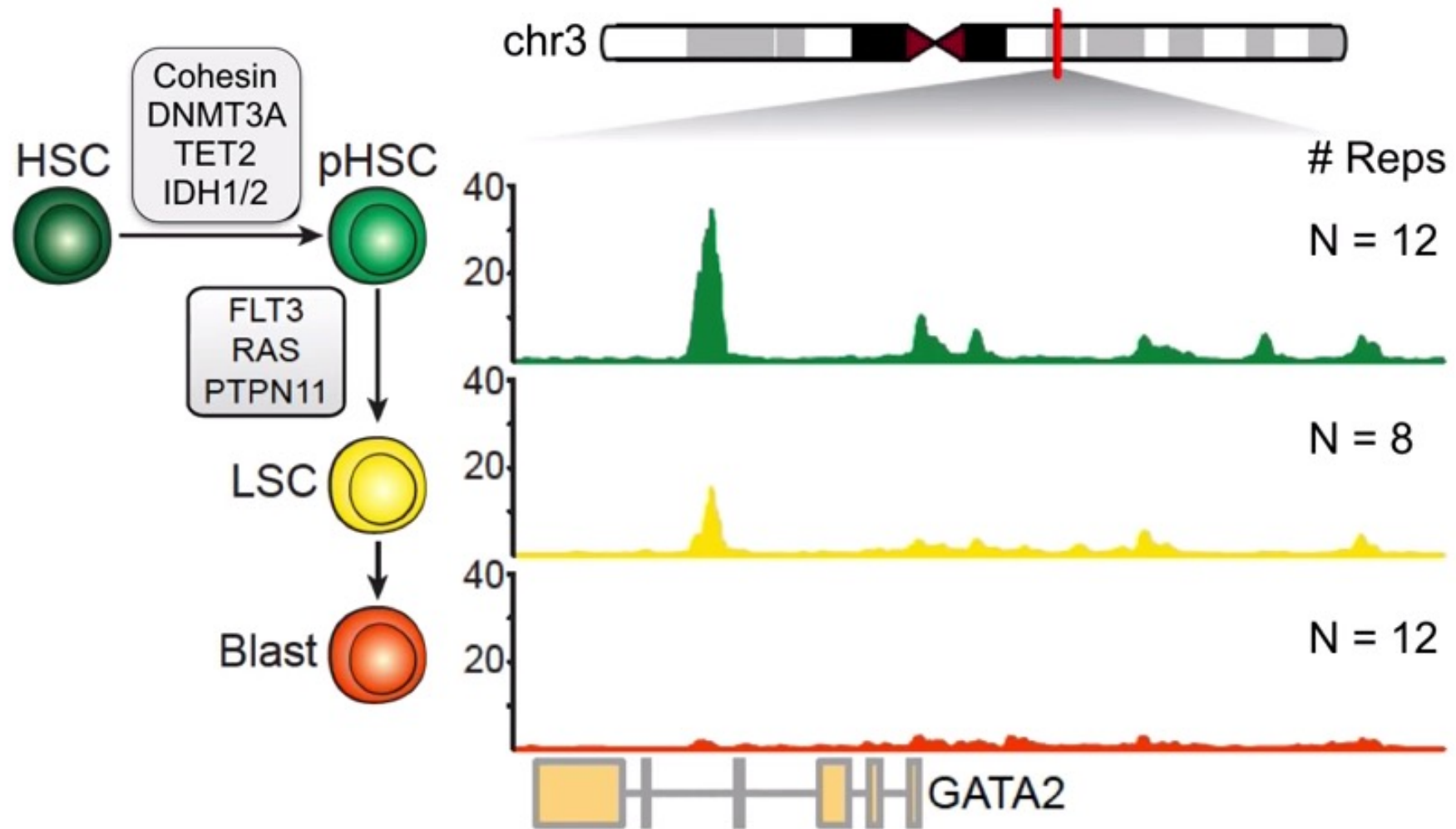
Data Integration: ATAC-seq Enhances Other NGS Assays

ATAC-seq correlates with RNA-seq and DNA methylation

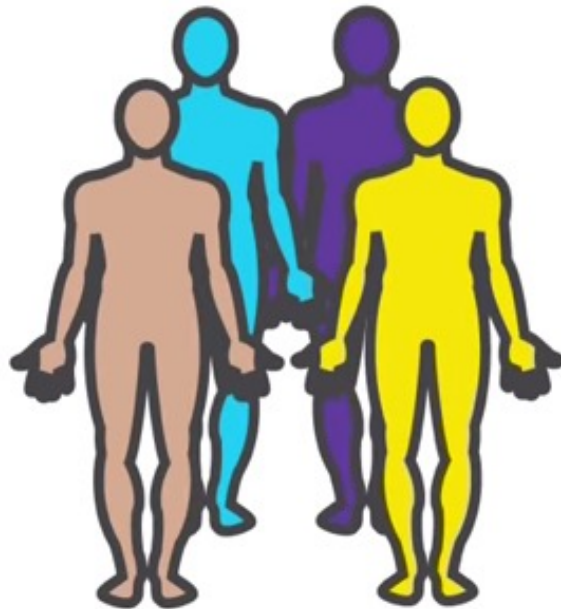


Beekman et al., 2018,
Nature Medicine

Regulome landscape of cancer evolution



Genetic risks for cancer: Faulty gene switches

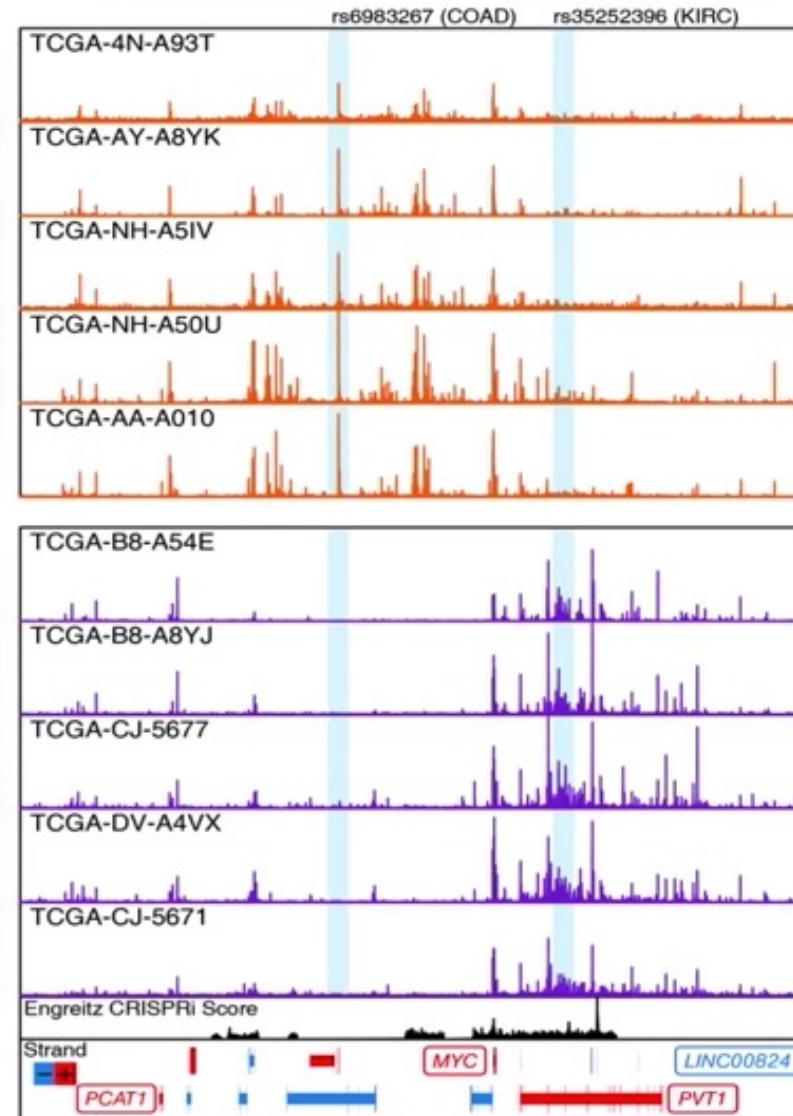


Upstream Elements

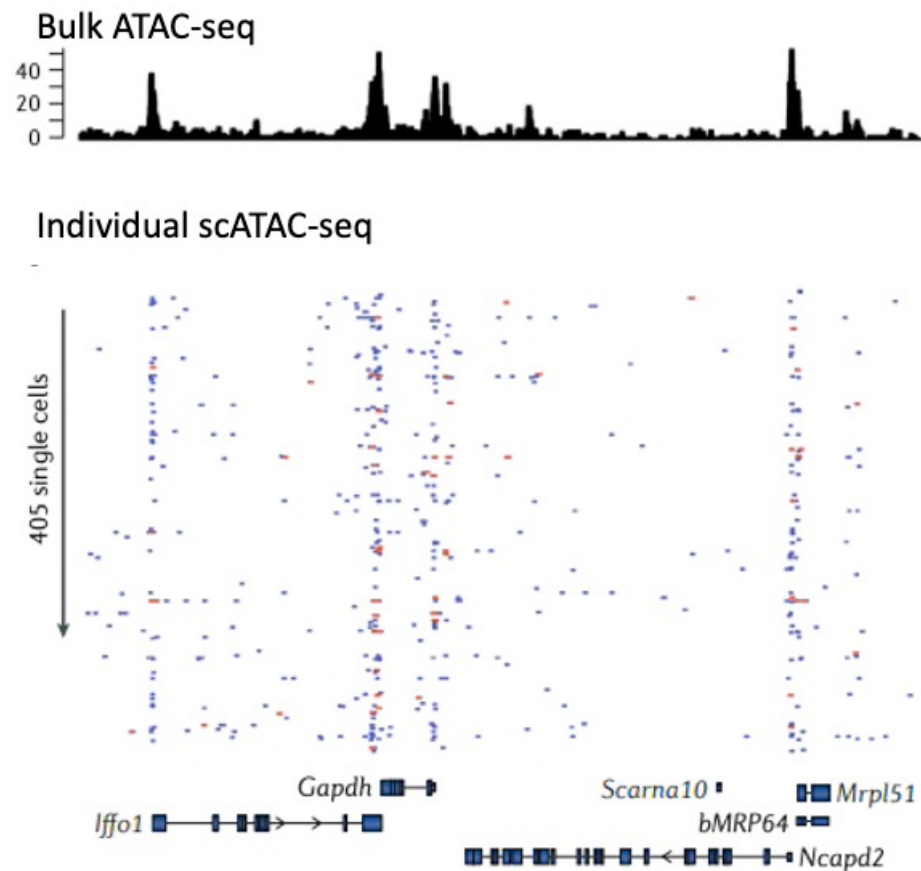
Downstream Elements

Colon cancer
predisposition

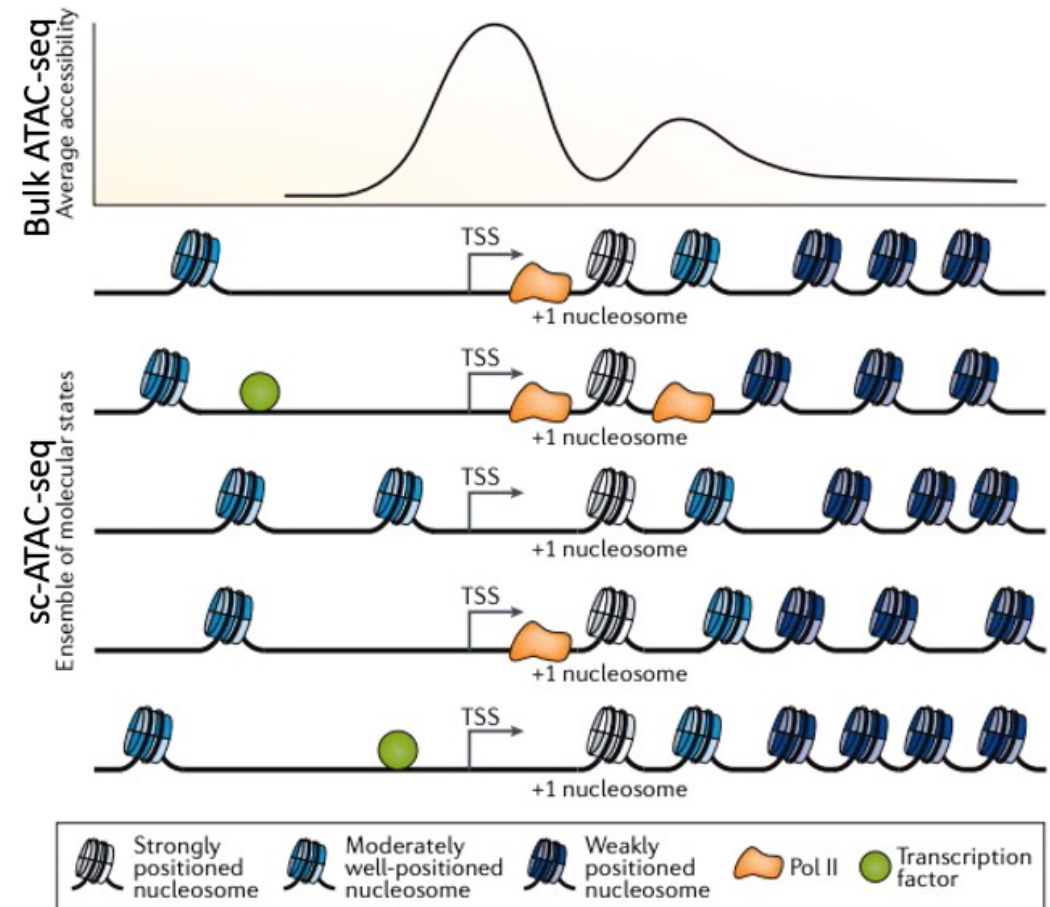
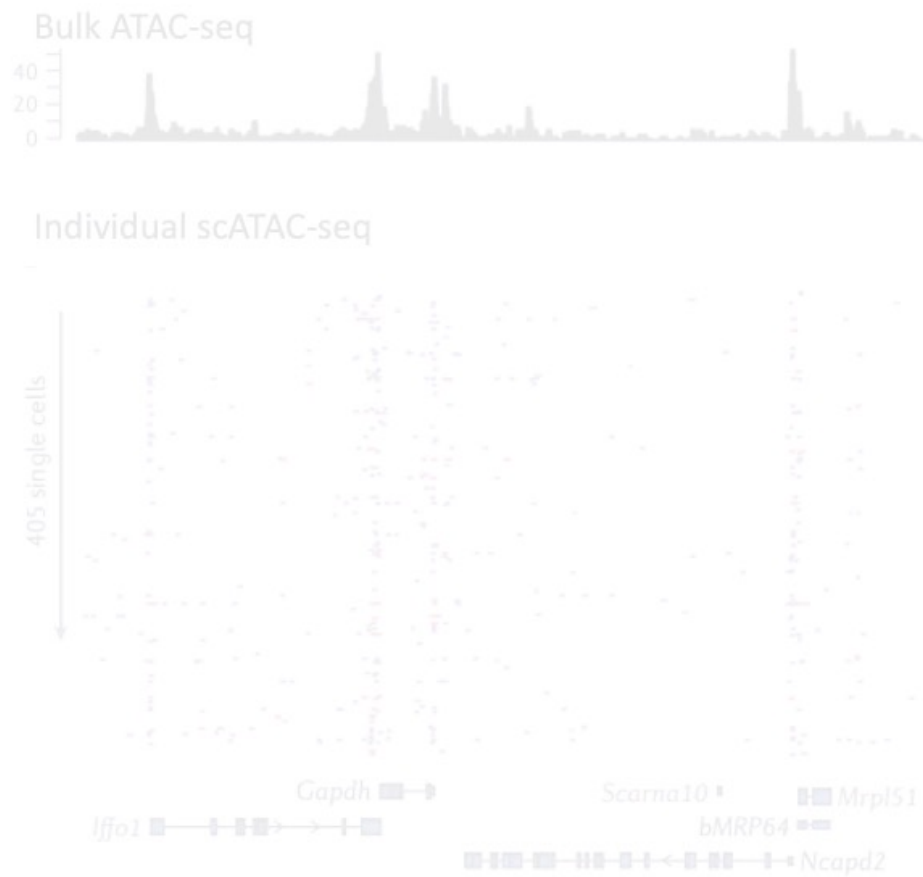
Kidney cancer
predisposition



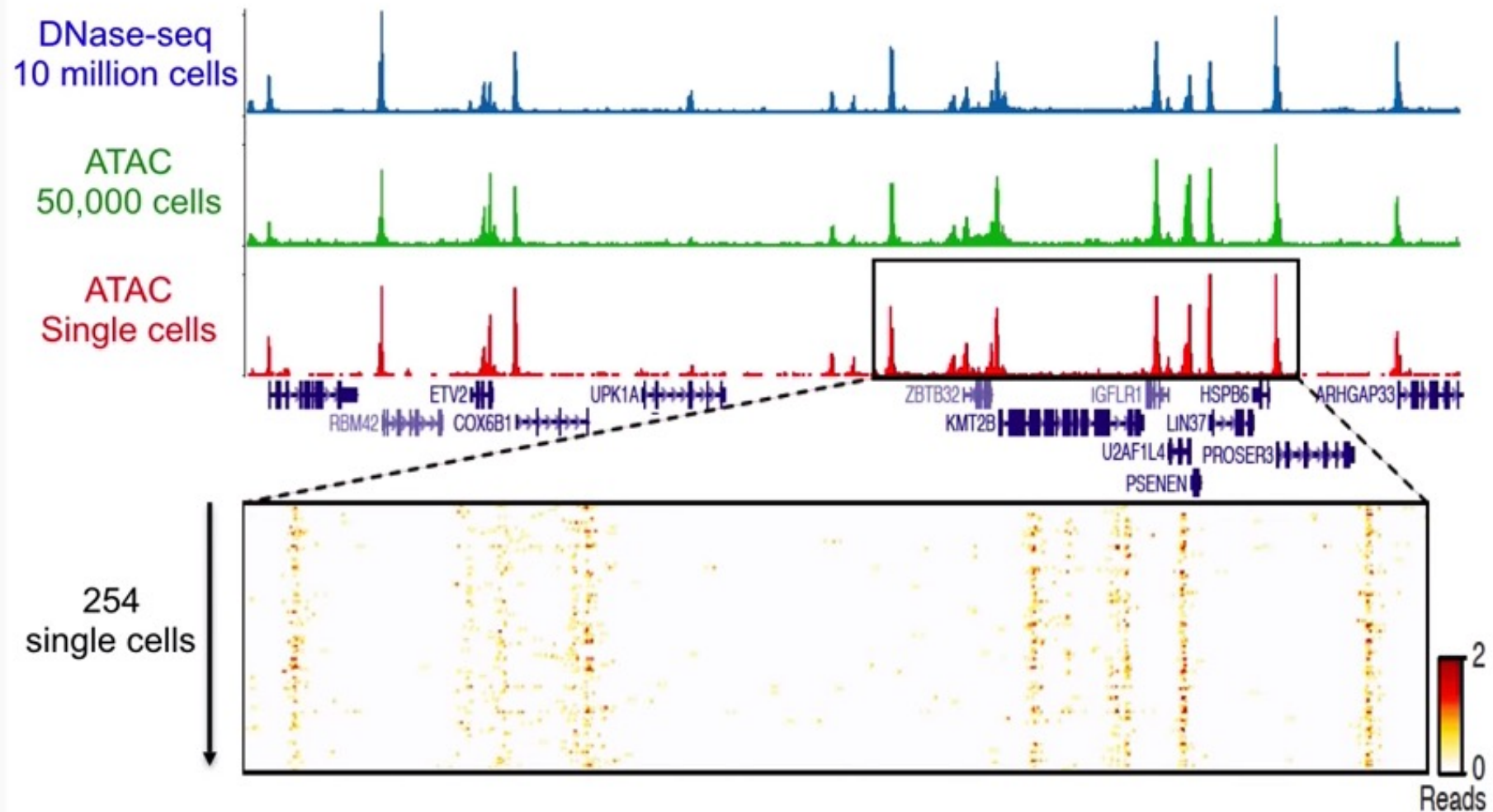
Chromatin accessibility is highly dynamic



Chromatin accessibility is highly dynamic



Active regulatory elements in single cells



Different scATAC-seq Techniques



Different scATAC-seq Techniques

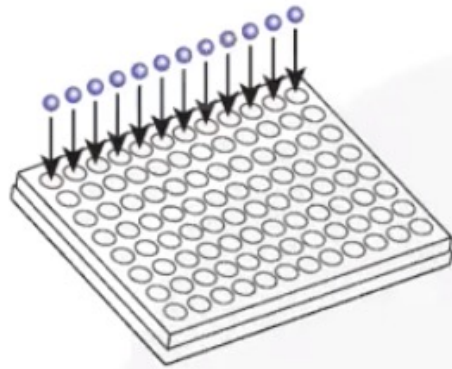
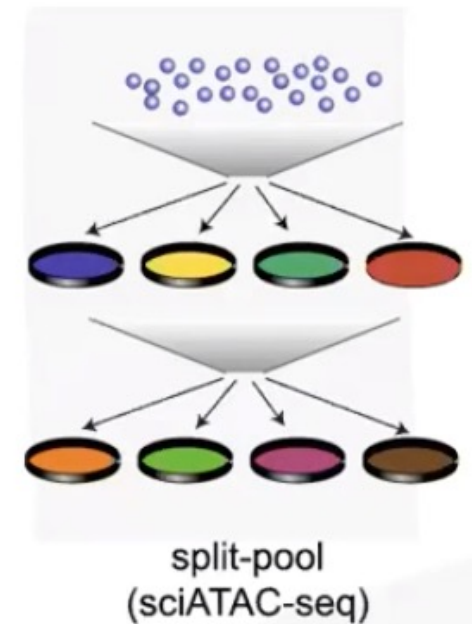
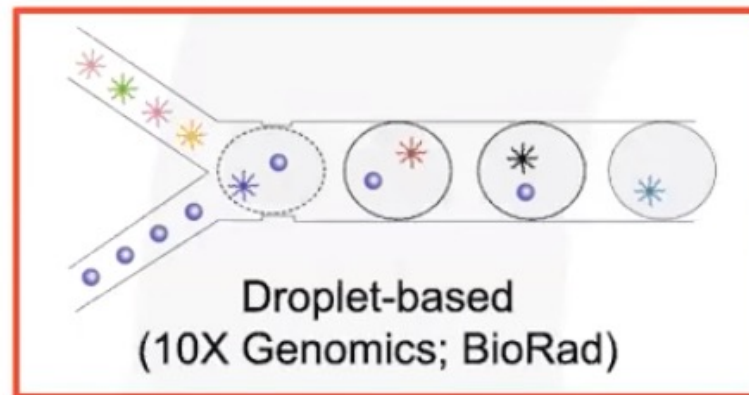


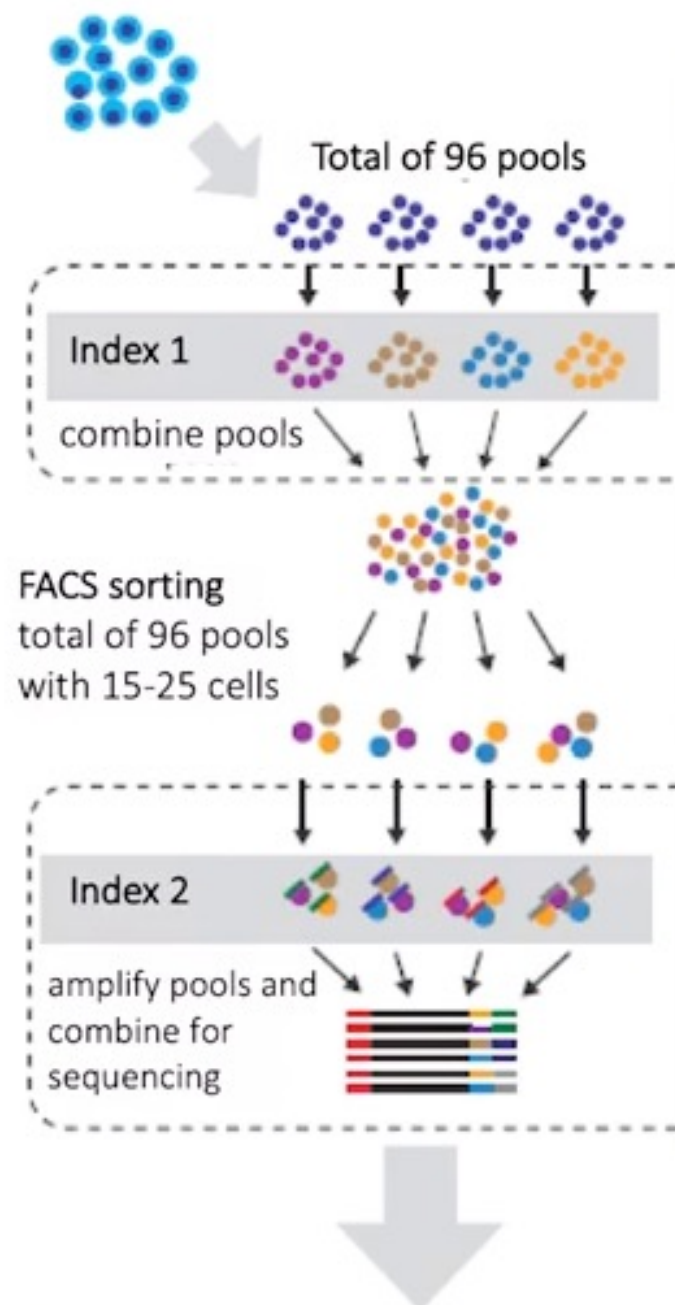
Plate or array
(ICELL8, Fluidigm C1)

Different scATAC-seq Techniques



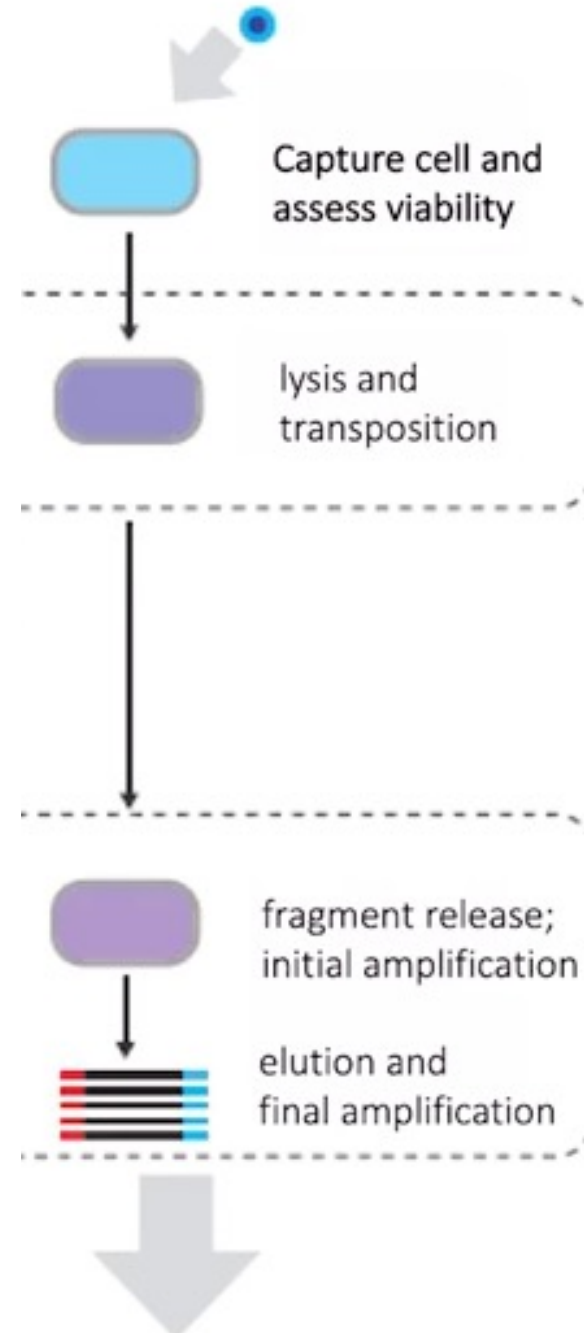
Cell Indexing Method

(Cusanovich et al)



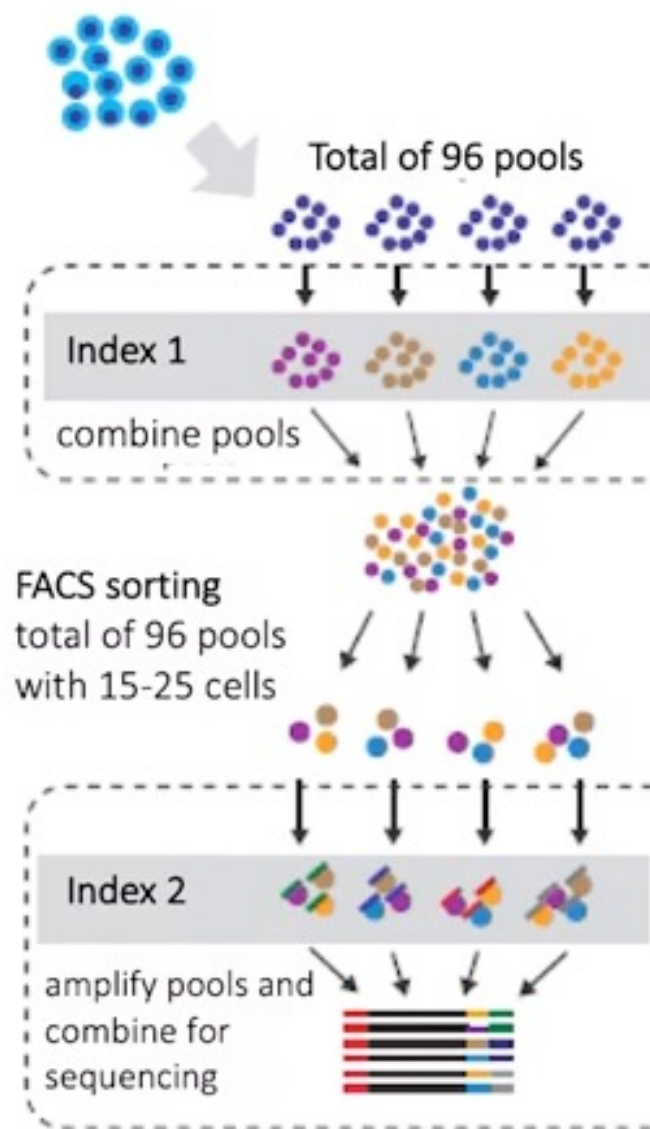
Microfluidic Method

(Buenrostro et al)

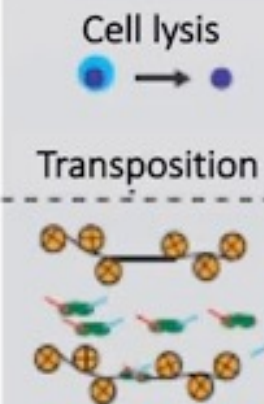


Cell Indexing Method

(Cusanovich et al)

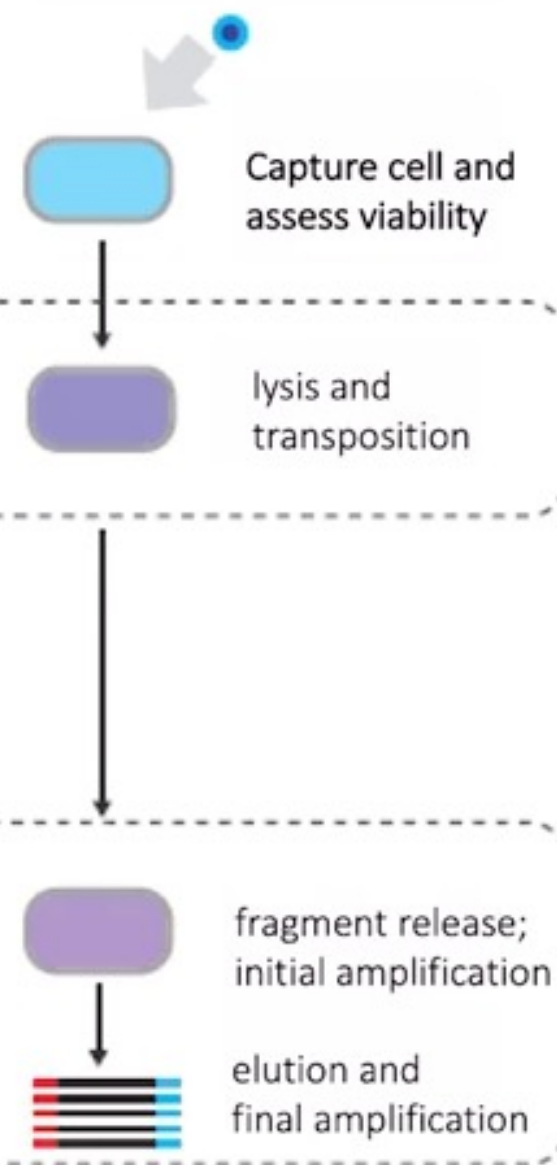


Key steps in ATAC-Seq



Microfluidic Method

(Buenrostro et al)



Single Cell ATAC-Seq using 10x Genomics technology

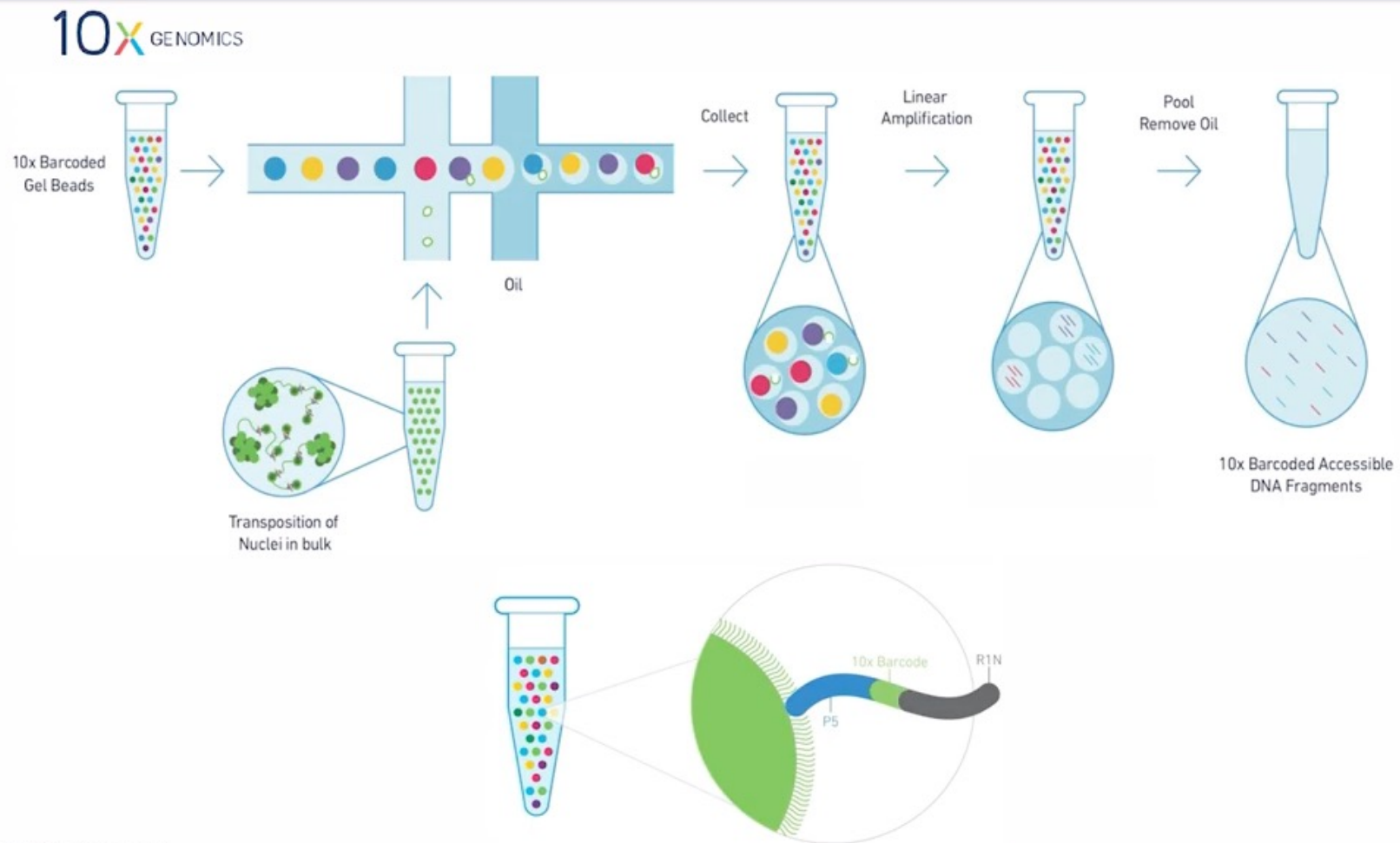
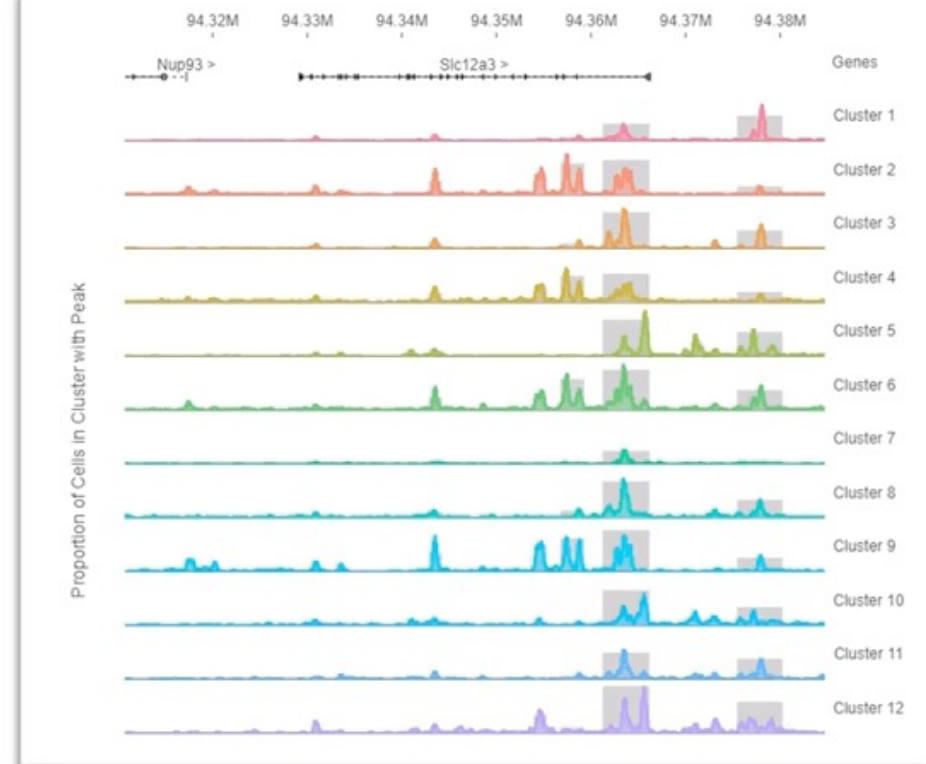
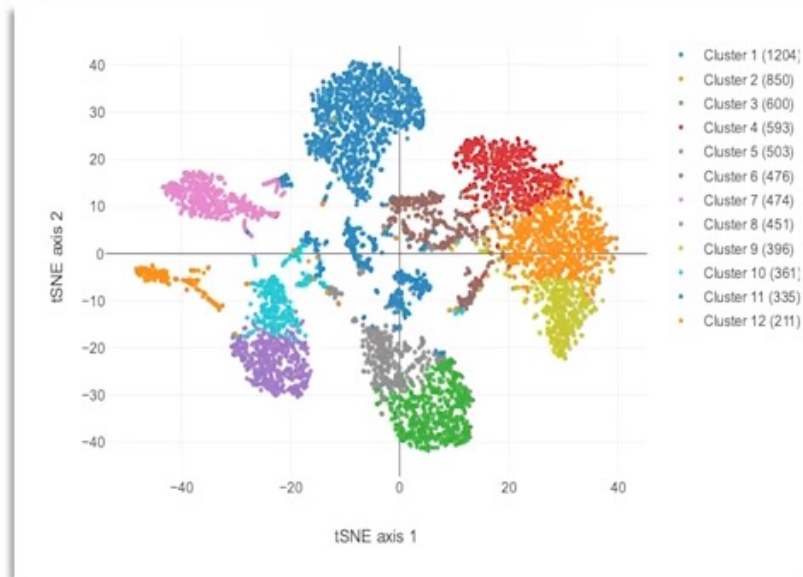
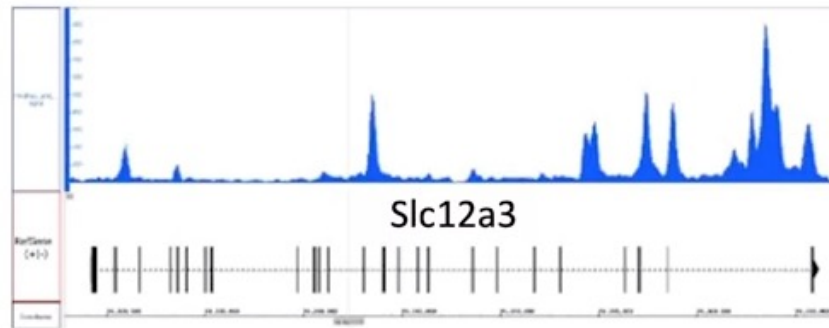


Image courtesy of 10x Genomics

scATAC-Seq data is represented using tSNE plots

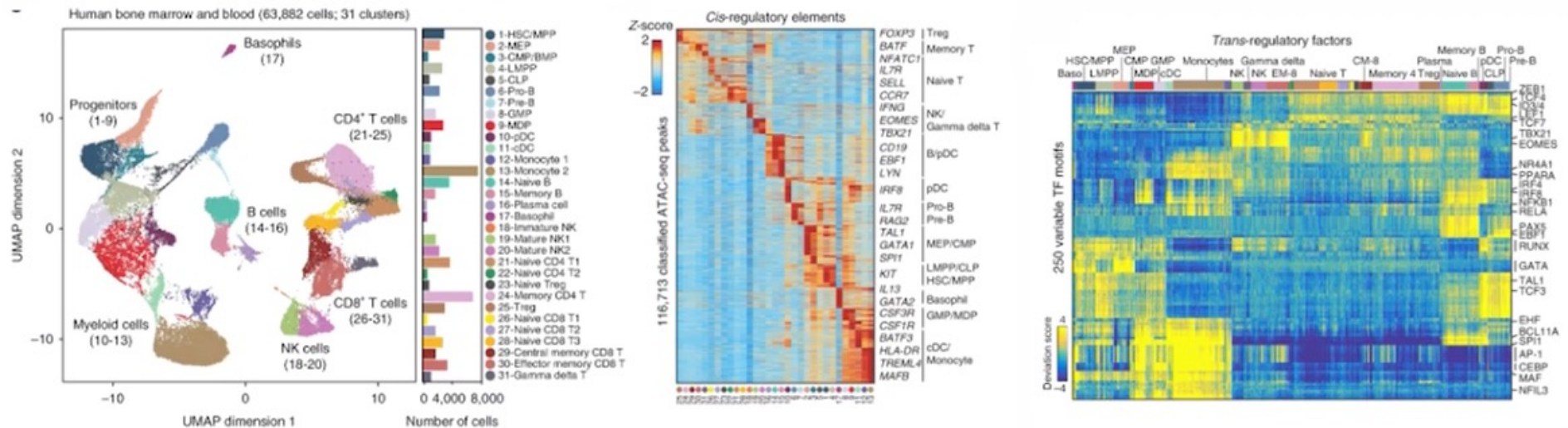


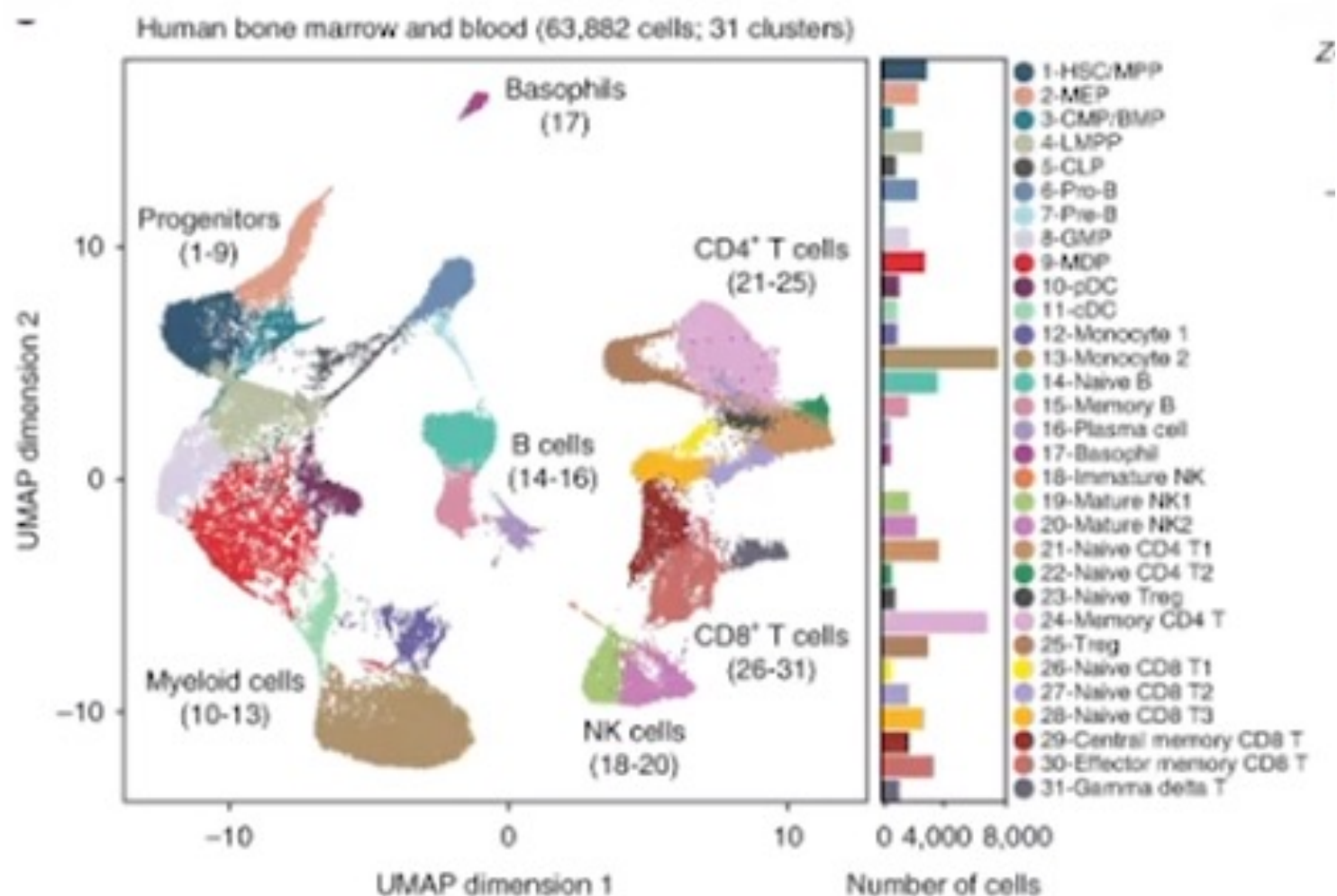
scATAC-Seq can help address many experimental questions

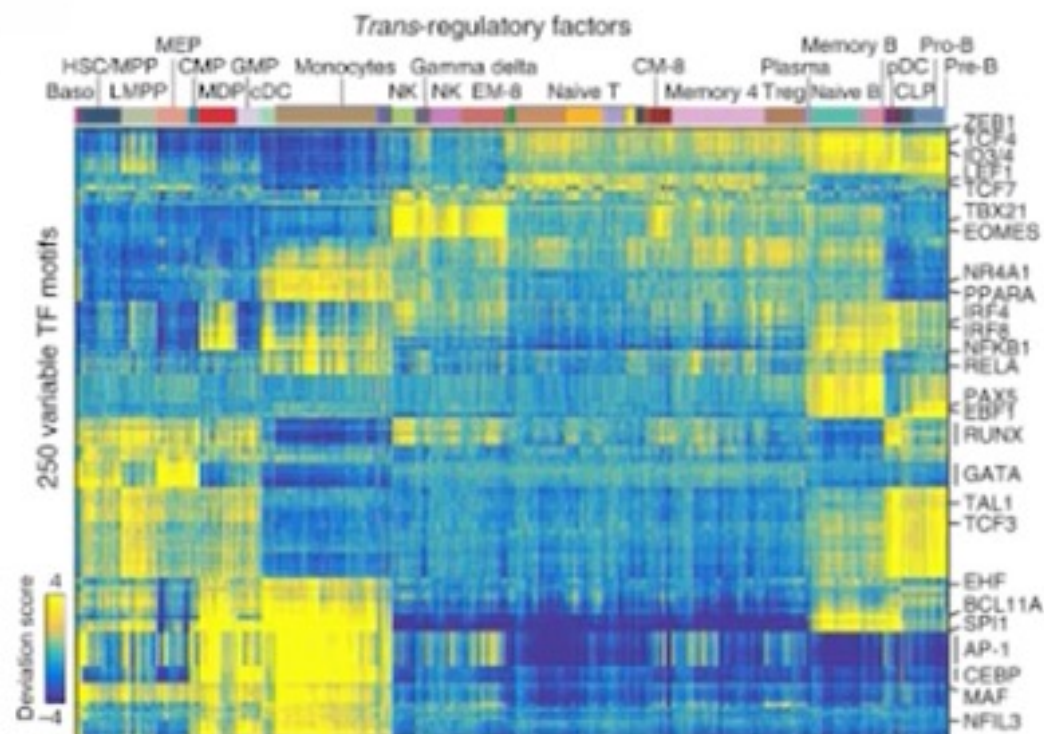
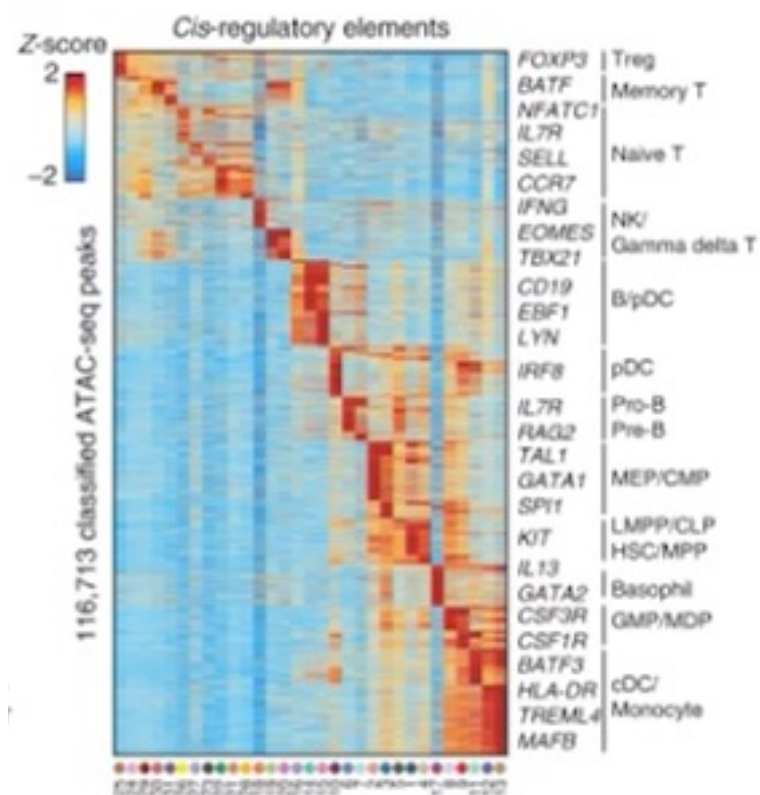
- Classify different cell populations in a heterogenous population
- Understand the tumor microenvironment
 - What cell population(s) are present?
 - What population(s) are driving drug response?
- Examine cell differentiation and cell trajectories during development
- Identify gene regulatory networks involved in development or disease

scATAC-Seq allows for marker-free identification of different cell types

- scATAC-Seq can identify chromatin states of progenitor cells in human peripheral blood and their cis- and trans- regulatory elements

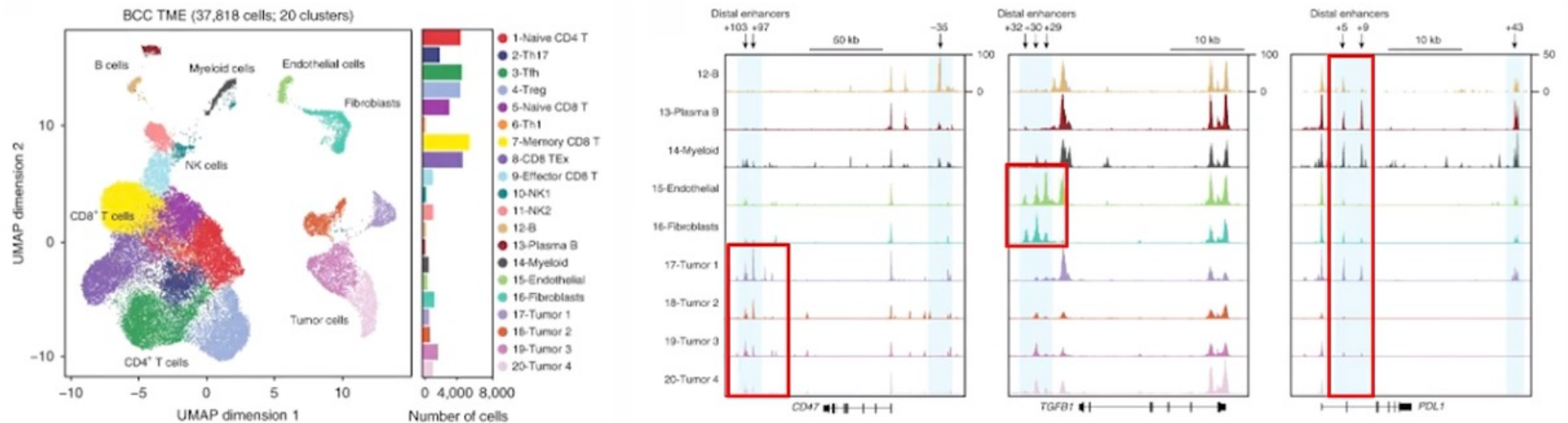


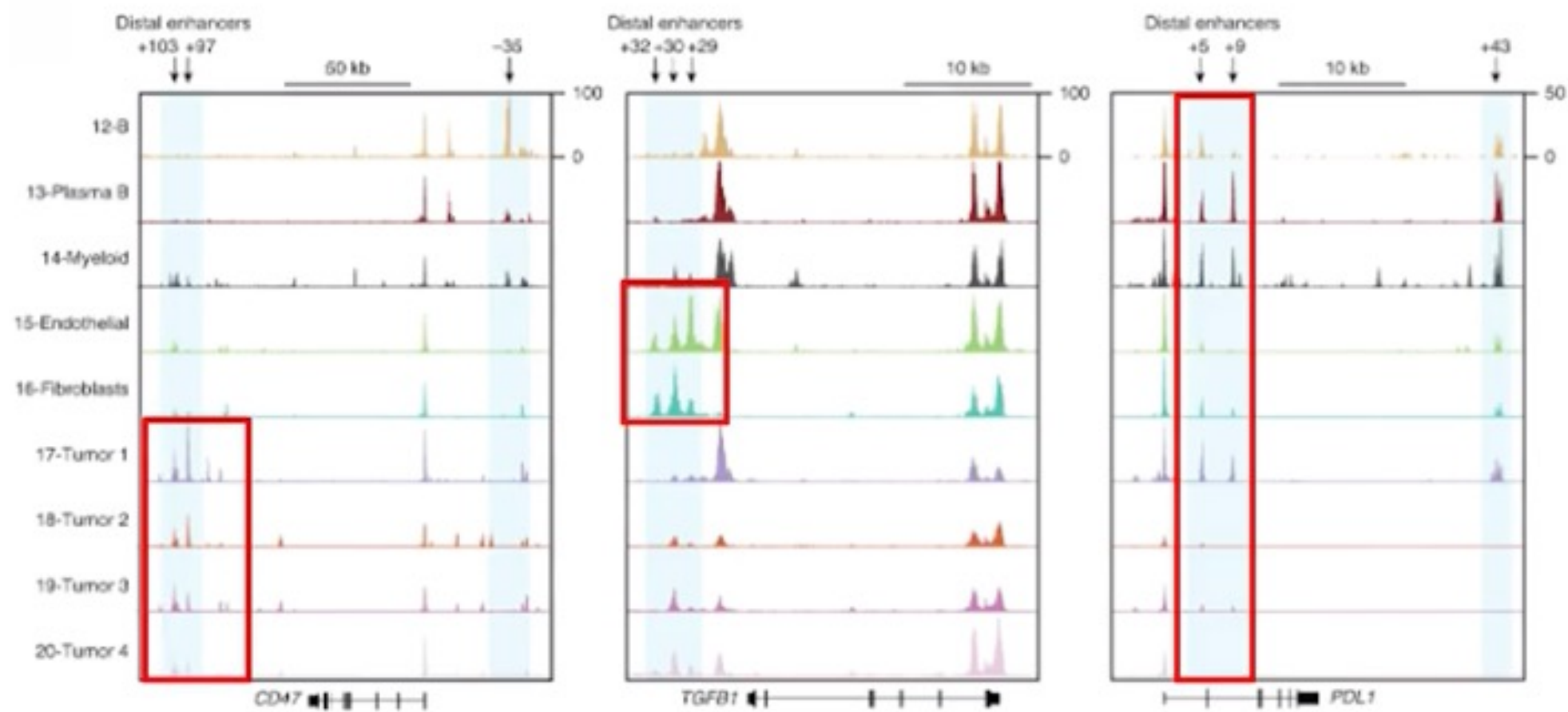




scATAC-Seq can help deconvolute the tumor microenvironment

- Biopsies from patients with basal cell carcinoma (BCC) receiving PD-1 immunotherapy (PD-1 blockade)





Summary

- scATAC-Seq allows for the interrogation of open chromatin at single cell resolution
- Single cell ATAC methodologies include cell indexing or the use of microfluidics
- Examining areas of open chromatin can provide mechanistic insight into how your pathway/cellular response/disease is regulated
- scATAC-Seq can identify what cell population(s) are driving disease or drug response
- scATAC-Seq can deconvolute heterogeneity of a mixed population

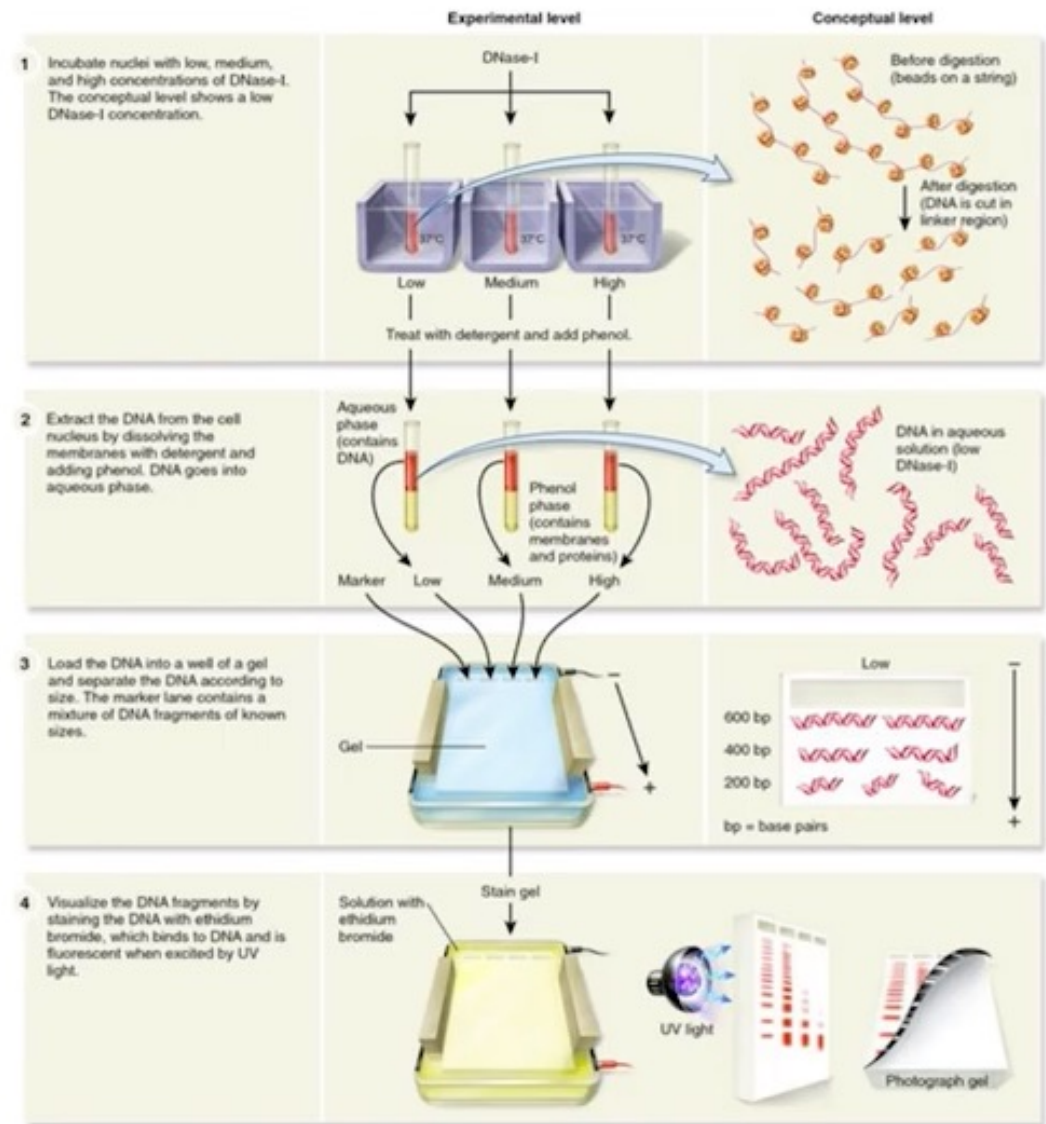
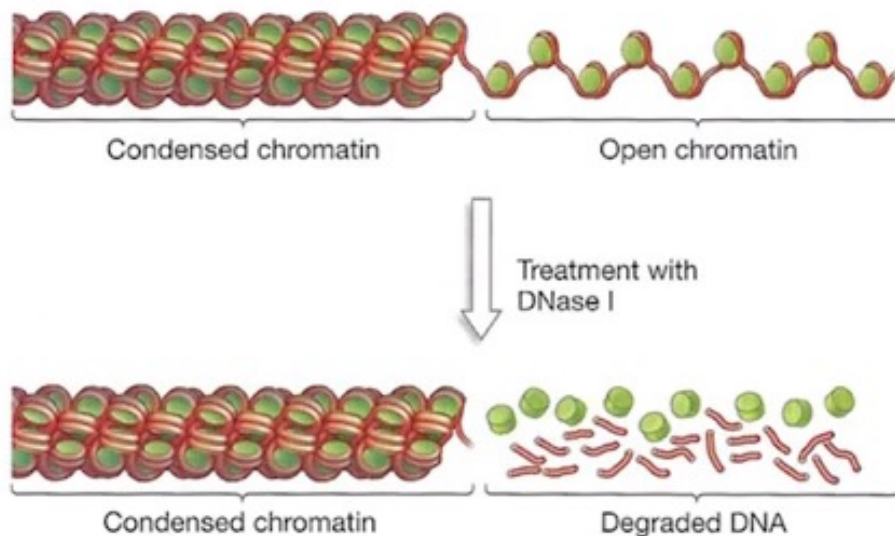


What information can open chromatin provide?

- Quickly determine if epigenetics is involved in response
 - » Logical choice if specific candidate histone modifications are unknown
- Infer which histone modification(s) are of interest based on correlation of open chromatin regions to genomic annotations
- Identify what transcription factors are driving cell fate, disease or response
- Mechanistic insight into how your pathway/cellular response/disease is regulated

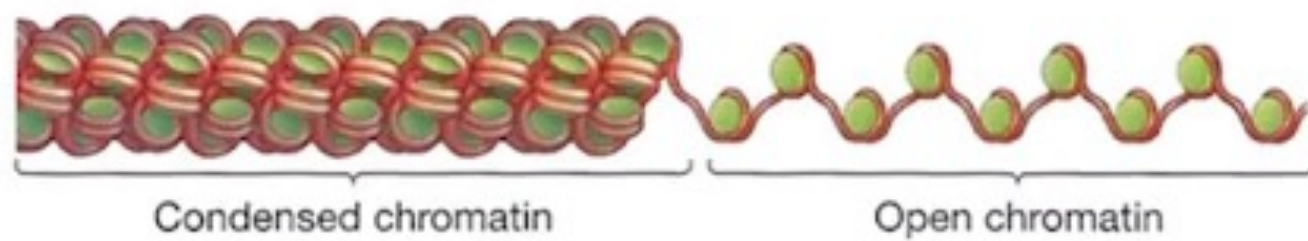
DNaseI-seq

Digestion of accessible DNA with DNaseI enzyme

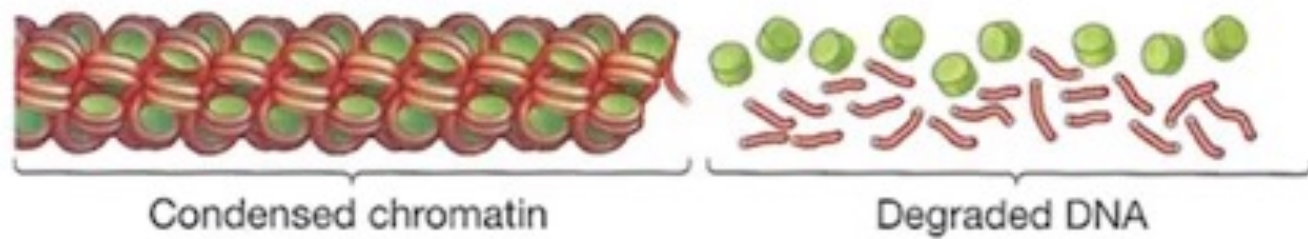


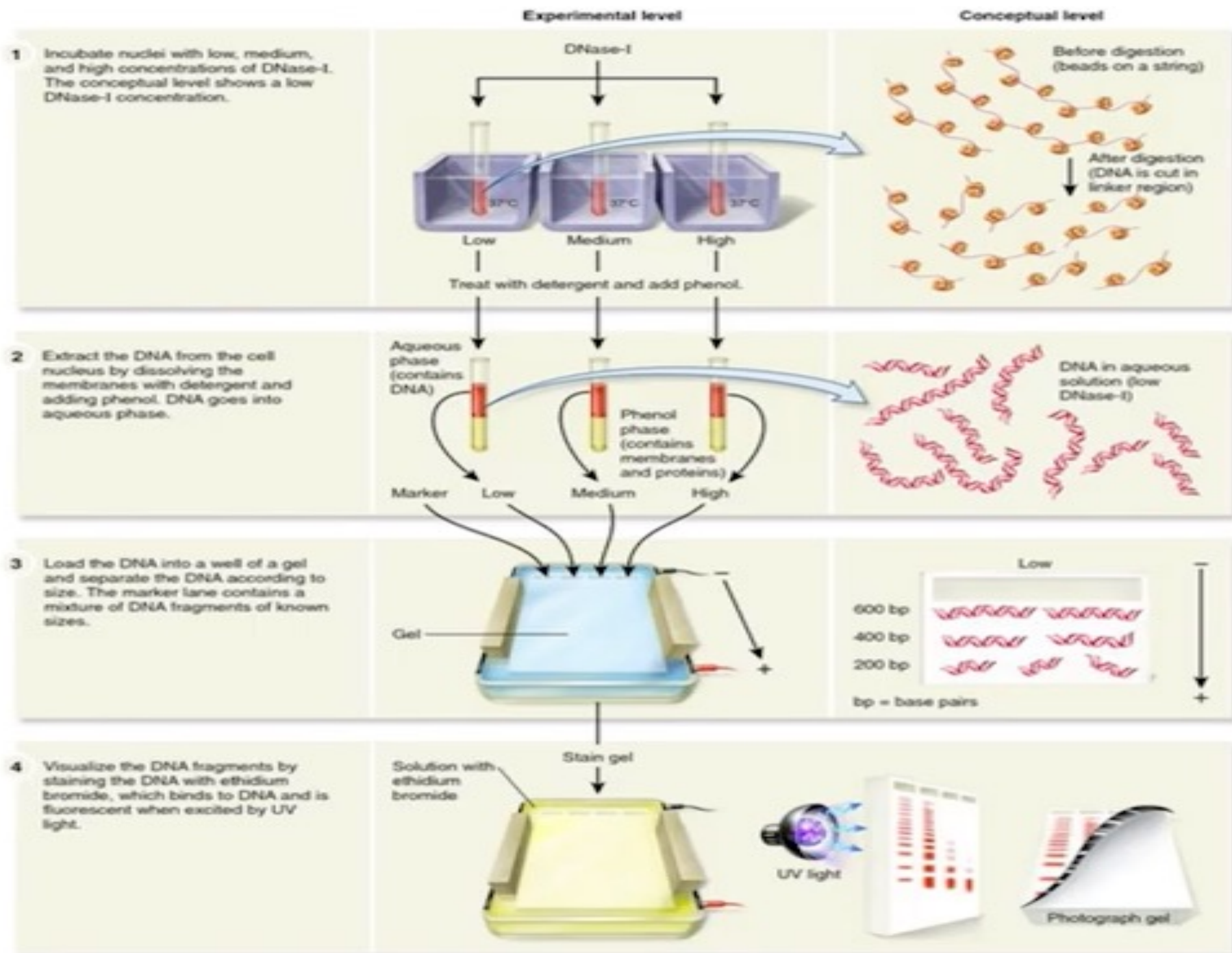
<http://www.discoveryandinnovation.com/BIOL202/notes/lecture18.html>

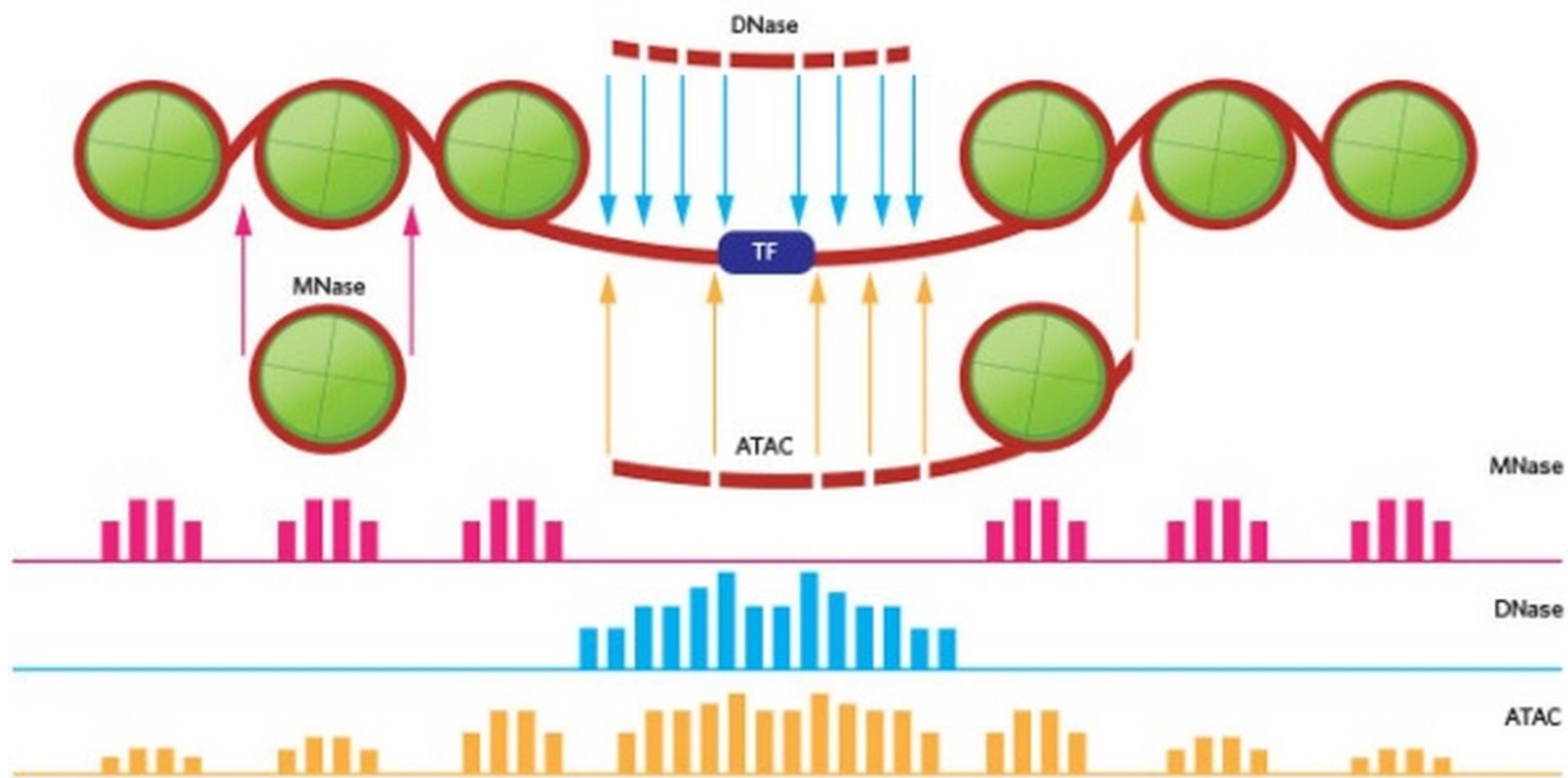
<http://biology-forums.com/index.php?action=gallery;sa=view;id=487>

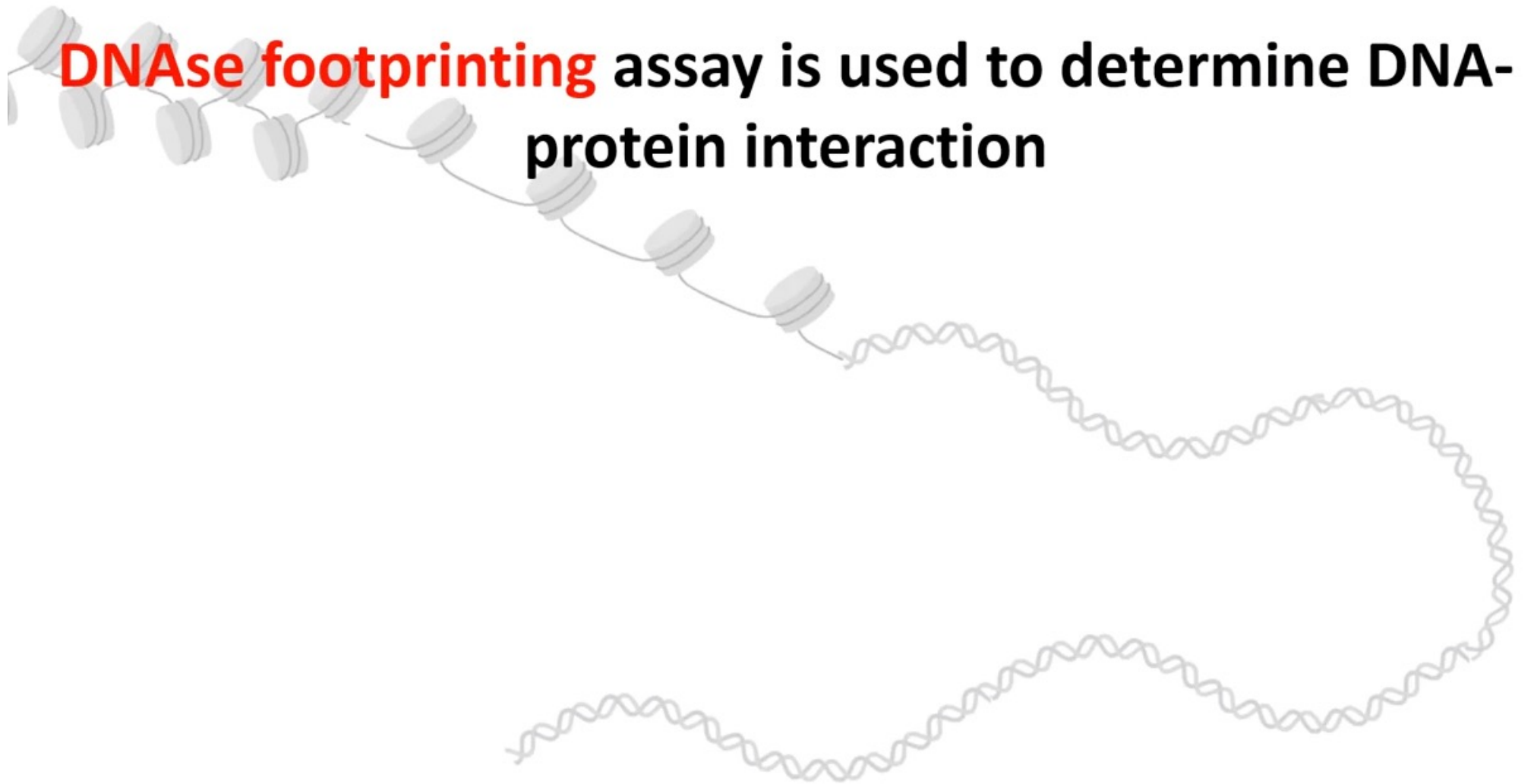


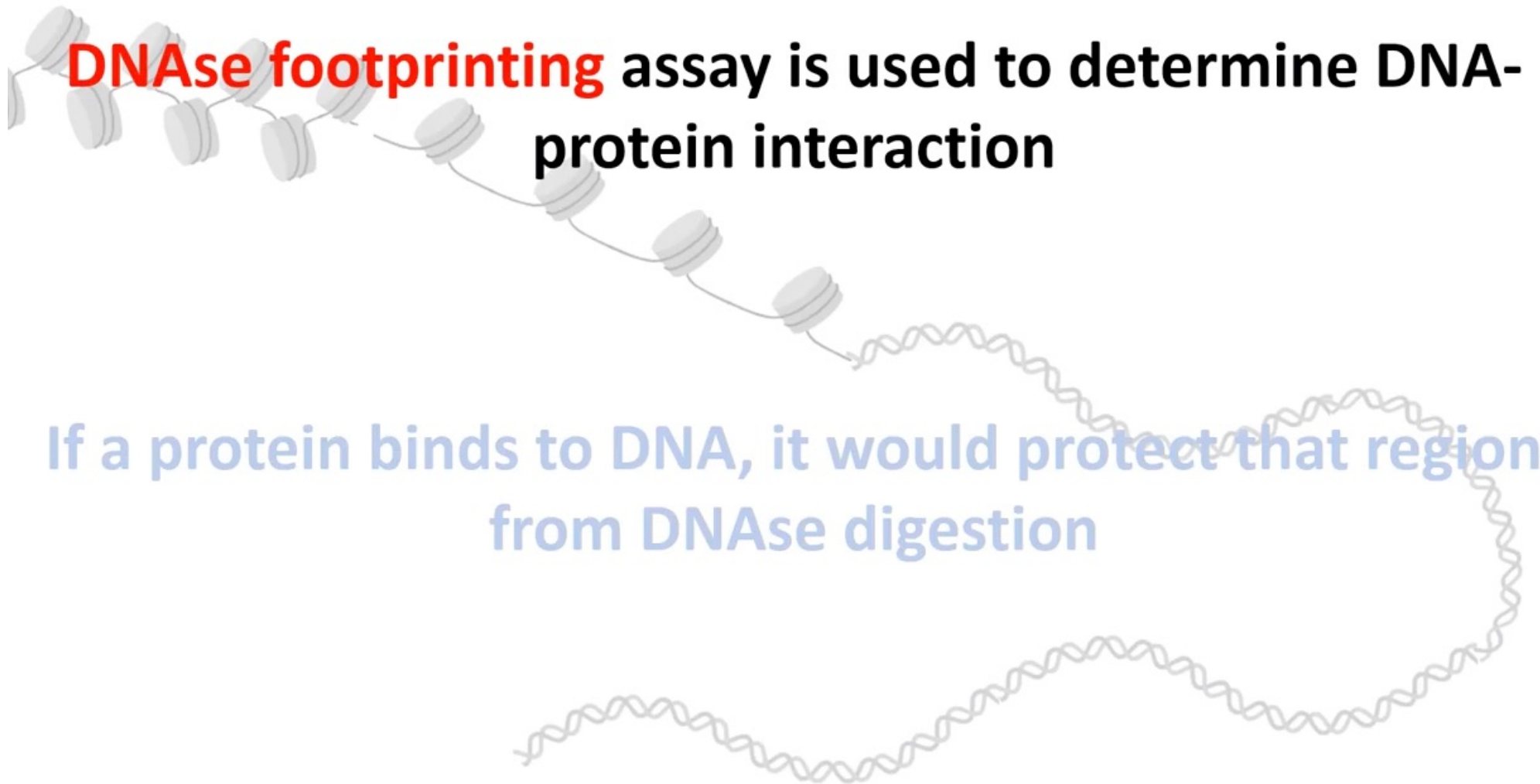
Treatment with
DNase I





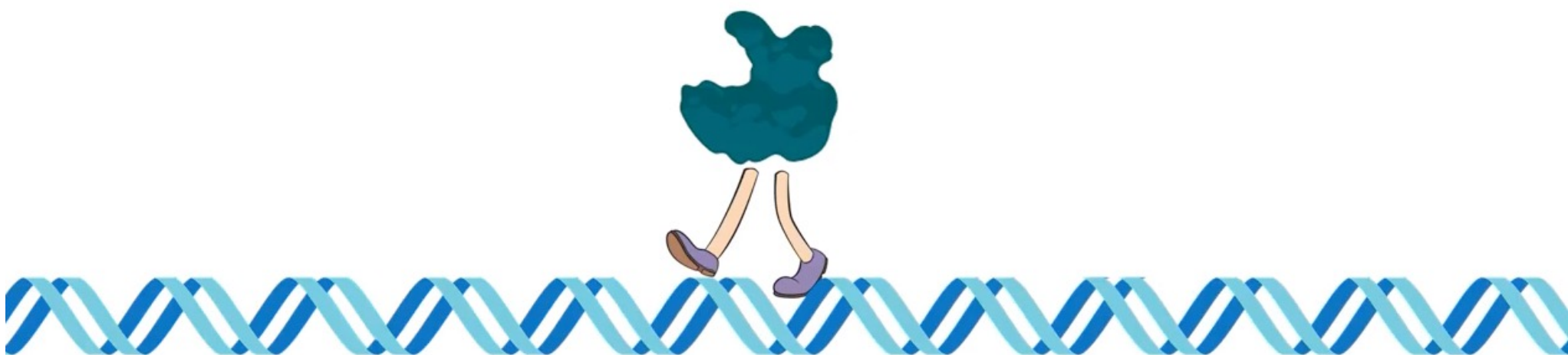


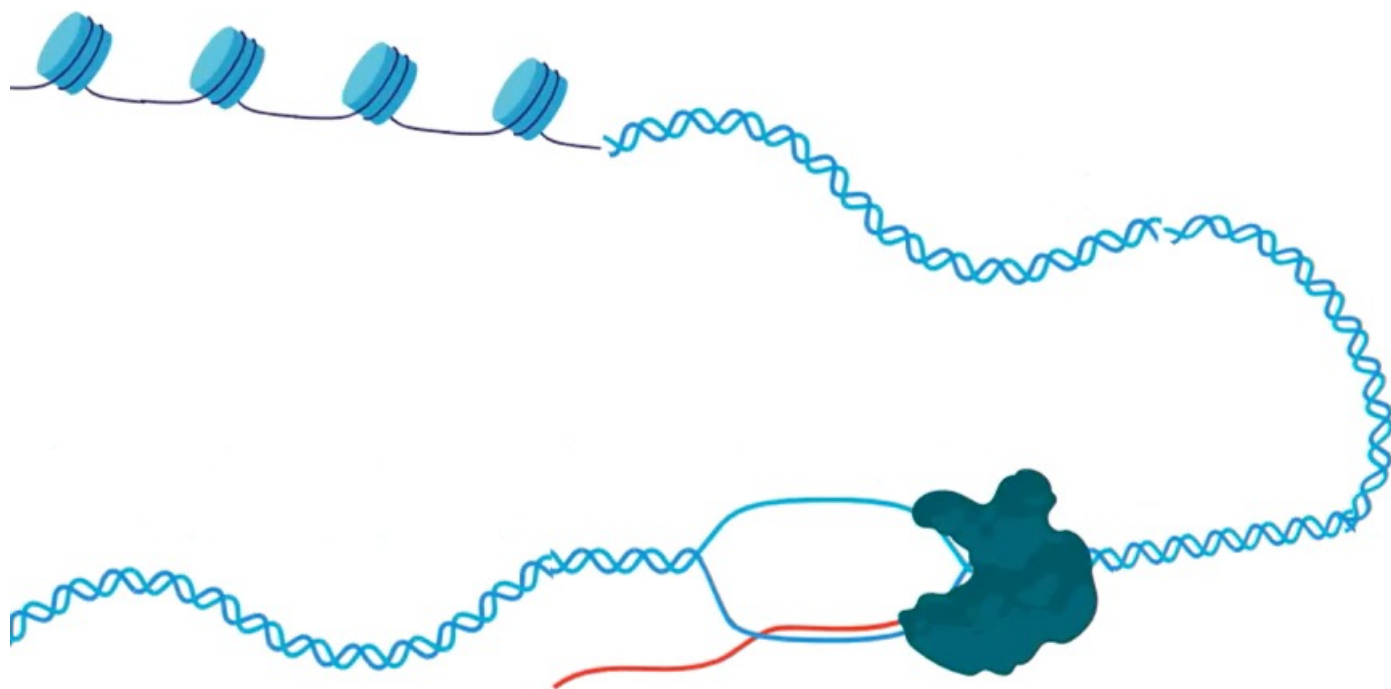




DNAse footprinting assay is used to determine DNA-protein interaction

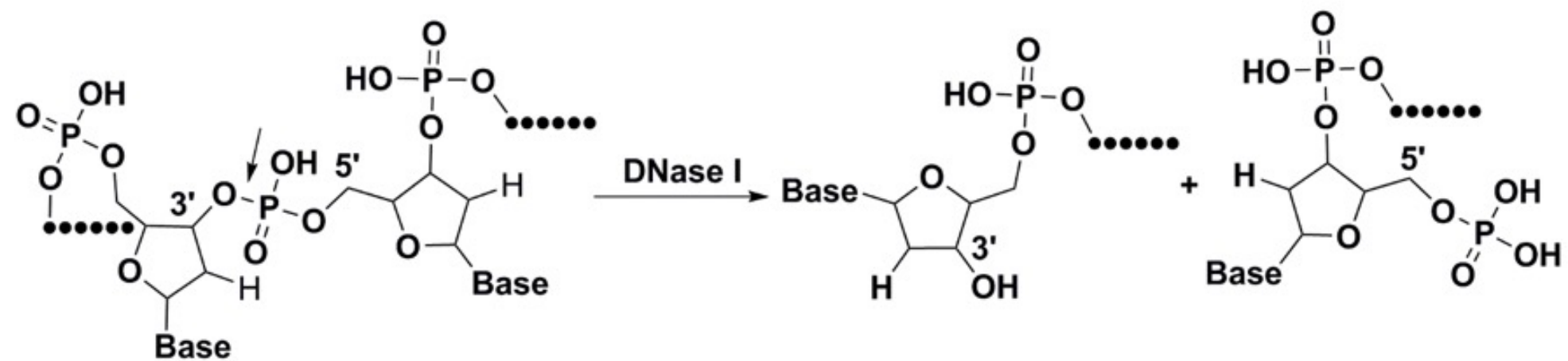
If a protein binds to DNA, it would protect that region from DNAse digestion

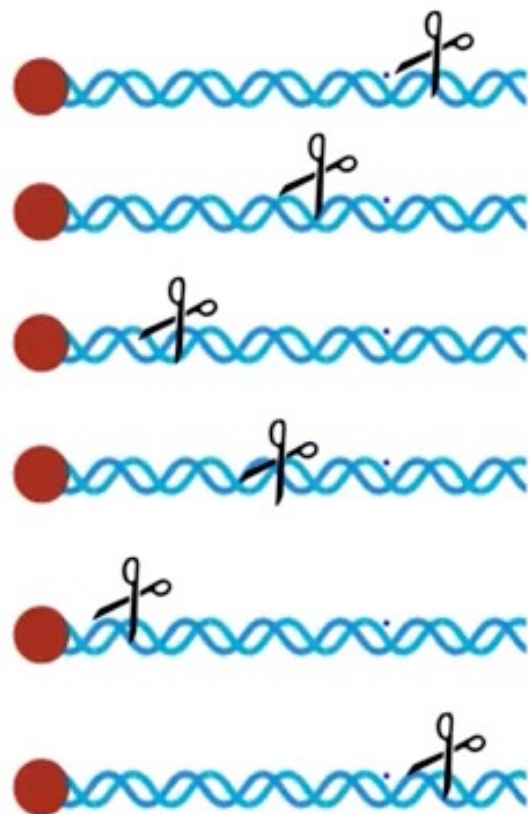




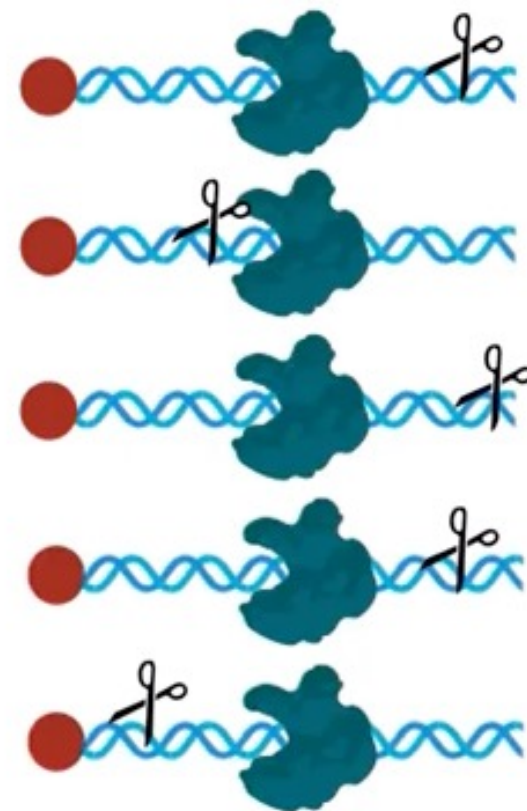
Why it is called foot printing?

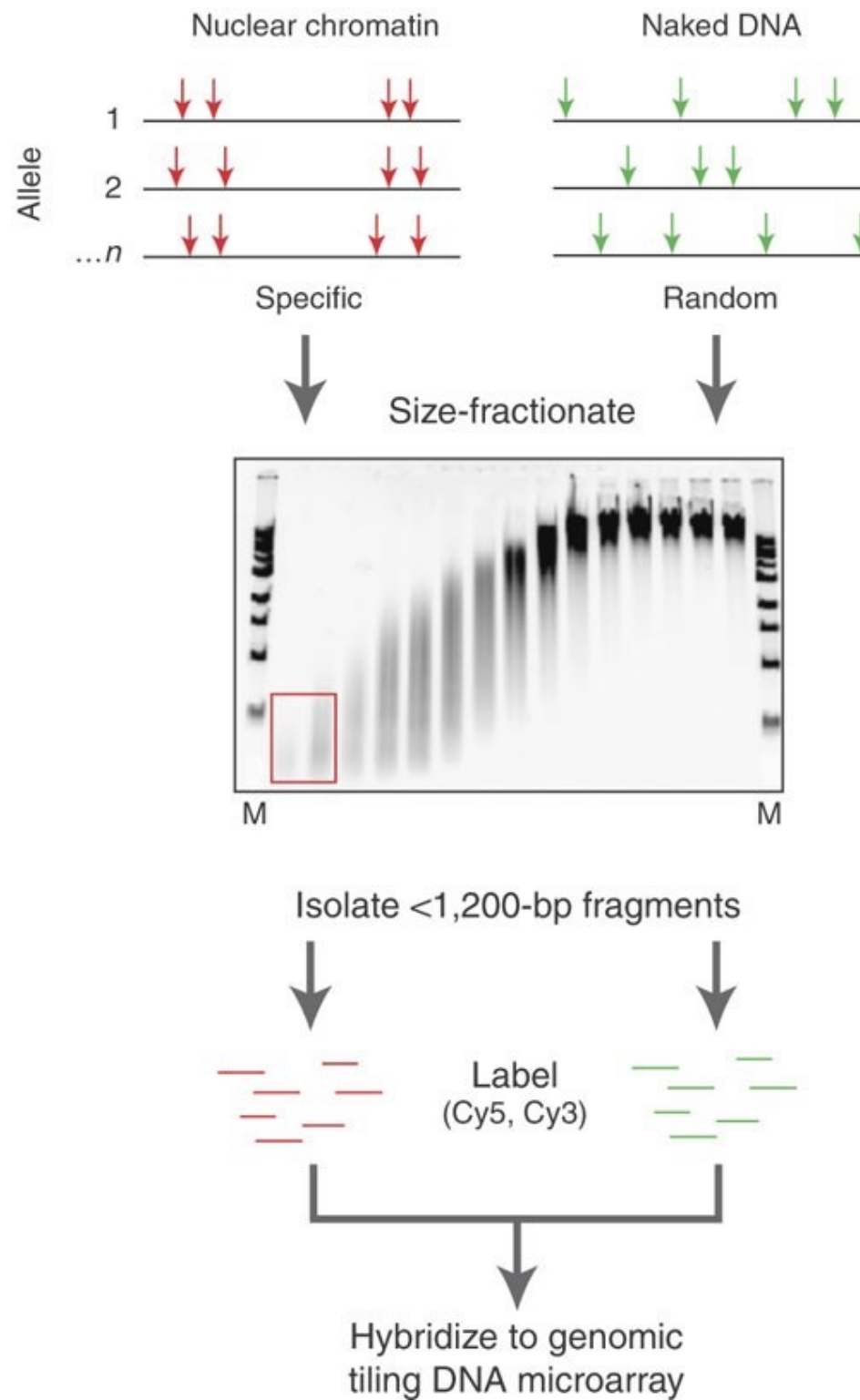


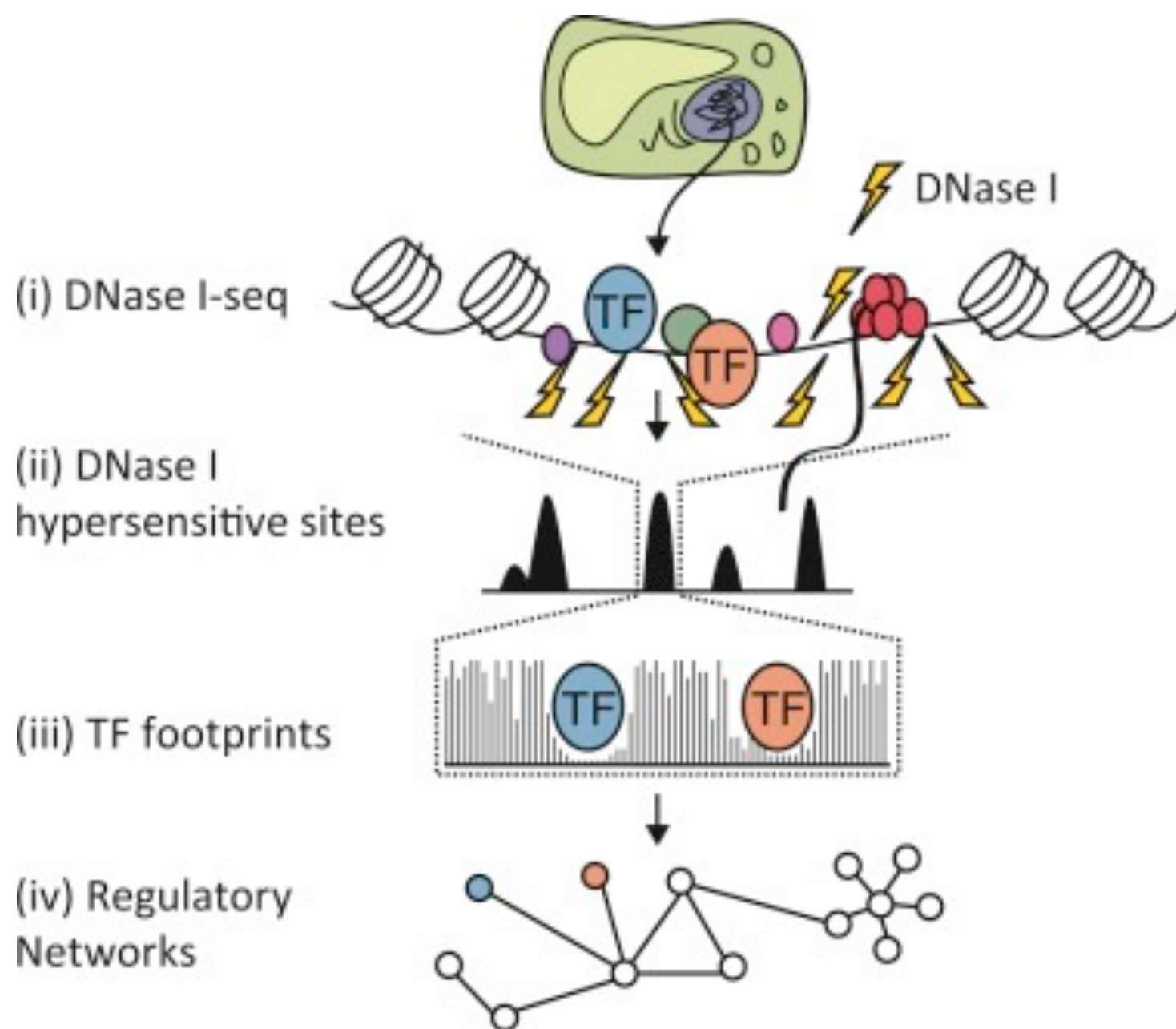


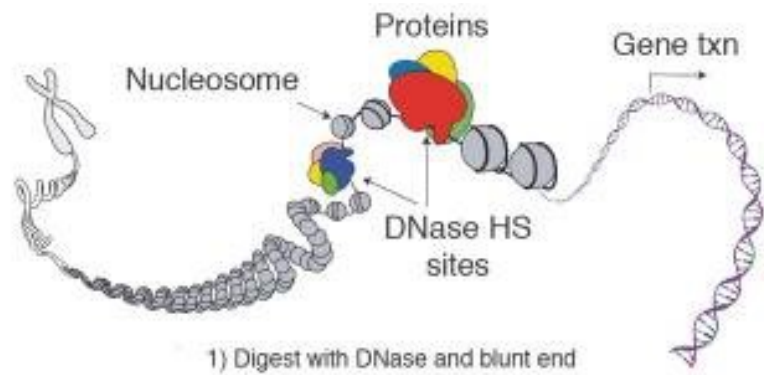


TCGAGGCAACGC AAACAACGTCCTTG
 (-44)
 (-21)



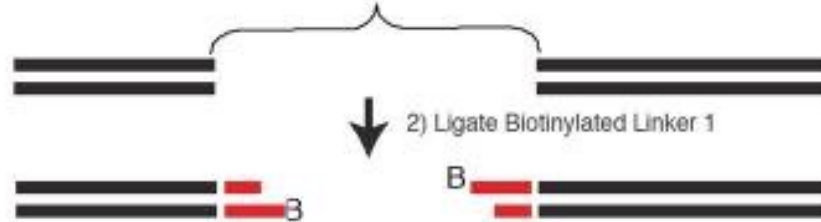






1) Digest with DNase and blunt end

DNase HS site



2) Ligate Biotinylated Linker 1



3) MmeI digested, bind to Dynal beads



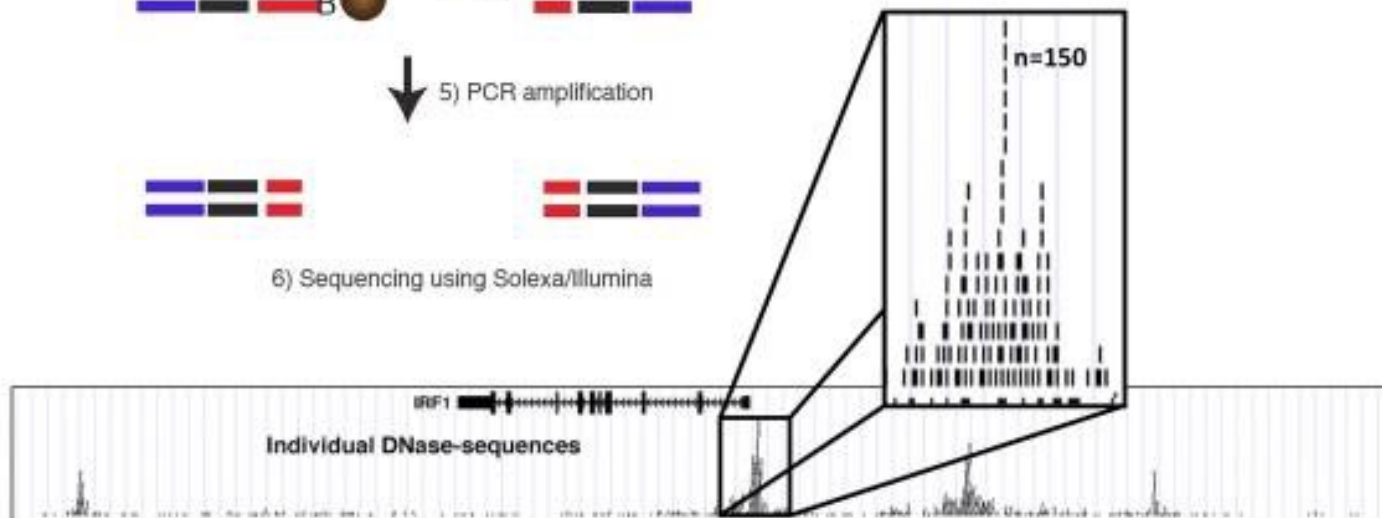
4) Ligate Linker 2



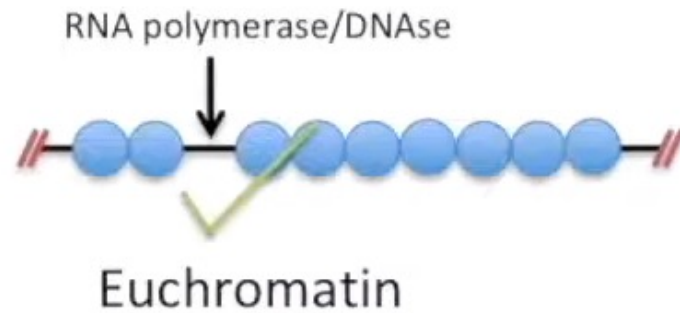
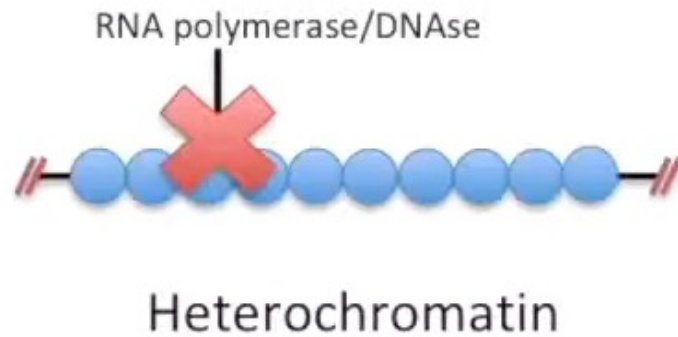
5) PCR amplification



6) Sequencing using Solexa/Illumina



Transcription in Chromatin?





↓ DNase



↓ Remove Protein



↓ RE



13 kb



↓ DNase



↓ Remove Protein

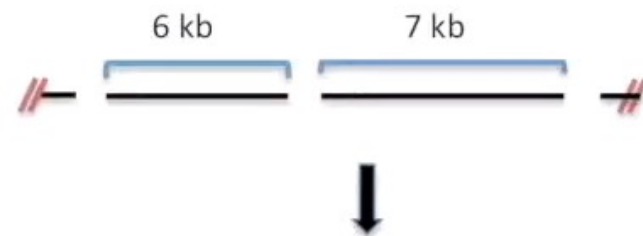
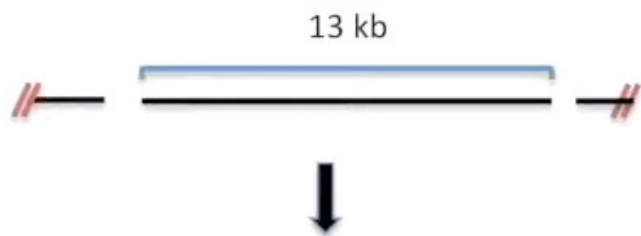


↓ RE

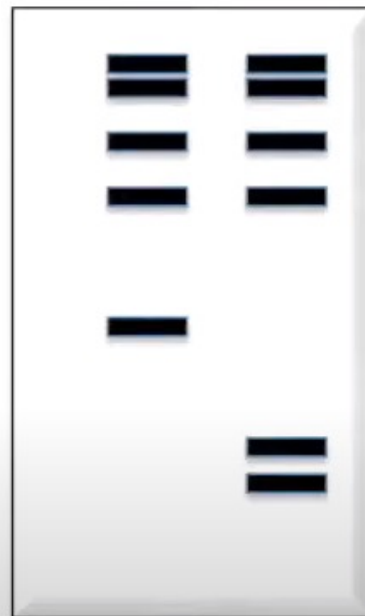


6 kb

7 kb



Gel Electrophoresis

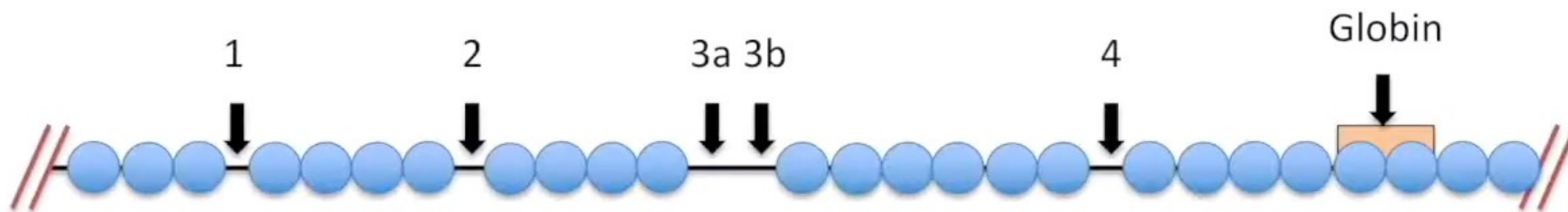


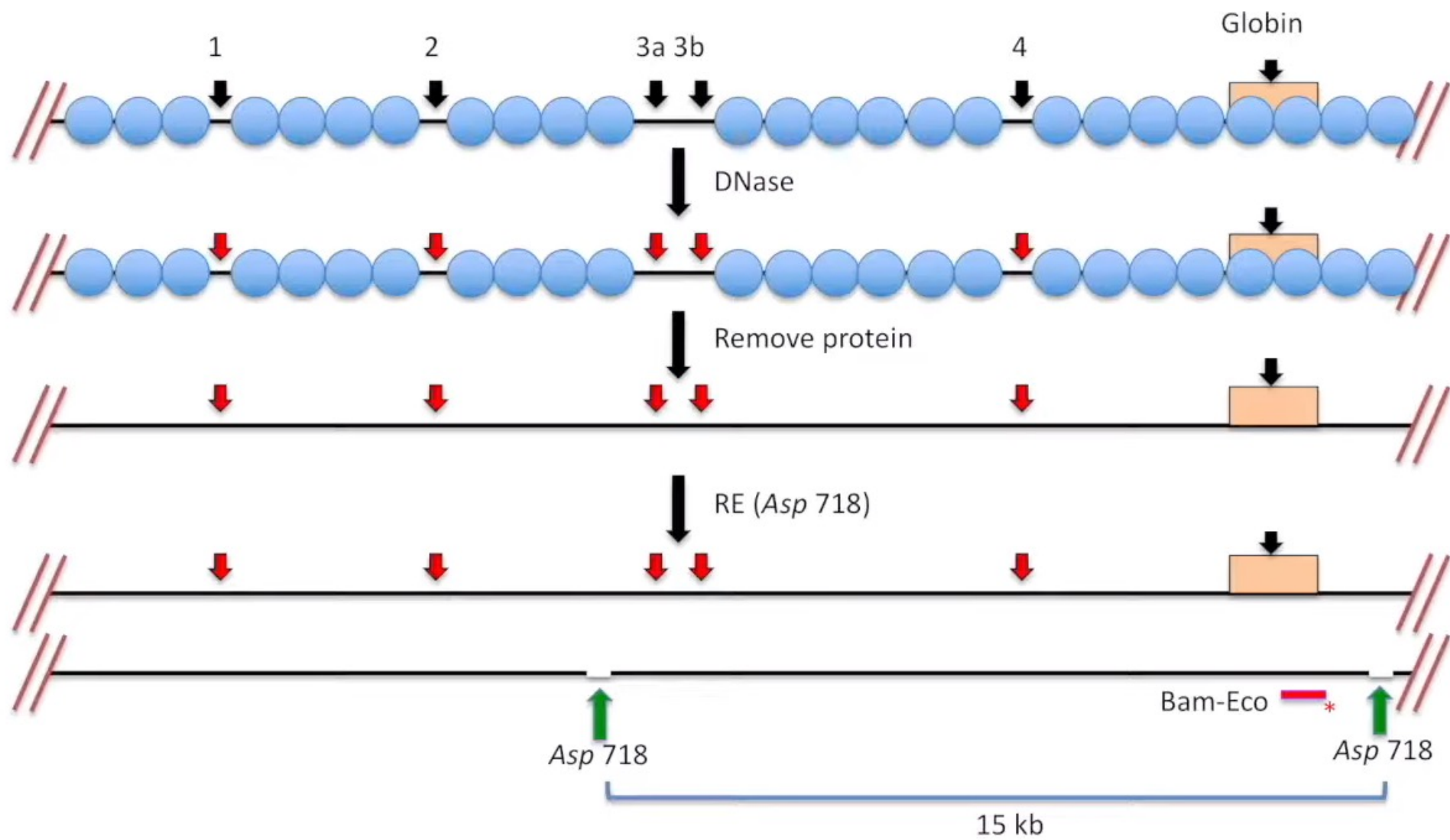
“Shown are just a few fragments to illustrate that there are multiple DNA fragments other than in your region of interest”

← 13 kb

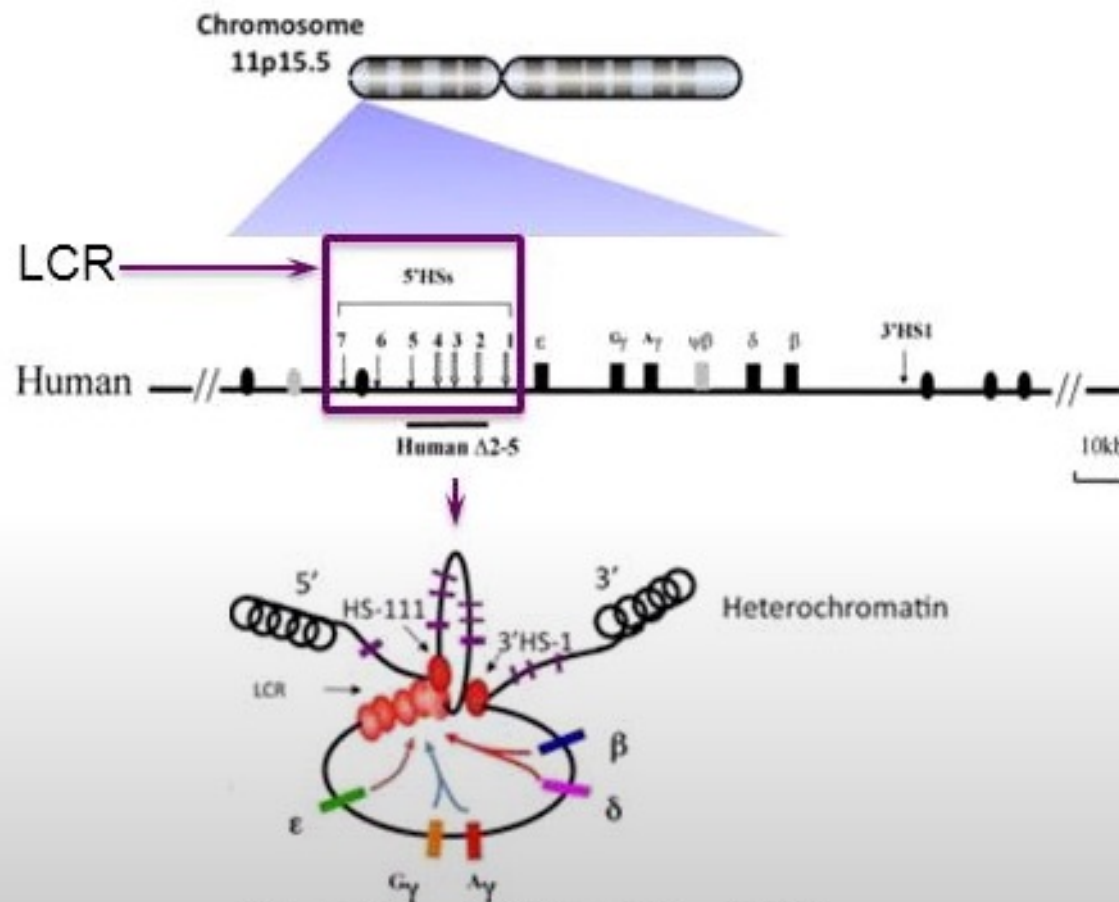
← 7 kb

← 6 kb

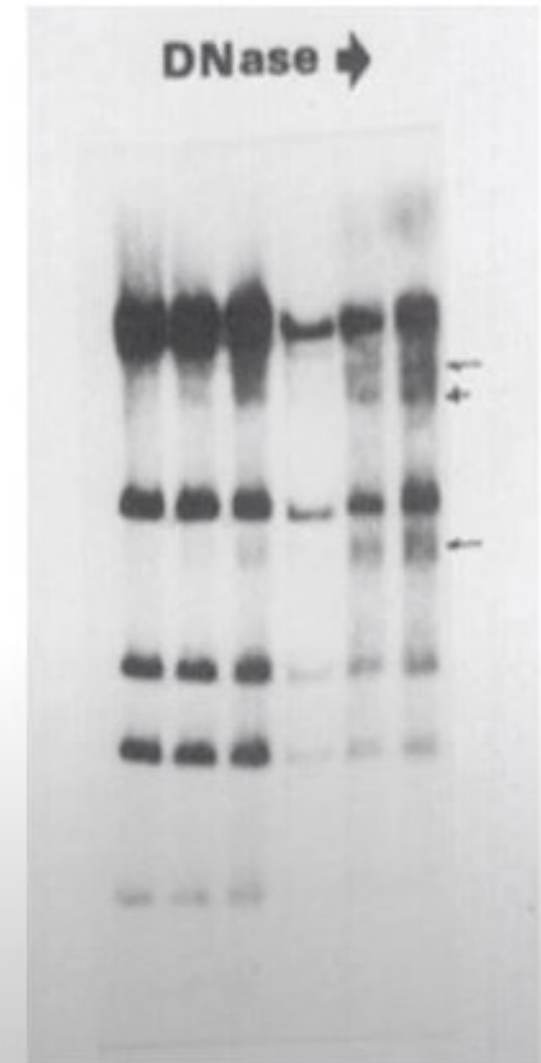




Distal cis-regulatory elements first identified using DNaseI Hypersensitivity in the β -globin locus

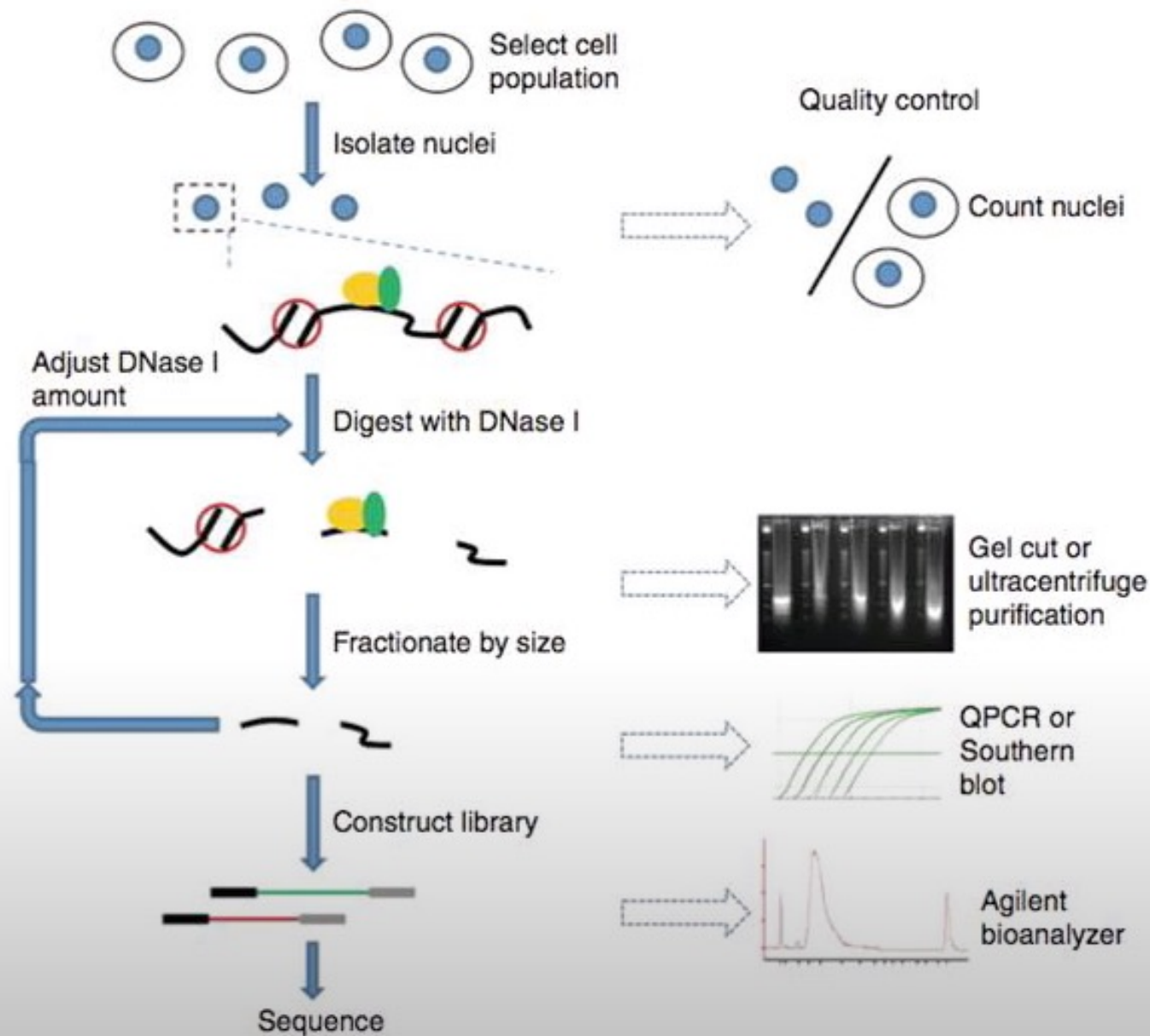


Adapted from Lii Q, et al Blood 2002



Stalder J et al, Cell 1980

DNase-seq experimental design



Information derived from DNase-seq experiments

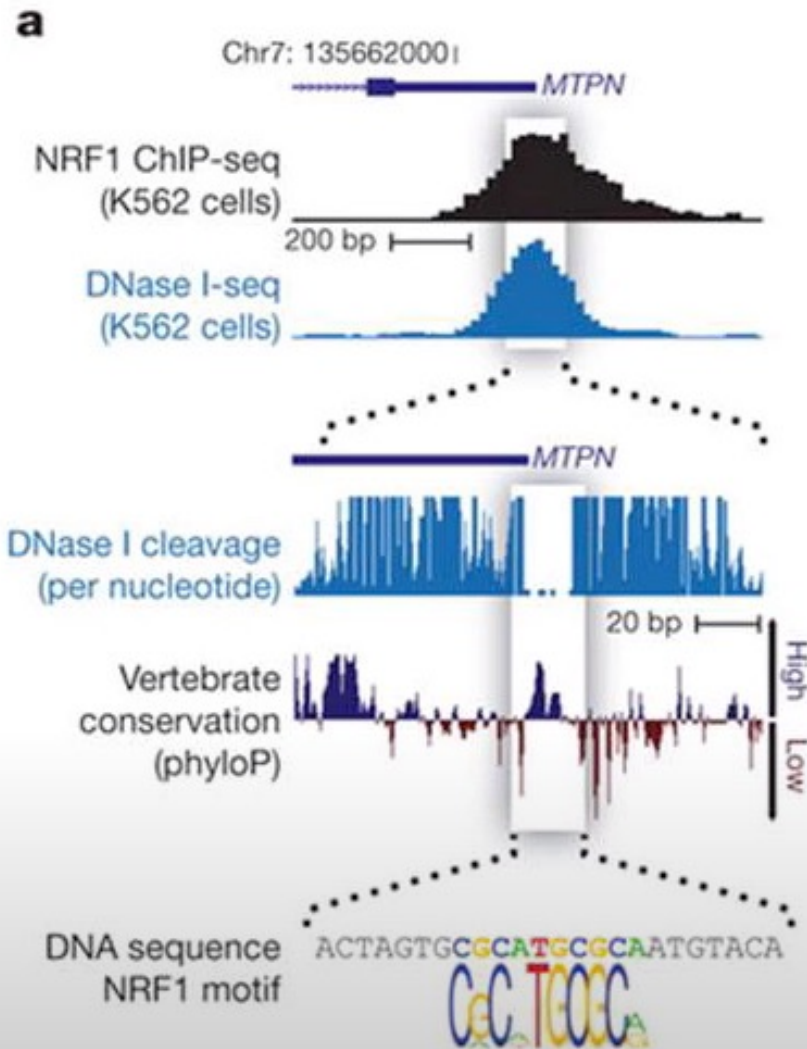
St

- Multiple states and profiles of chromatin accessible elements (promoters, enhancers, silencers) can be identified.
- Binding sites of multiple transcription factors can be derived from DNase-seq data (DNaseI footprinting).

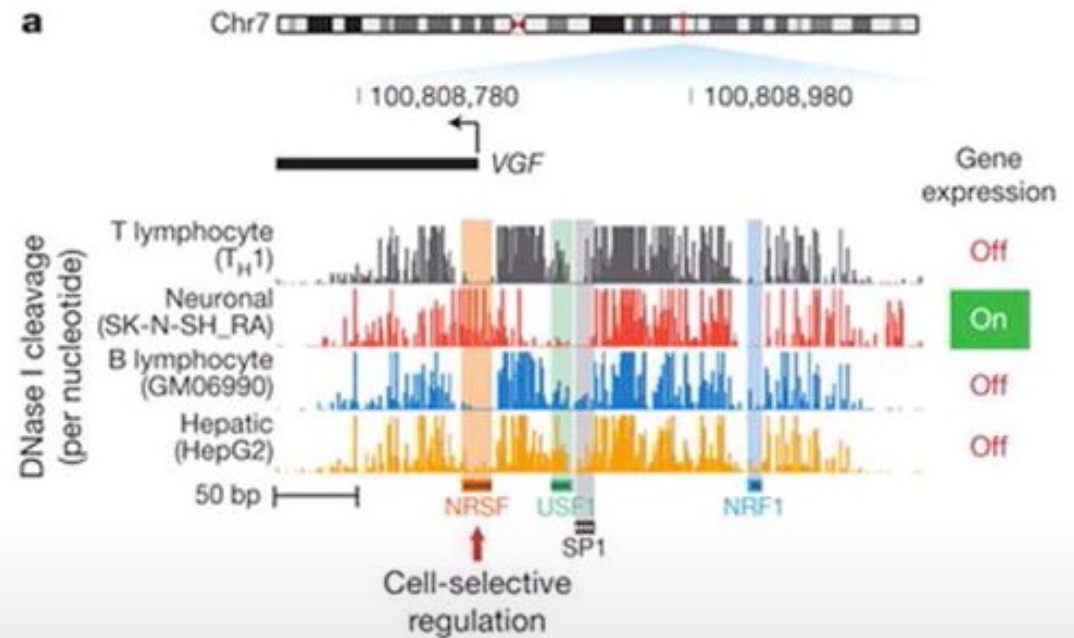
DNaseI footprinting



Summer School

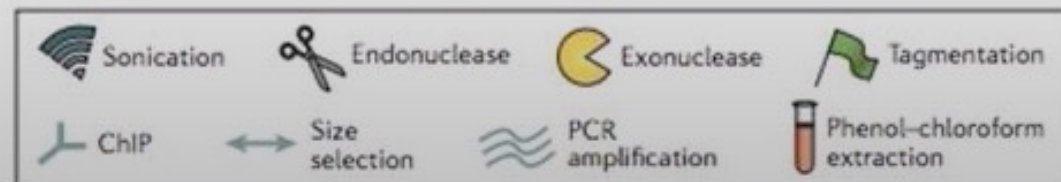
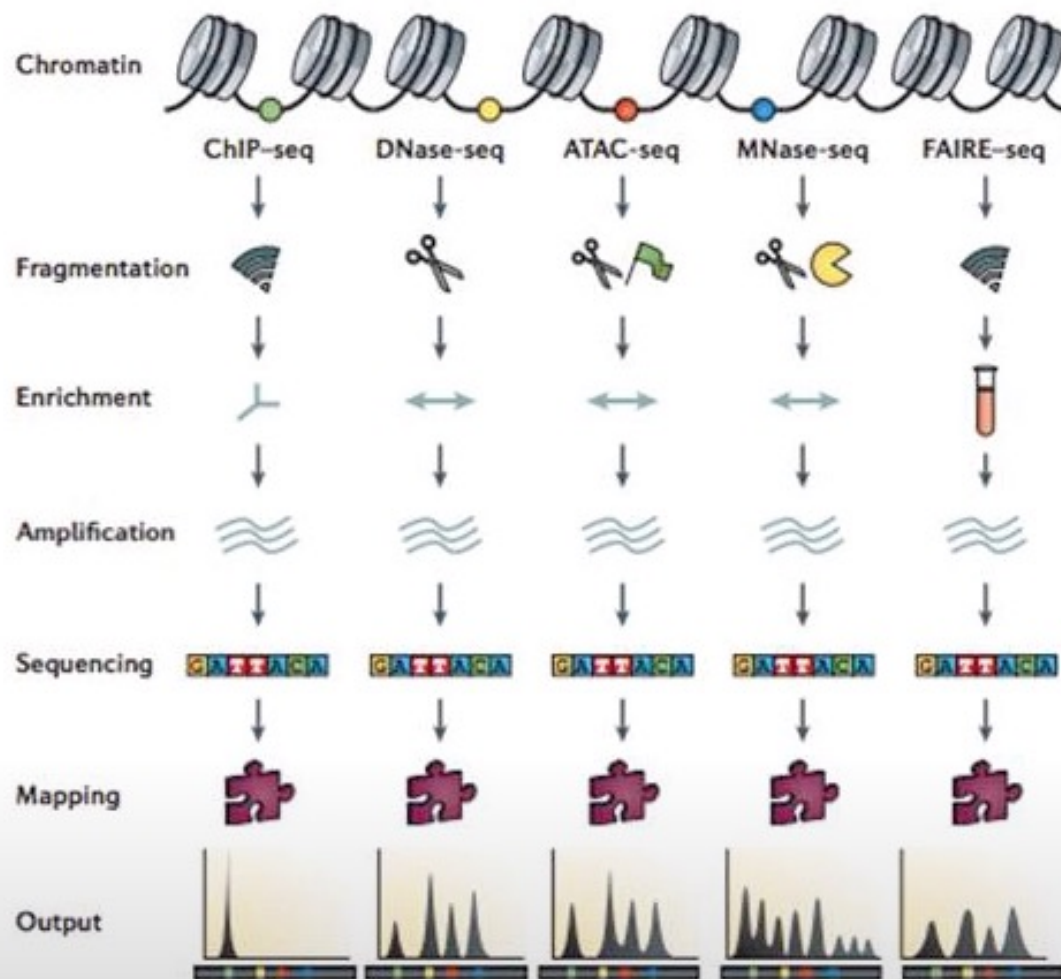


Neph et al 2012 Nature

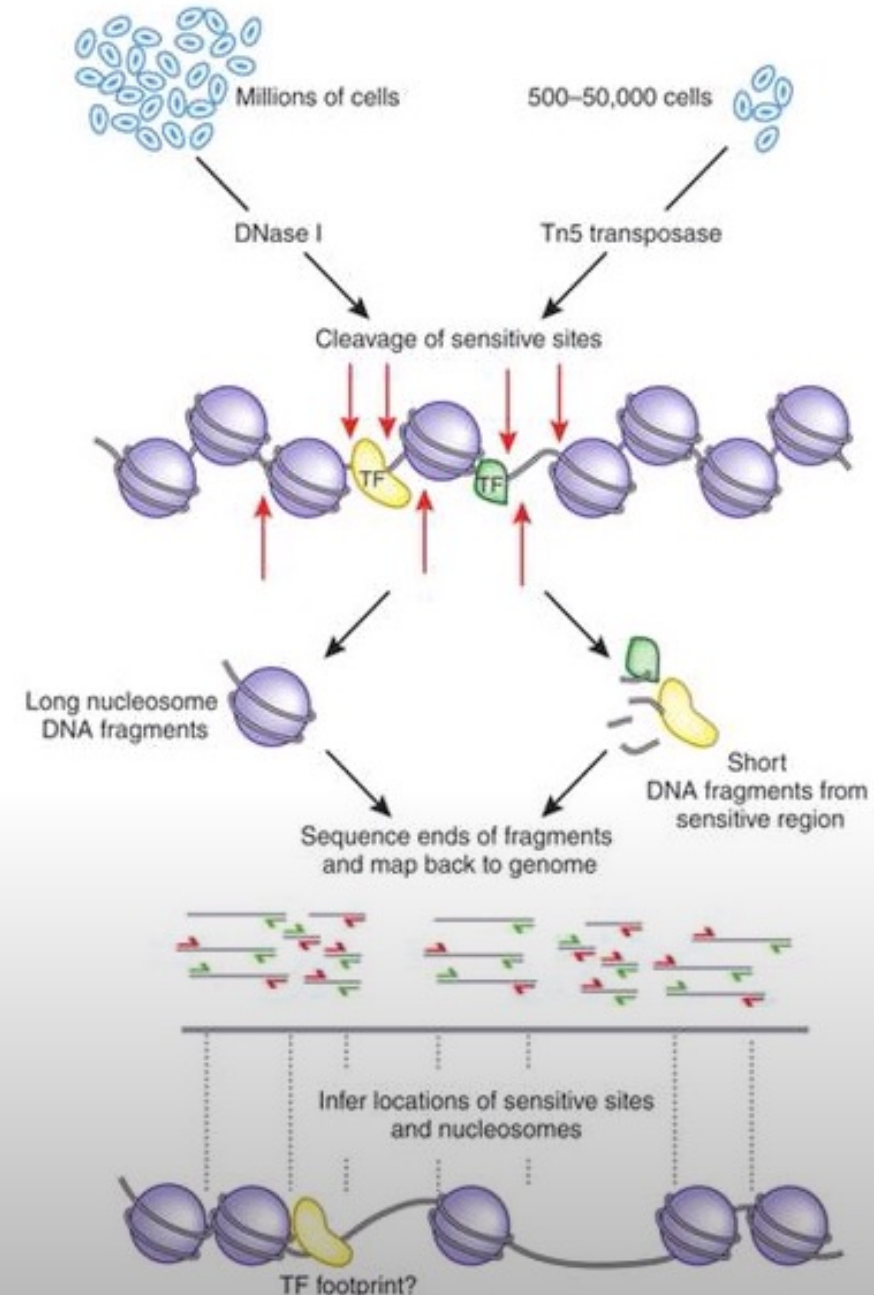


Neph et al 2012 Nature

Surveying chromatin dynamics genome-wide



Meyer et al, Nature Reviews Genetics 2014



Raj et al, Nature Methods 2013

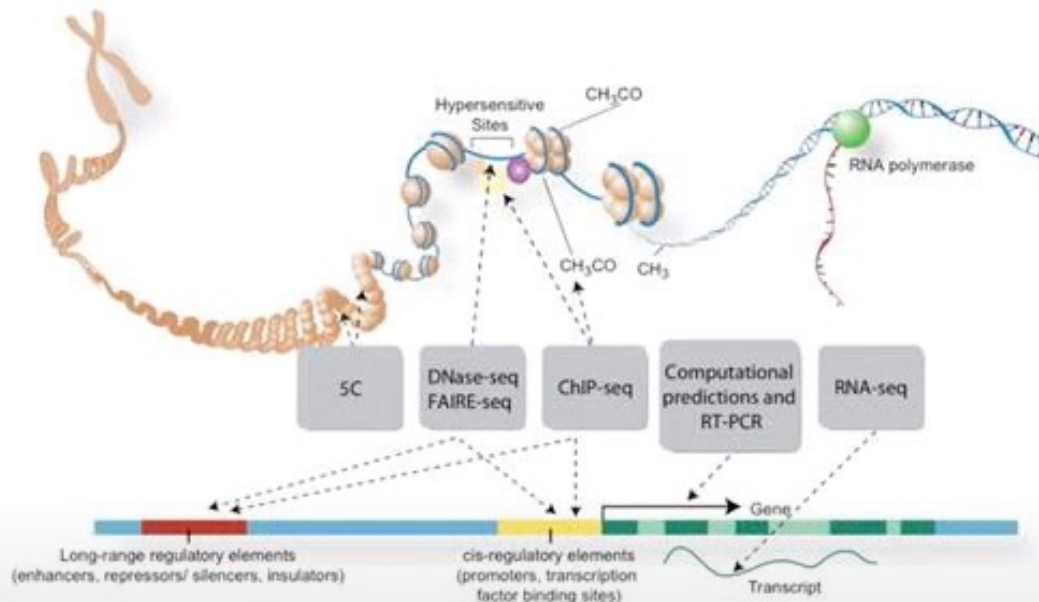
Chromatin accessibility genome-wide



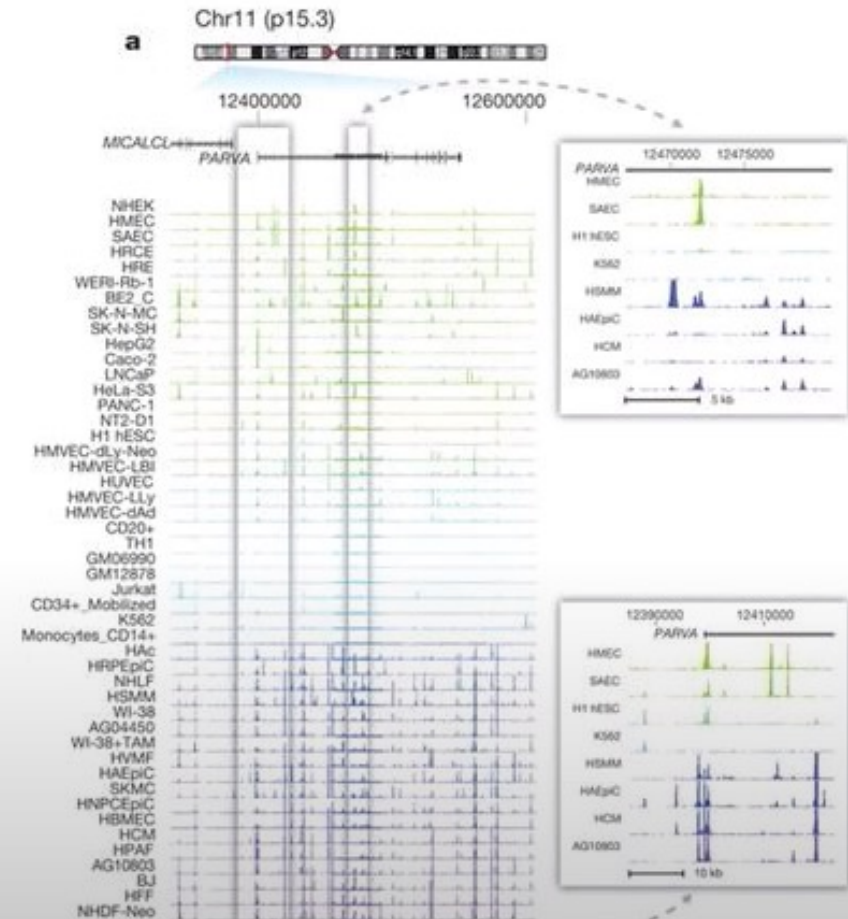
Summer School

ENCyclopedia Of DNA Elements (ENCODE)

1% of the genome is DNaseI Hypersensitive



ENCODEproject.org



Thurman et al, Nature 2012